MECHANICAL LEGEND

T	THERMOSTAT
H	HUMIDITY SENSOR
S1 / 300	AIR TERMINAL TAG WITH AIR FLOW (LPS)
300x250	RECTANGULAR DUCT DIMENSIONS (MILLIMETERS)
300ø	ROUND DUCT DIMENSION (MILLIMETERS)
	FLEXIBLE ROUND DUCT
	DOWNWARD DIRECTION OF SLOPED PIPING
	SUPPLY DUCT, SUPPLY DUCT DOWN/UP
	RETURN DUCT, RETURN DUCT DOWN/UP
	GATE VALVE
	GLOBE VALVE
	BUTTERFLY VALVE
	MANUAL FLOW CONTROL BALANCING VALVE
	CHECK VALVE
	AUTOMATIC FLOW BALANCING VALVE
	PRESSURE REDUCING VALVE
	SQUARE HEAD BALANCING COCK
	DRAIN VALVE WITH HOSE CONNECTION
	2-WAY CONTROL VALVE
	3-WAY CONTROL VALVE
	RELIEF VALVE
	BACKFLOW PREVENTER
	MANUAL AIR VENT
	AUTOMATIC AIR VENT
	VACUUM BREAKER
	WYE STRAINER
	VALVE IN VERT. PIPING
	SPRINKLER HEAD
	STRAINER W/ BLOWDOWN VALVE
C.O.	CLEAN OUT
	ECCENTRIC REDUCER
	PIPE REDUCER
	UNION
	THERMOMETER IN PIPE WELL
	PRESSURE GAUGE W/ SHUTOFF COCK
	PIPE CAP
	FLANGE
	FLEXIBLE PIPE CONNECTOR
SP	STATIC PRESSURE SENSOR
DPS	DIFFERENTIAL PRESSURE SENSOR
HGD	HYDROGEN GAS DETECTION SENSOR
CAV	CLEAN AGENT VENTILATION SWITCH
E	HVAC EMERGENCY SHUTOFF SWITCH
	DUCTWORK WITH ACOUSTICAL LINING
	RETURN/EXHAUST GRILLE

MD — MOTORIZED DAMPER

FD — FIRE DAMPER

AD — ACCESS DOOR

VOLUME DAMPER

MECHANICAL SYMBOLS

---(NAME)---	PIPING TO BE REMOVED
——A——	COMPRESSED AIR PIPING
——CDR——	CONDENSER WATER RETURN PIPING
——CDS——	CONDENSER WATER SUPPLY PIPING
——CWR——	CHILLED WATER RETURN PIPING
——CWS——	CHILLED WATER SUPPLY PIPING
——D——	CONDENSATE DRAIN PIPING
——DCW——	DOMESTIC COLD WATER PIPING
——DTS——	DUAL TEMPERATUR SUPPLY PIPING
——DTR——	DUAL TEMPERATURE RETURN PIPING
——G——	NATURAL GAS PIPING
——HG——	HOT GAS BYPASS PIPING
——HPS——	HIGH PRESSURE STEAM PIPING
——HWR——	HOT WATER RETURN PIPING
——HWS——	HOT WATER SUPPLY PIPING
——LPC——	LOW PRESSURE CONDENSATE PIPING
——LPS——	LOW PRESSURE STEAM PIPING
—— RL ——	REFRIGERANT LIQUID PIPING
—— RS ——	REFRIGERANT SUCTION PIPING
——VAC——	VACUUM (AIR)

GENERAL MECHANICAL NOTES

1. THESE GENERAL MECHANICAL NOTES APPLY TO THE ALL MECHANICAL DRAWINGS IN THIS SET. SHEET SPECIFIC NOTES ARE INDICATED ON EACH SHEET UNDER GENERAL SHEET NOTES.
2. PATCH ALL HOLES FROM REMOVAL OF PIPING, DUCTWORK, OR EQUIPMENT TO MATCH ADJACENT CONSTRUCTION.
3. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
4. INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, THIS CONTRACT'S DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.
5. PROVIDE VIBRATION ISOLATORS FOR ALL MECHANICAL EQUIPMENT TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.
6. PROVIDE ACCESS PANELS IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVE FIRE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT THAT REQUIRES ROUTINE MAINTENANCE. ACCESS PANELS BE LARGE ENOUGH TO ALLOW PROPER CLEARANCE FOR ACCESS DOORS.
7. ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC., SHALL BE FIRE STOPPED.
8. EXISTING DUCT AND PIPING SIZES INDICATED ARE APPROXIMATE. FIELD VERIFY ALL DIMENSIONS.
9. LOCATE ALL NEW HORIZONTAL DUCTWORK ABOVE THE SUSPENDED CEILINGS UNLESS OTHERWISE INDICATED.
10. ALL 90 DEGREE RECTANGULAR MITERED DUCT ELBOWS IN THE SUPPLY AND RETURN DUCTS SHALL HAVE TURNING VANES.
11. PROVIDE PIPE SLEEVES FOR ALL PIPING PENETRATIONS THRU WALLS AND FLOORS. EACH PIPE SHALL BE SEPARATELY SLEEVED. PROVIDE CONTINUOUS SEALANT JOINT AROUND PIPE SLEEVE, BOTH SIDES OF WALL, WATERTIGHT.
12. PROVIDE P-TRAPS IN CONDENSATE DRAIN PIPING AT EACH AHU UNLESS THE AHU IS EQUIPPED WITH AN INTERNAL TRAP.
13. AFTER PRELIMINARY OPERATIONAL TESTING OF ALL BOILERS OR DIRECT FIRED DOMESTIC WATER HEATERS WITH A CAPACITY GREATER THAN 120 KILOWATTS, THE BOILER SHALL BE INSPECTED AND AN OPERATIONAL TEST SHALL BE PERFORMED BY A NAVFAC FACILITIES ENGINEERING COMMAND BOILER INSPECTOR. ALLOW 15 DAYS ADVANCE NOTICE FOR SCHEDULING AN INSPECTOR TO CONDUCT AN ACCEPTANCE OPERATIONAL TEST AND INSPECTION.
14. MOUNT THERMOSTATS 1.2 METERS ABOVE THE FINISHED FLOOR UNLESS OTHERWISE INDICATED. REFER TO ELECTRICAL DRAWINGS FOR LOCATION OF LIGHT SWITCHES AND COORDINATE.
15. AFTER THE PIPING IS INSULATED, PROVIDE PIPE LABELS, WITH DIRECTION OF FLOW ARROWS, ON THE FOLLOWING PIPING, AT 10 FOOT ON CENTER ON THE PIPING MAINS AND 1.5 METERS ON CENTER ON THE BRANCH RUNOUTS FROM MAINS.

MECHANICAL ABBREVIATIONS

ø	DIAMETER
°	DEGREES
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AP	ACCESS PANEL
APD	AIR PRESSURE DROP
APPROX	APPROXIMATELY
AS	AIR SEPARATOR
BF	BELOW RAISED FLOOR
BT	BUFFER TANK
C	COMMON/CELSIUS
CH	AIR-COOLED CHILLER
CLG	CEILING
CRAC	COMPUTER ROOM AIR HANDLING UNIT
CU	CONDENSING UNIT
Cv	LOSS COEFFICIENT
CWP	CHILLED WATER PUMP
DB	DRY BULB
DDC	DIRECT DIGITAL CONTROL
DEG	DEGREE
DIA	DIAMETER
DNXX	NOMINAL PIPE SIZE
DX	DIRECT EXPANSION
EA	EXAUST AIR
EAT	ENTERING AIR TEMPRATURE
EER	ENERGY EFFICIENCY RATIO
EF	EXHAUST FAN
ESP	EXTERNAL STATIC PRESSURE
ET	EXPANSION TANK
EWT	ENTERING WATER TEMPERATURE
FLA	FULL LOAD AMPS
HGT	HEIGHT
HW	HOT WATER
IPLV	INTEGRATED PART LOAD VALUE
KG	KILOGRAM
kPA	KILOPASCAL
kW	KILOWATT
L	LOUVER
LAT	LEAVING AIR TEMPERATURE
LPS	LITERS PER SECOND
LWT	LEAVING WATER TEMPERATURE
M	METER
MAX	MAXIMUM
MCA	MINIMUM CIRCUIT AMPS
MERV	MINIMUM EFFICIENCY REPORTING VALUE
MFS	MAXIMUM FUSE SIZE
MIN	MINIMUM
mm	MILLIMETER
MOCP	MAXIMUM OVER CURRENT PROTECTION
MPS	METERS PER SECOND
NC	NOISE CRITERIA/NORMALLY CLOSED
NO	NUMBER/NORMALLY OPEN
OA	OUTSIDE AIR
PA	PASCAL
PD	PRESSURE DROP
PH	PHASE
RA	RETURN AIR
RAD	RADIUS
REF	REFRIGERANT
RPM	REVOLUTIONS PER MINUTE
RS	REFRIGERANT SUCTION
SA	SUPPLY AIR
SEER	SEASONAL ENERGY EFFICIENCY RATING
SP	STATIC PRESSURE
SQ	SQUARE
SSIU	DUCTLESS SPLIT SYSTEM INDOOR UNIT
SSOU	DUCTLESS SPLIT-SYSTEM HEAT PUMP OUTDOOR UNIT
TYP	TYPICAL
UL	UNDERWRITERS LABORATORY
UON	UNLESS OTHERWISE NOTED
V	VOLTS
VAV	VARIABLE AIR VOLUME
VFD	VARIABLE FREQUENCY DRIVE
WB	WET BULB
WPD	WATER PRESSURE DROP

SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'
FOR COMMANDER NAVFAC

ACTIVITY D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES OLS | DRW OLS | CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION
NORFOLK VA

VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

MECHANICAL GENERAL NOTES, SYMBOLS, AND LEGEND

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157176
SHEET 114 OF 170

M-001

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

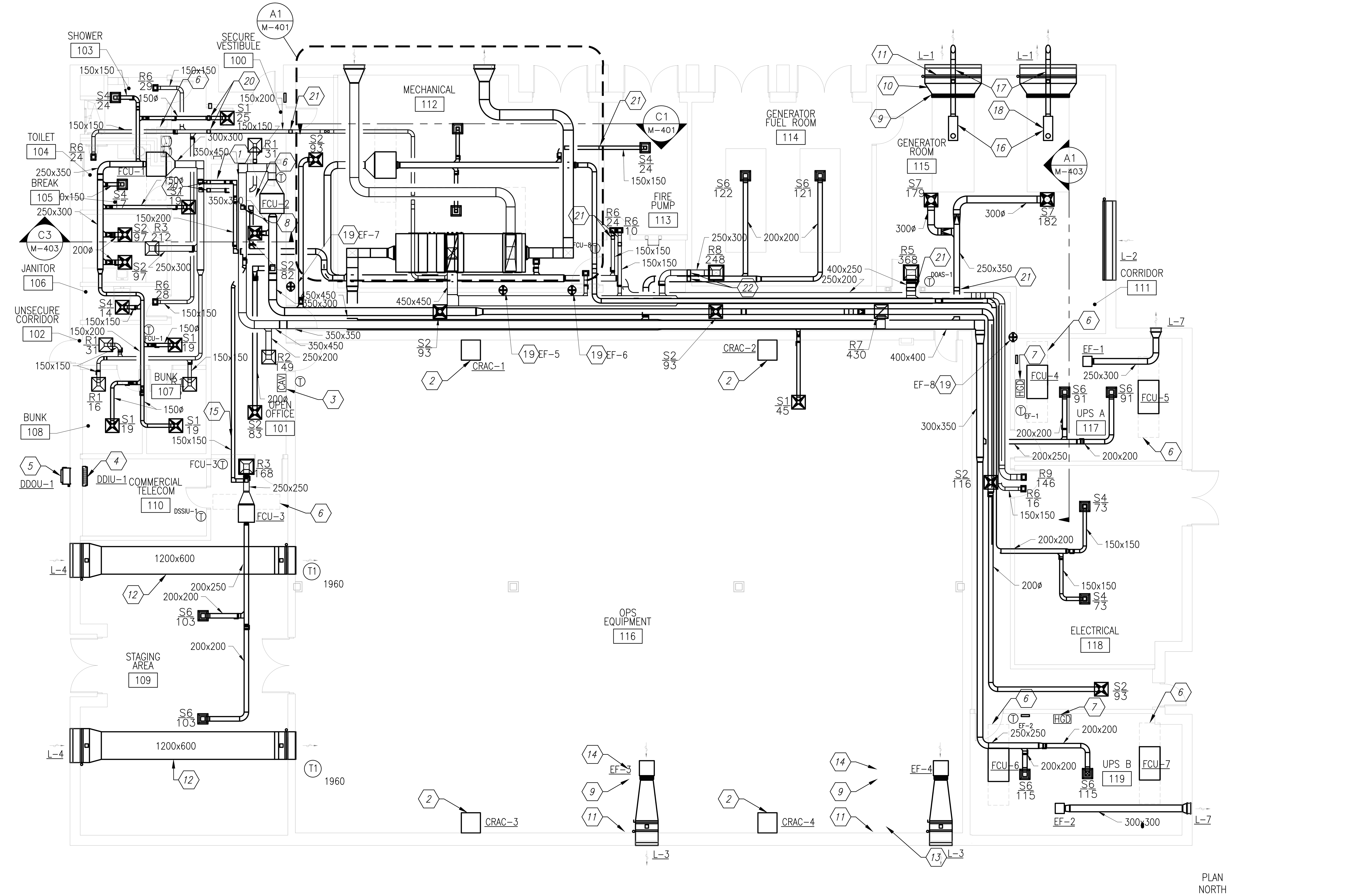
FILE NAME: C:\Users\jennifer.a.klein\AppData\Local\Temp\AcPublish_21216\P1360_1783264_M-001.dwg LAYOUT NAME: M-001 PLOTTED: Tuesday, January 20, 2026 - 3:09pm USER: jennifer.a.klein

GENERAL NOTES

1. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATION OF LOUVERS ABOVE GRADE/FINISHED FLOOR.
2. REFER TO SHEET M-001 LEGEND, GENERAL NOTES, AND ABBREVIATIONS.
3. SUPPLY AND RETURN DUCT RUN OUT DIMENSIONS SHALL MATCH DIFFUSER AND GRILLE NECK DIMENSIONS UNLESS OTHERWISE INDICATED.
4. INSULATE DOAS DUCTWORK IN ACCORDANCE WITH SPECIFICATIONS.
5. PROVIDE MANUAL BALANCING DAMPERS ON ALL BRANCH DUCT RUNS.
6. PROVIDE MITERED ELBOWS WHERE SPATIAL CONSTRAINTS DO NOT ALLOW FOR SMOOTH RADIUS ELBOWS.
7. ALL INTERIOR DUCTWORK TO BE PROVIDED WITH 50MM OF INSULATION IN ALL DIRECTIONS.

SHEET KEYNOTES

1. BALANCE 80 LPS OF OUTDOOR AIR FROM DOAS-1 TO RETURN OF FCU-1
2. FLOOR-MOUNTED COMPUTER ROOM AIR HANDLING UNIT.
3. CLEAN AGENT VENTILATION SWITCH
4. WALL-MOUNTED DUCTLESS SPLIT SYSTEM INDOOR UNIT.
5. WALL-MOUNTED DUCTLESS SPLIT SYSTEM OUTDOOR UNIT.
6. CLEARANCE REQUIRED FOR MAINTENANCE, TYPICAL.
7. HYDROGEN DETECTION SYSTEM.
8. BALANCE 40 LPS OF OUTDOOR AIR FROM DOAS-1 TO RETURN OF FCU-2.
9. 150 WIDE FLEXIBLE DUCT CONNECTOR, TYP.
10. TRANSITION DUCT FROM FULL SIZE OF GENERATOR DISCHARGE TO FULL SIZE OF EXHAUST LOUVER OPENING, TYP.
11. FULL SIZED CONTROL DAMPER, MOTORIZED (LOW VOLTAGE), THERMALLY INSULATED, LOW LEAKAGE, WITH FLANGE FOR WALL-MOUNTING. EQUIP WITH FAST ACTING ACTUATOR WITH NEMA 3 ENCLOSURE.
12. BOTTOM OF DUCT 2950 mm ABOVE FINISHED FLOOR.
13. ACCESS DOOR, TYPICAL
14. EXHAUST FAN AND DUCTWORK ARE MOUNTED BELOW DROP CEILING.
15. BALANCE 40 LPS OF OUTDOOR AIR FROM DOAS-1 TO RETURN OF FCU-3.
16. POWER GENERATOR SILENCER, CRITICAL GRADE, BOTTOM INLET, END OUTLET. PROVIDE FLEXIBLE CONNECTOR AT GENERATOR.
17. PRE-MANUFACTURED WALL THIMBLE, DESIGNED IN ACCORDANCE WITH NFPA, TO PROTECT WALL FROM EXHAUST PIPE HEAT.
18. PRE-MANUFACTURED EXHAUST PIPE WITH OPEN 45 DEGREE OPENING PROTECTED WITH EXPANDED METAL BIRDSCREEN. OPEN MUST FACE DOWN TOWARD GRADE.
19. RADON FANS ABOVE CEILING. SEE PLUMBING FOR RADON PIPING.
20. SMOKE/FIRE DAMPER
21. 2 HOUR FIRE DAMPER
22. 3 HOUR FIRE DAMPER



A1 FLOOR PLAN - MECHANICAL HVAC
SCALE: 1:100

GRAPHIC SCALE

1:100 0 1000 5000 10000mm

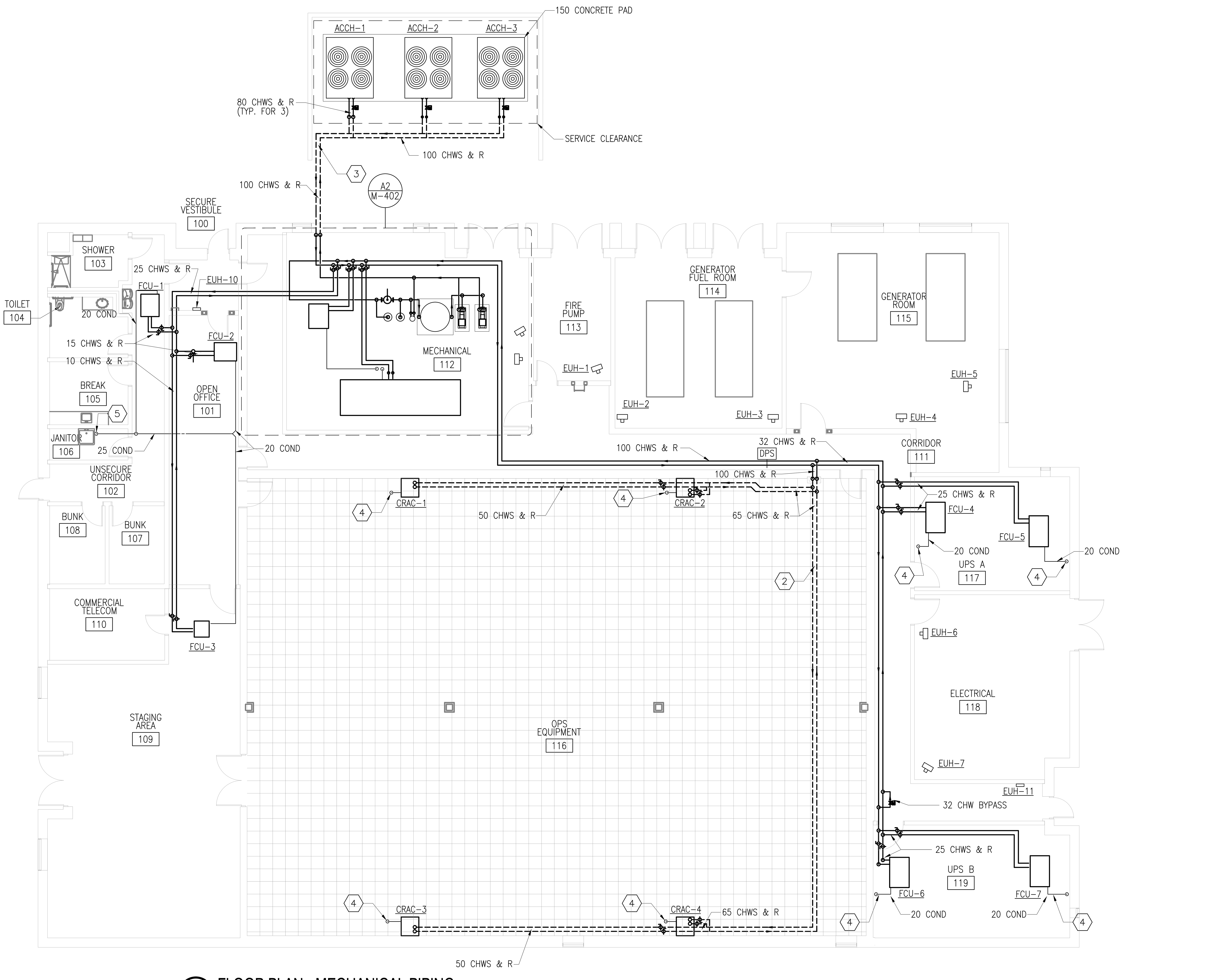
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY
FLOOR PLAN - MECHANICAL HVAC

NAVFAC

APPROVED: Wm B Forbes — Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'
FOR COMMANDER NAVFAC
ACTIVITY: D. EVANS VIA EMAIL
SATISFACTORY TO / DATE: 01/15/2026
DES JAK DRW JAK CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN
SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157177
SHEET 115 OF 170
M-101

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2016



A1 FLOOR PLAN - MECHANICAL PIPING

SCALE: 1:100

PLAN NORTH

GENERAL NOTES

1. REFER TO SHEET M-001 LEGEND, GENERAL NOTES, AND ABBREVIATIONS.

SHEET KEYNOTES

1. NOT USED
2. CHILLED WATER PIPING BELOW RAISED FLOOR IN THIS SPACE.
3. UNDERGROUND CHILLED WATER PIPING.
4. PIPE COOLING COIL CONDENSATE TO NEAREST FLOOR DRAIN.
5. PIPE COOLING COIL CONDENSATE TO FLOOR SINK BELOW.

GRAPHIC SCALE

1:100 0 1000 5000 10000mm

SYM	DESCRIPTION	DATE	APPR

NAVFAC

Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'

D. EVANS VIA EMAIL

SATISFACTORY TO: DATE 01/15/2026

DES RER DRW RER CHK FISH

PM/DM MJD/ATP

BRANCH MANAGER SCE

DES PROD DIR WBF

FIRE PROTECTION ENGINEER DSN

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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
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NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

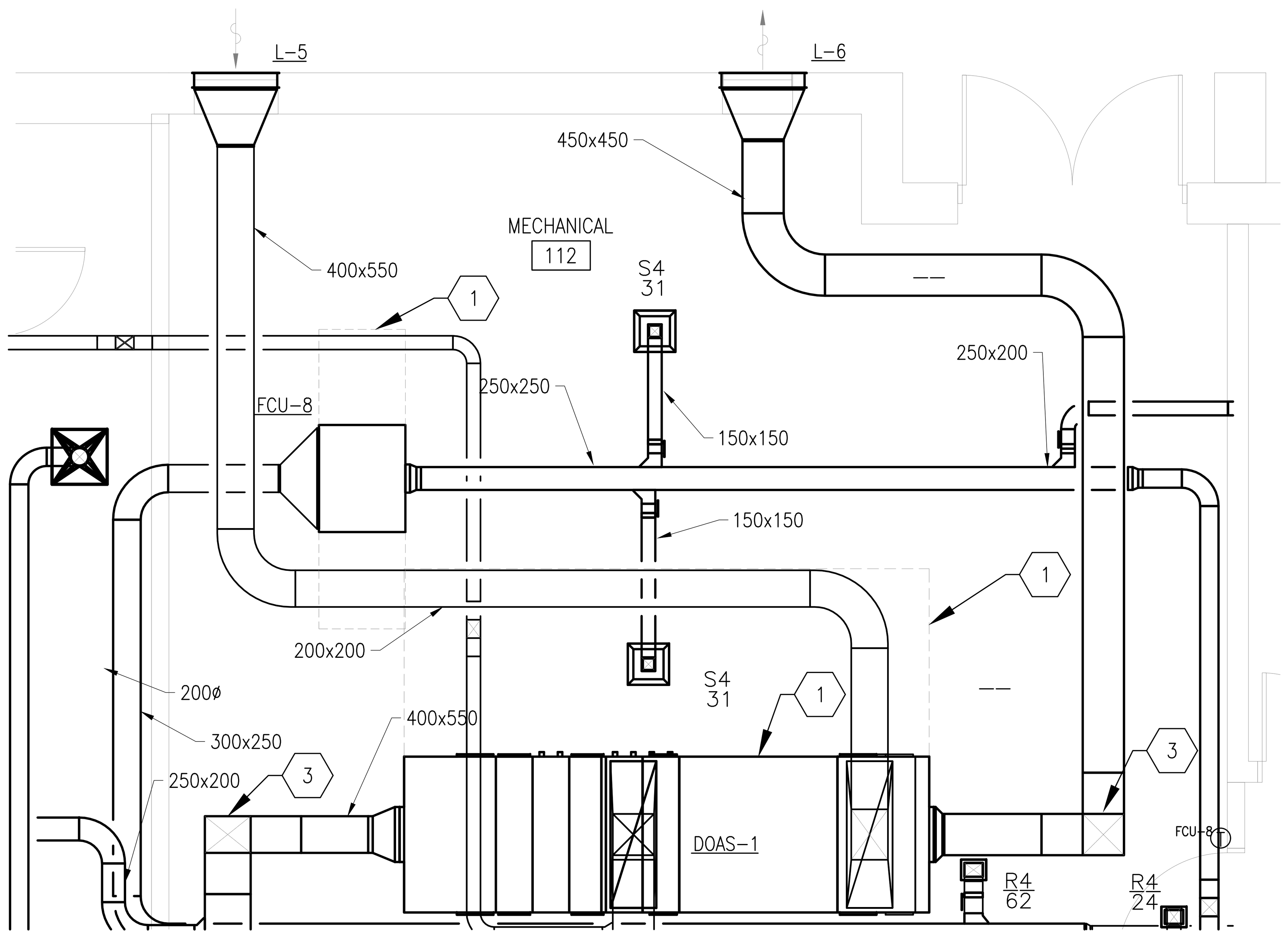
JOINT MARITIME FACILITY
FLOOR PLAN - MECHANICAL PIPING

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157178
SHEET 116 OF 170
M-102

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

SHEET KEYNOTES

1. EQUIPMENT CLEARANCE REQUIRED FOR MAINTENANCE.
2. 100% DEDICATED OUTDOOR AIR SYSTEM.
3. DUCT TURN UP. SEE SHEET M-403 SECTION C1 FOR CLARIFICATION.



A1 ENLARGED FLOOR PLAN - MECHANICAL ROOM DUCTWORK
SCALE: 1:50

PLAN NORTH

GRAPHIC SCALE

1:50 0 1000 3000 5000mm

APPR
DATE
DESCRIPTION
SYM

A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'
FOR COMMANDER NAVFAC

ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES JAK	DRW JAK	CHK OSA
PM/DM	MJD/ATP	
BRANCH MANAGER	SCE	
DES PROD DIR	WBF	
FIRE PROTECTION ENGINEER	DSN	

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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
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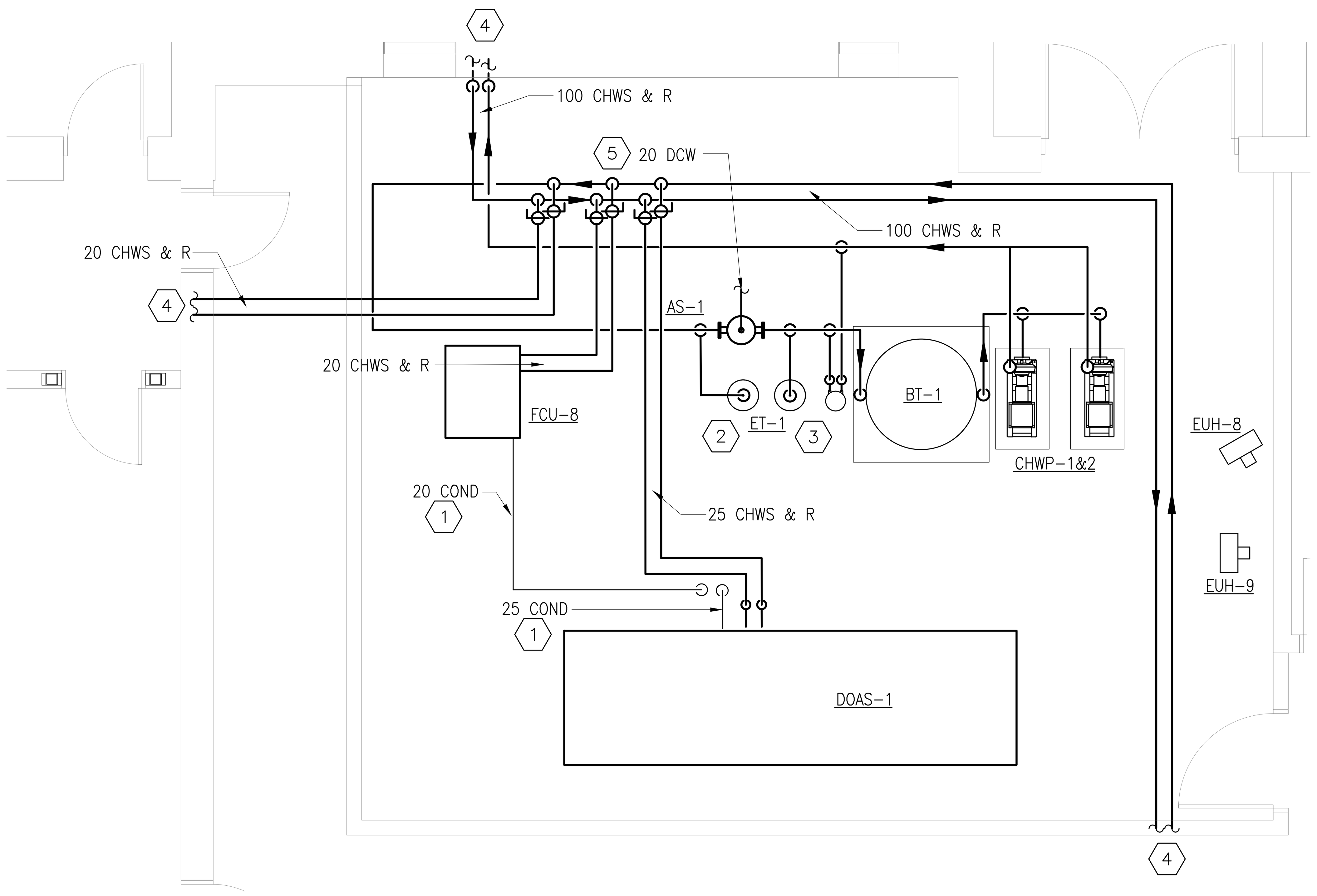
PLANNING, DESIGN AND CONSTRUCTION
NORFOLK VA
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

ENLARGED FLOOR PLAN - MECHANICAL ROOM DUCTWORK

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157179
SHEET 117 OF 170

M-401



A2 ENLARGED FLOOR PLAN - MECHANICAL ROOM PIPING
SCALE: 1:50

PLAN NORTH

GENERAL NOTES

1. REFER TO SHEET M-001 LEGEND, GENERAL NOTES, AND ABBREVIATIONS.

SHEET KEYNOTES

1. PIPE COOLING COIL CONDENSATE TO NEAREST FLOOR DRAIN.
2. PROPYLENE GLYCOL TANK AND PUMPING SYSTEM.
3. CHEMICAL SHOT FEEDER.
4. CONTINUES TO SHEET M-102
5. DOMESTIC COLD WATER MAKEUP. SEE PLUMBING SHEETS FOR CONTINUATION

GRAPHIC SCALE

1:50 0 1000 3000 5000mm

SYM	DESCRIPTION	DATE	APPR

A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'
FOR COMMANDER NAVFAC

ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES RER DRW RER CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION NORFOLK VA
VARIOUS LOCATIONS VARIOUS LOCATIONS

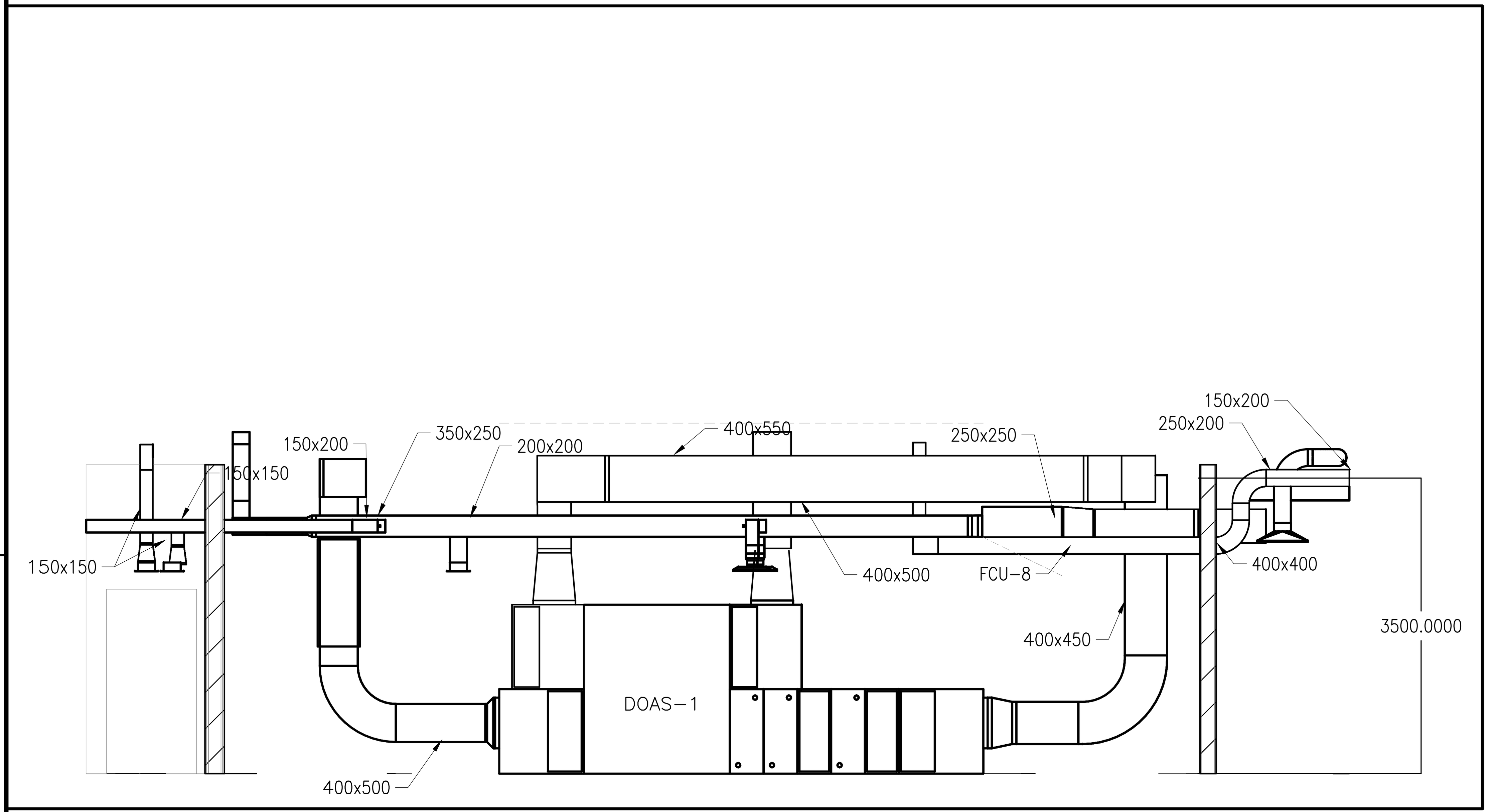
JOINT MARITIME FACILITY

ENLARGED FLOOR PLAN - MECHANICAL ROOM PIPING

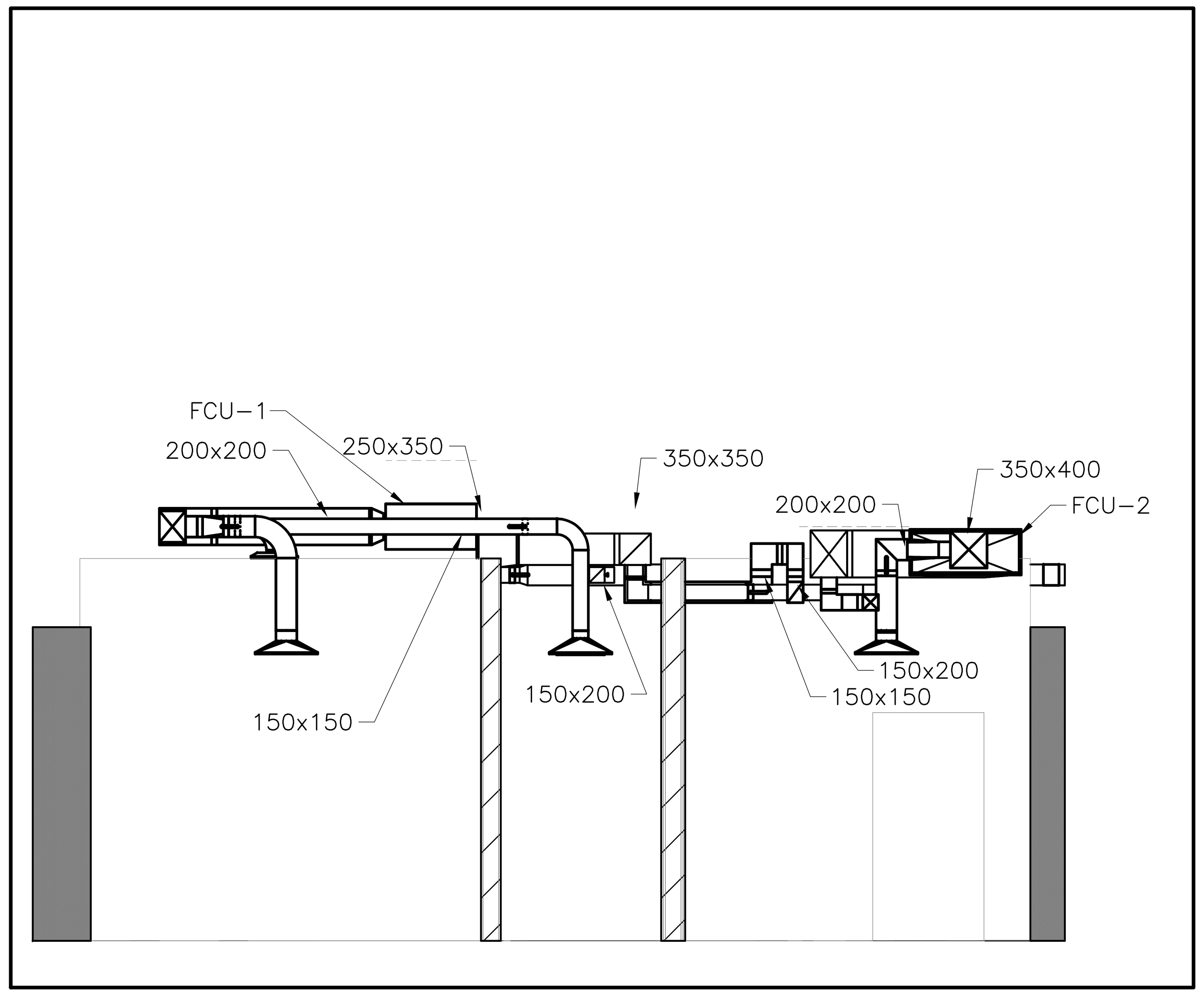
SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157180
SHEET 118 OF 170

M-402

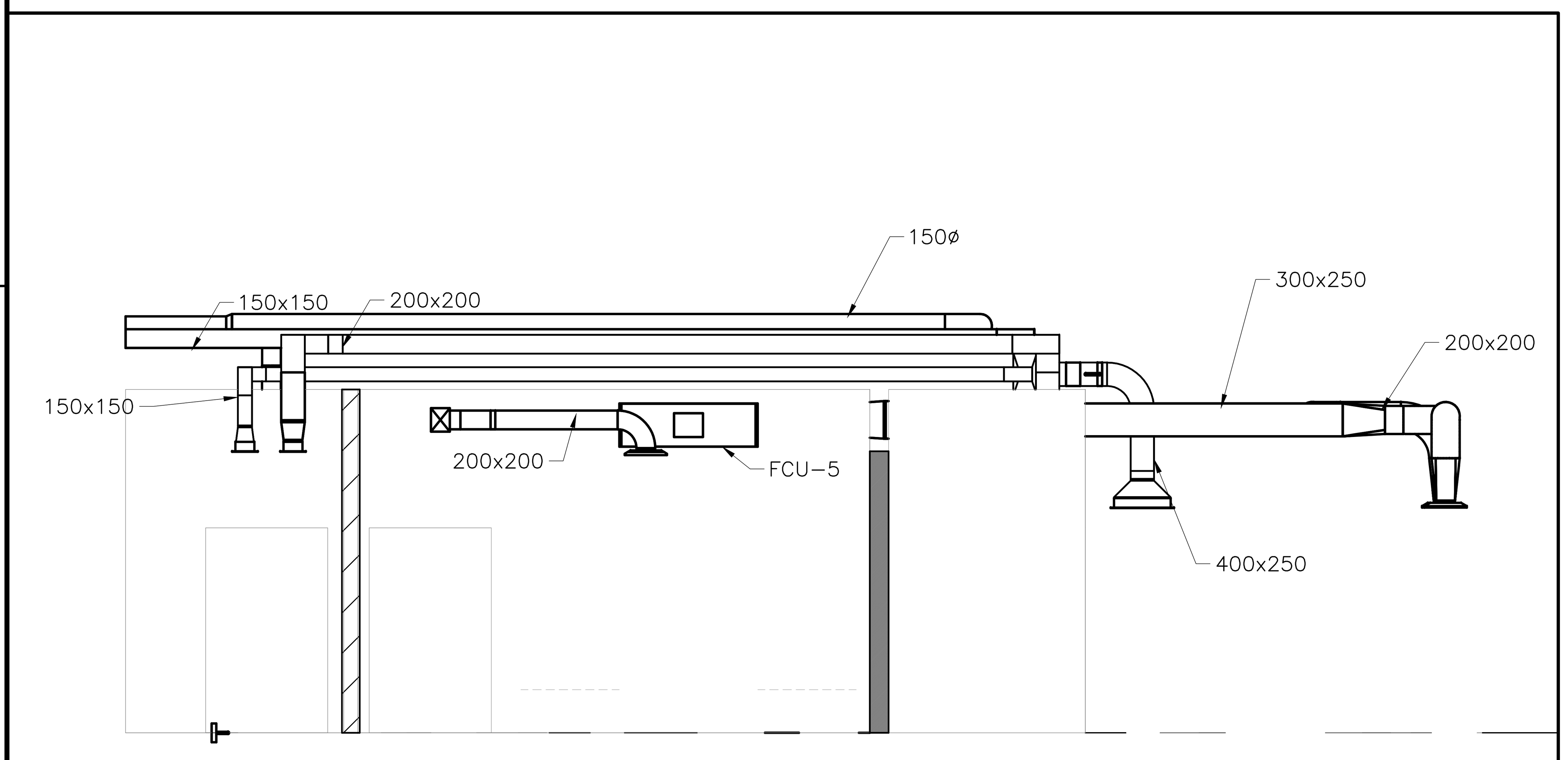
NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018



C1 MECHANICAL ROOM SECTION
SCALE: 1:50



C3 MECHANICAL SECTION
SCALE: 1:50



A1 MECHANICAL SECTION
SCALE: 1:50

GRAPHIC SCALE

1:50 0 1000 3000 5000mm

SYM	DESCRIPTION	DATE	APPR

NAVFAC

SEAL

A/E INFO

APPROVED Wm B Forbes Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'
FOR COMMANDER NAVFAC

ACTIVITY
D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026
DES JAK DRW JAK CHK OSA
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS VARIOUS LOCATIONS

JOINT MARITIME FACILITY

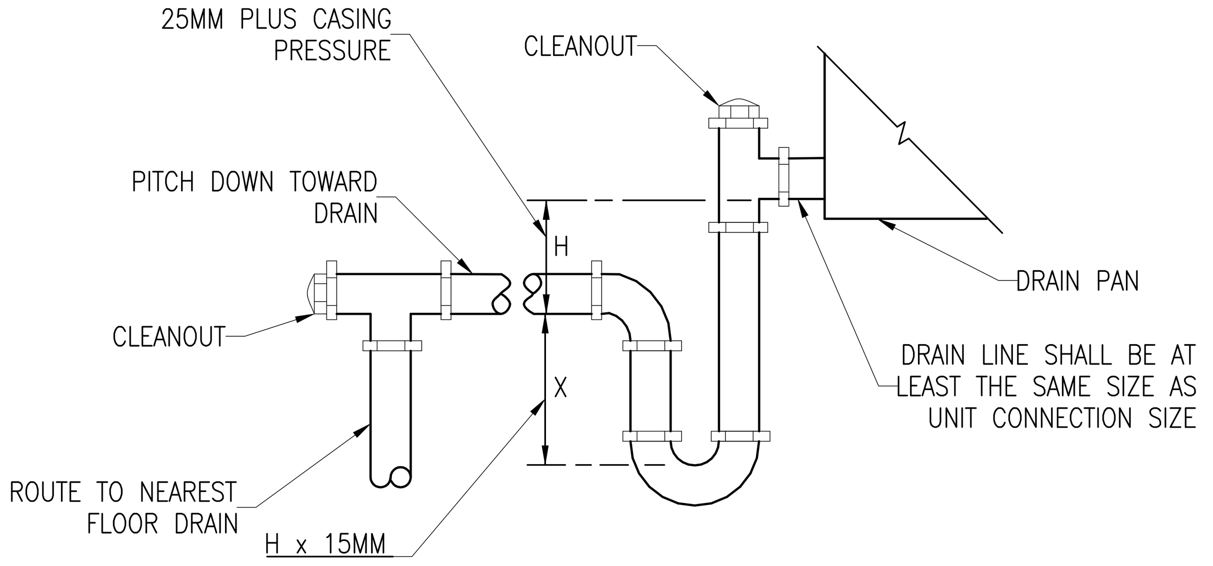
MECHANICAL SECTION PLANS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157181
SHEET 119 OF 170

M-403

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

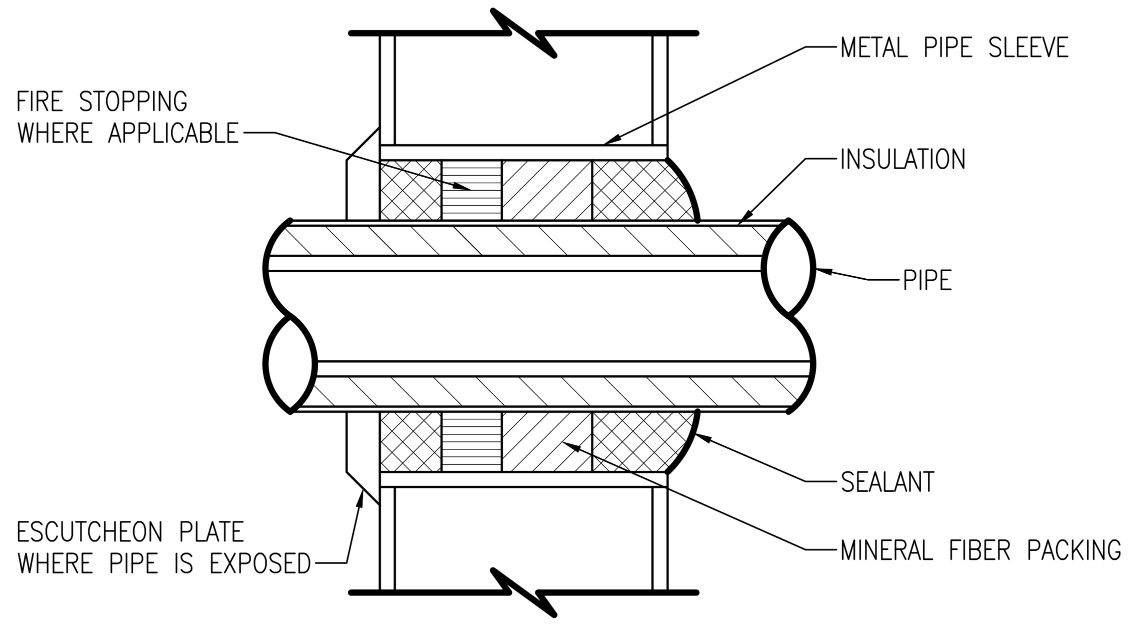
FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-403.dwg LAYOUT NAME: M-403 PLOTTED: Tuesday, January 20, 2026 - 3:10pm USER: jennifer.a.klein



TOTAL TRAP HEIGHT = X + H + (1.5 X PIPE DIAMETER)

C1 CONDENSATE P-TRAP PIPING DETAIL

SCALE: NOT TO SCALE M–102

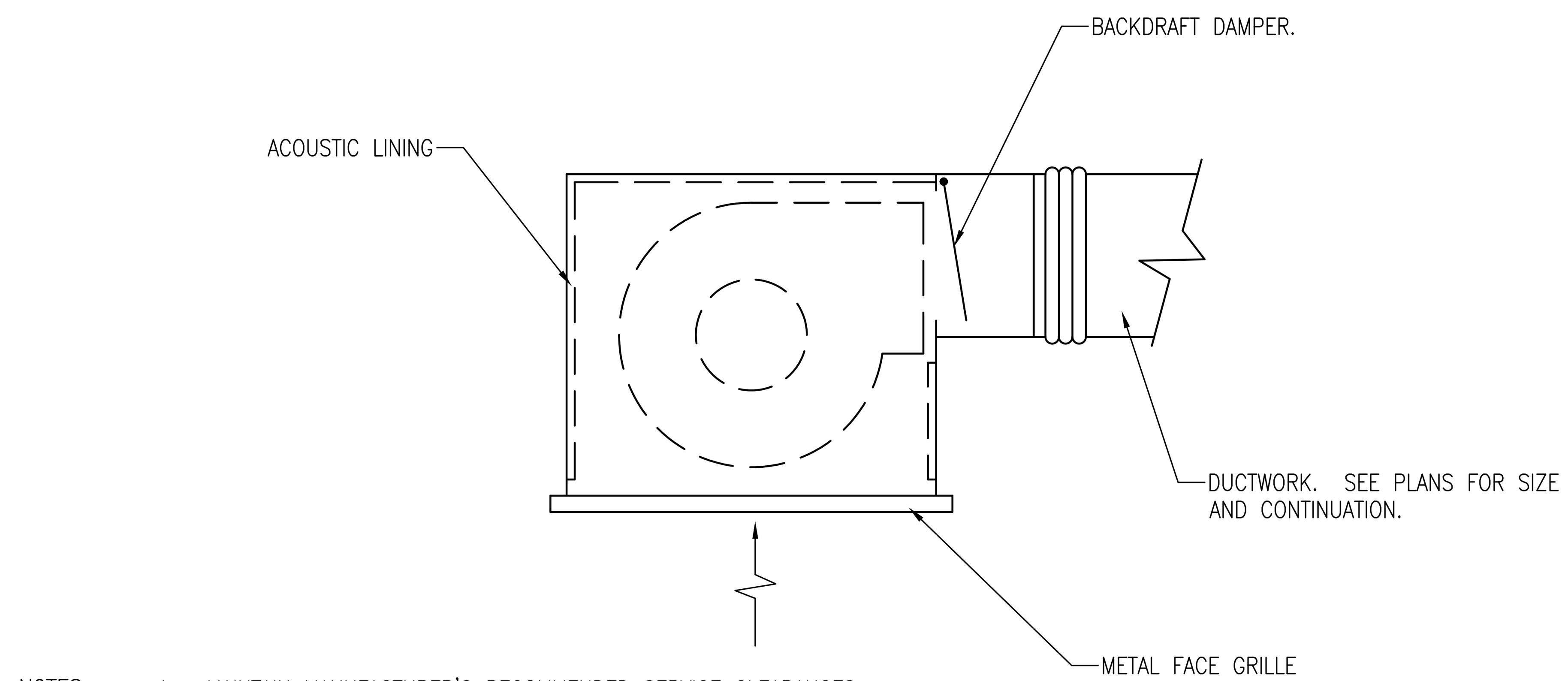


NOTES:

1. REFER TO FIRE PROTECTION SPECIFICATIONS FOR FIRE STOPPING SYSTEM REQUIREMENTS WHERE PIPE PENETRATES RATED WALL OR PARTITION.
2. PROVIDE APPROXIMATELY 13 GAP BETWEEN INSULATION AND METAL PIPE SLEEVE.
3. SUPPORT PIPE ON BOTH SIDES OF WALL. CONTACT WITH SLEEVE, WALL OR FRAMING IS NOT ALLOWED.

C2 WALL PIPE SLEEVE - INTERIOR DETAIL

SCALE: NOT TO SCALE M–102

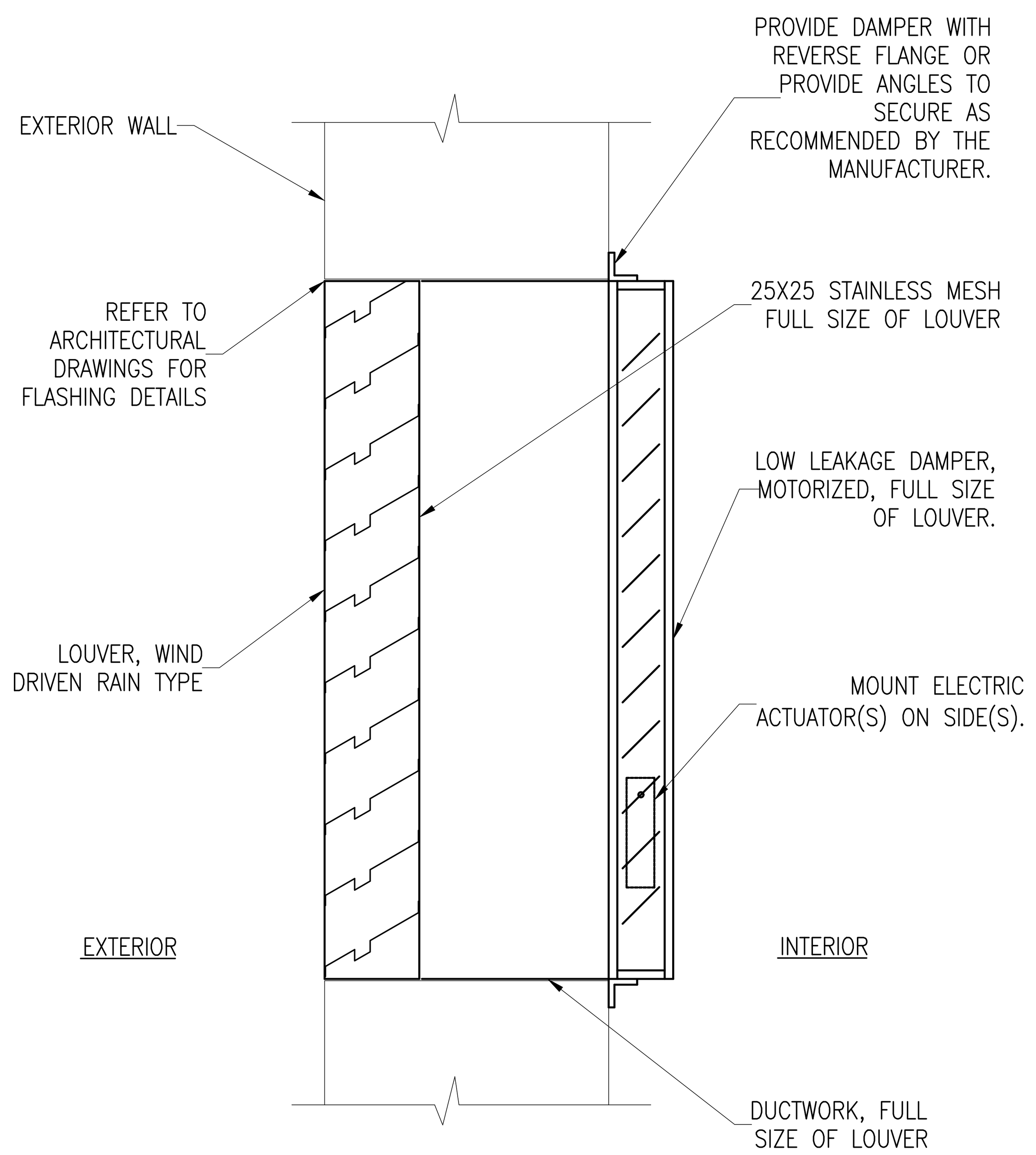


NOTES:

1. MAINTAIN MANUFACTURER'S RECOMMENDED SERVICE CLEARANCES.
2. SUPPORT EQUIPMENT TO THE STRUCTURE.

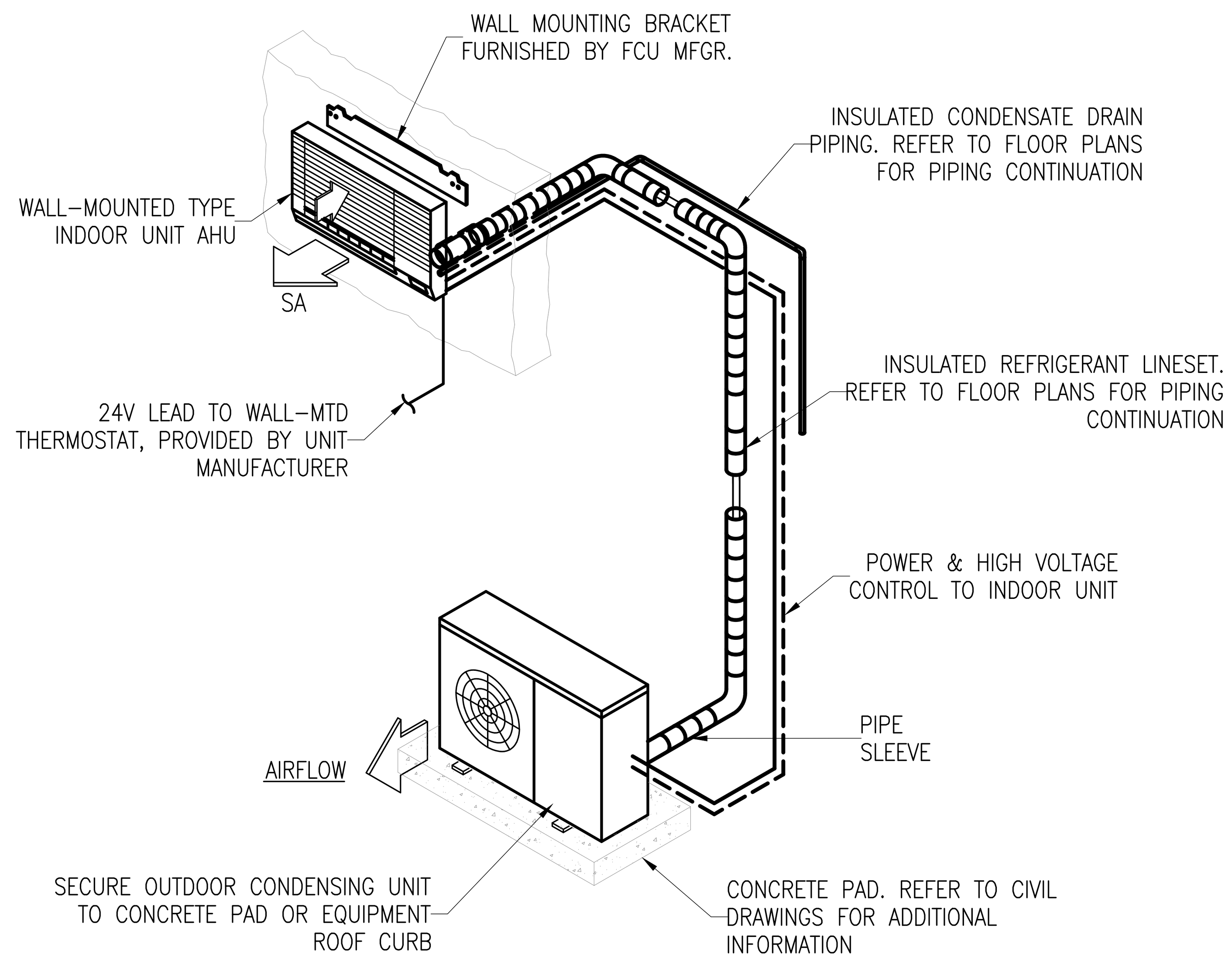
C4 CABINET CEILING FAN DETAIL

SCALE: NOT TO SCALE M–101



A1 LOUVER INSTALLATION DETAIL

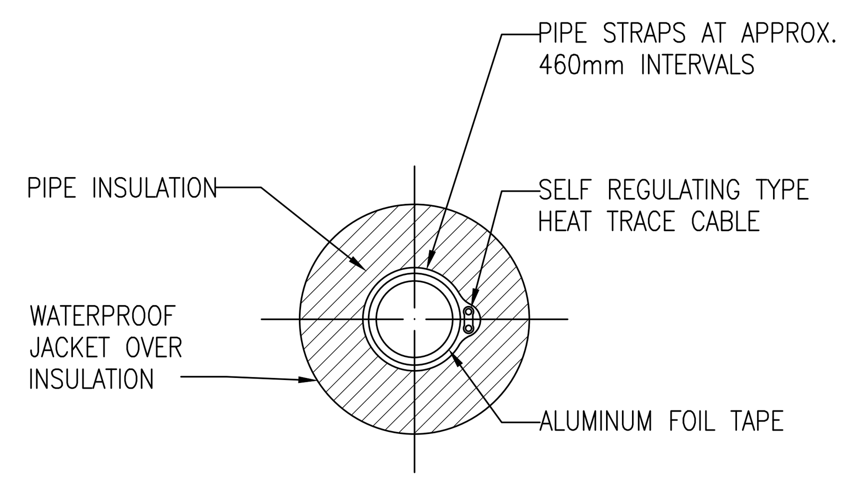
SCALE: NOT TO SCALE M–101



NOTES: 1. MOUNT BOTTOM OF WALL–MOUNTED INDOOR UNIT 2.5 METERS AFF.

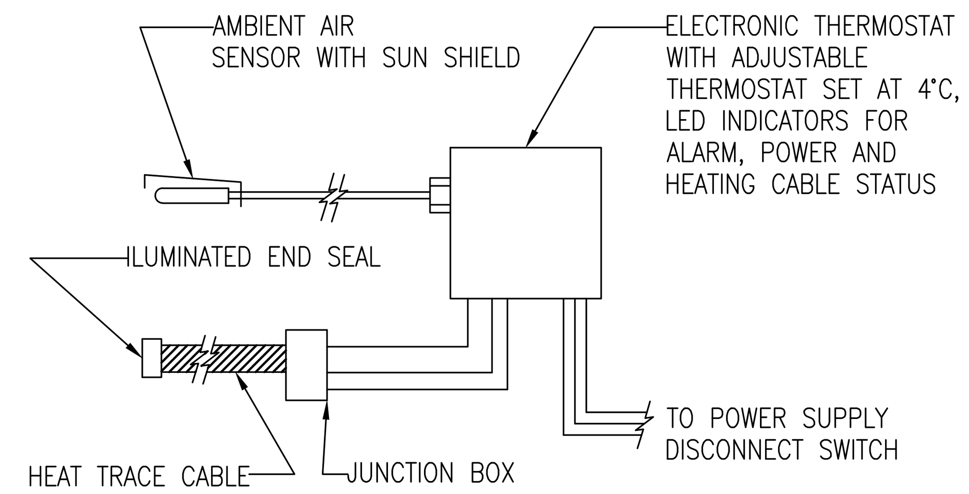
A2 DUCTLESS SPLIT SYSTEM PIPING DETAIL

SCALE: NOT TO SCALE M–102



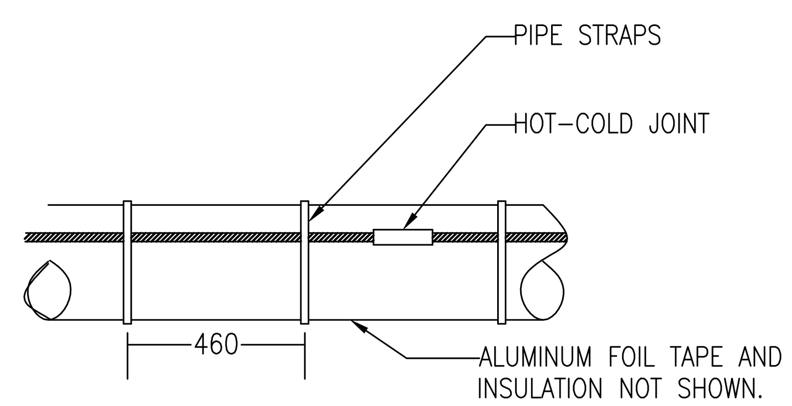
SECTION THRU PIPE

NOT TO SCALE



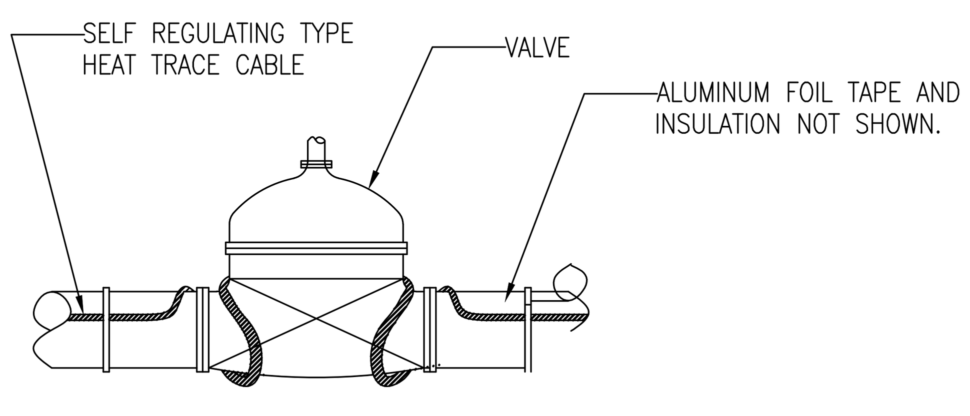
HEAT TRACE CONTROL BOX DIAGRAM

NOT TO SCALE



INSTALLATION DETAIL

NOT TO SCALE



INSTALLATION AT VALVES

NOT TO SCALE

NOTES: 1. PROVIDE WEATHER RESISTANT LABELING ON EXTERIOR OF JACKET IDENTIFYING PIPES AS BEING "ELECTRIC TRACED" WITH VOLTAGE INDICATED.

A4 HEAT TRACING INSTALLATION DETAIL

SCALE: NOT TO SCALE M–102

SYM DESCRIPTION DATE APPR

SEAL

A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:00-05'00'
FOR COMMANDER NAVFAC

ACTIVITY D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES	OLS	DRW	OLS	CHK	FISH
PM/DM	MJD/ATP				
BRANCH MANAGER	SCE				
DES PROD DIR	WBF				
FIRE PROTECTION ENGINEER	DSN				

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
PLANNING, DESIGN AND CONSTRUCTION NORFOLK VA
VARIOUS LOCATIONS VARIOUS LOCATIONS

JOINT MARITIME FACILITY

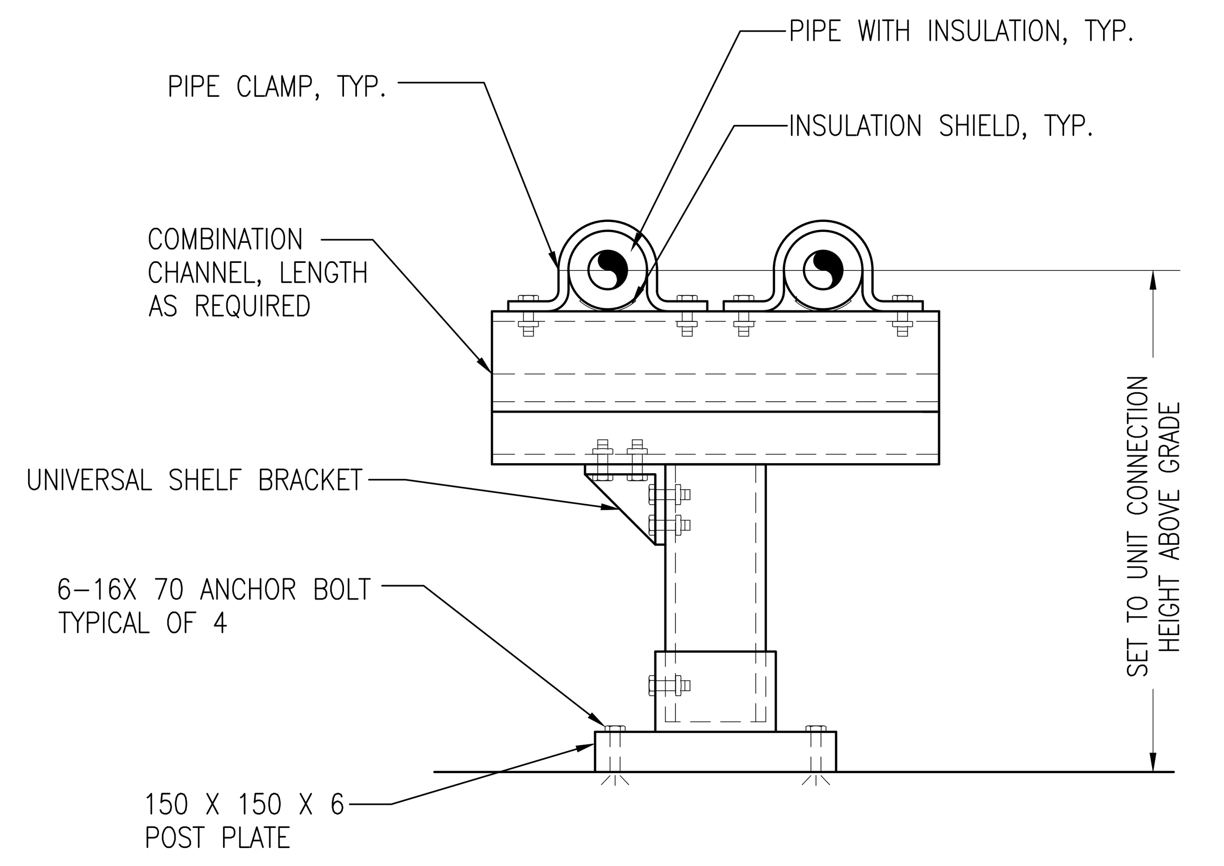
MECHANICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157182
SHEET 120 OF 170

M-501

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

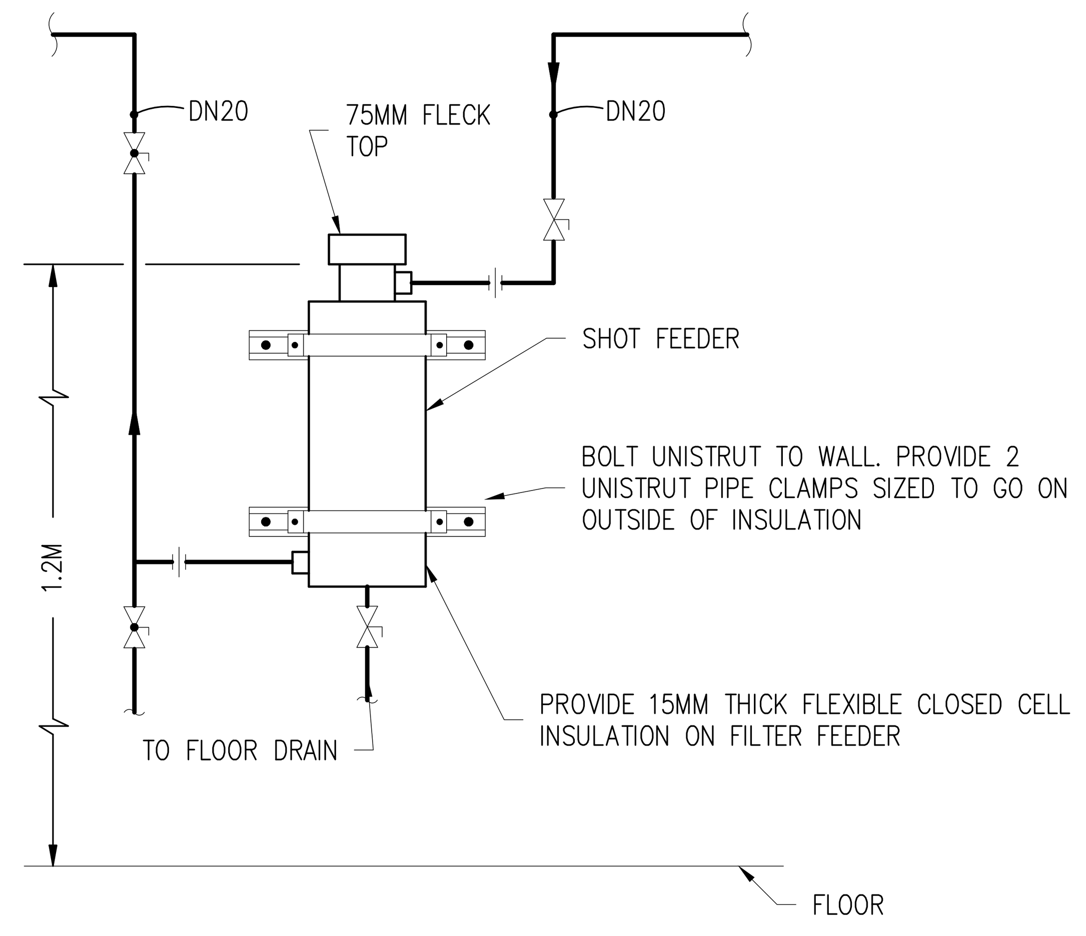
FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJM\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-501.dwg LAYOUT NAME: M-501 PLOTTED: Tuesday, January 20, 2026 – 3:11pm USER: jennifer.a.klein



NOTES:
1. CHANNELS, HANGERS, FRAMES, THREADED RODS, WASHERS, AND NUTS AND ALL ASSOCIATED APPURTENANCES SHALL BE HOT DIPPED GALVANIZED STEEL.
2. PRE-MANUFACTURED PIPE SUPPORT.

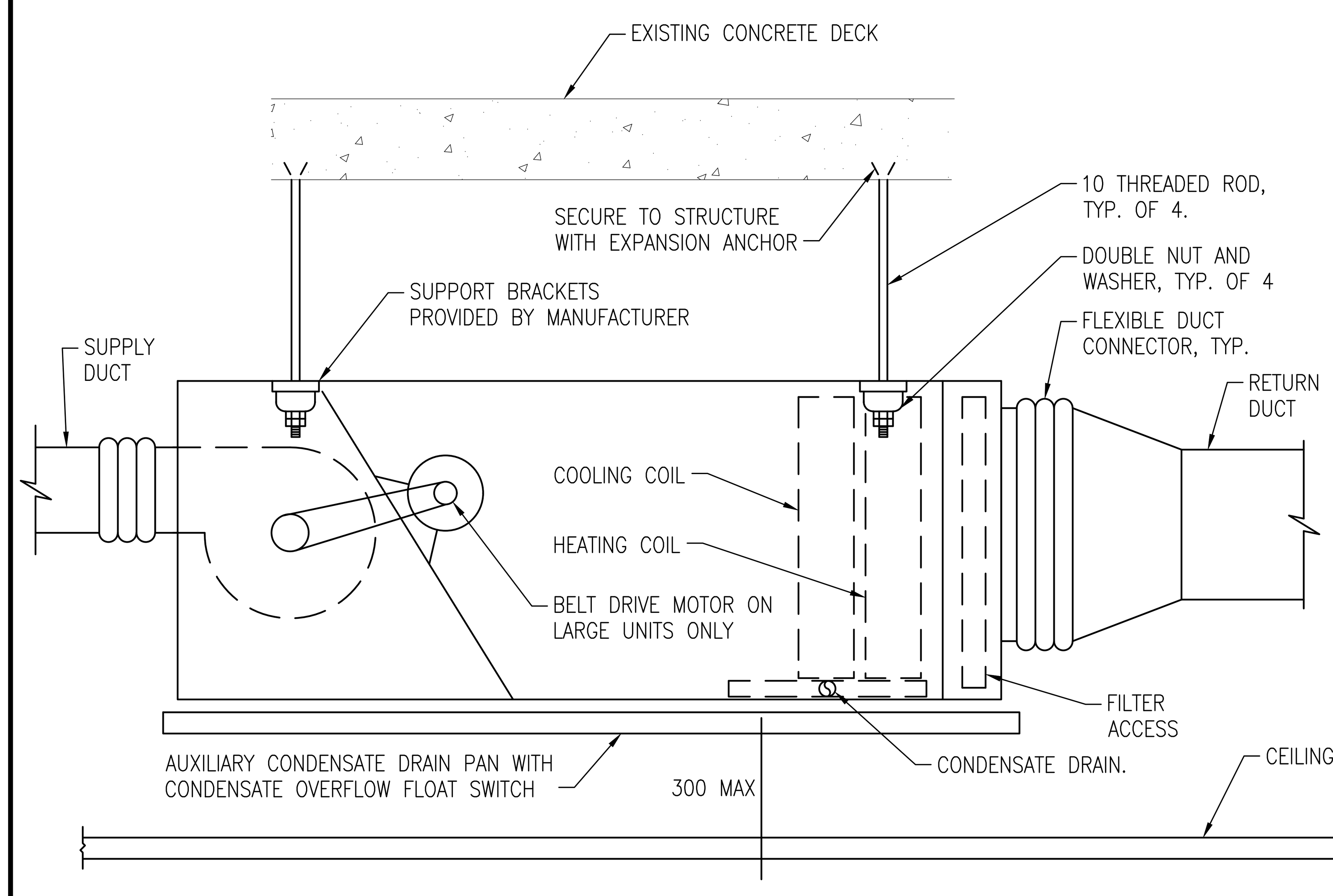
C1 PIPE SUPPORT DETAIL

SCALE: NOT TO SCALE M-102



C2 CHEMICAL SHOT FEEDER PIPING DETAIL

SCALE: NOT TO SCALE M-102

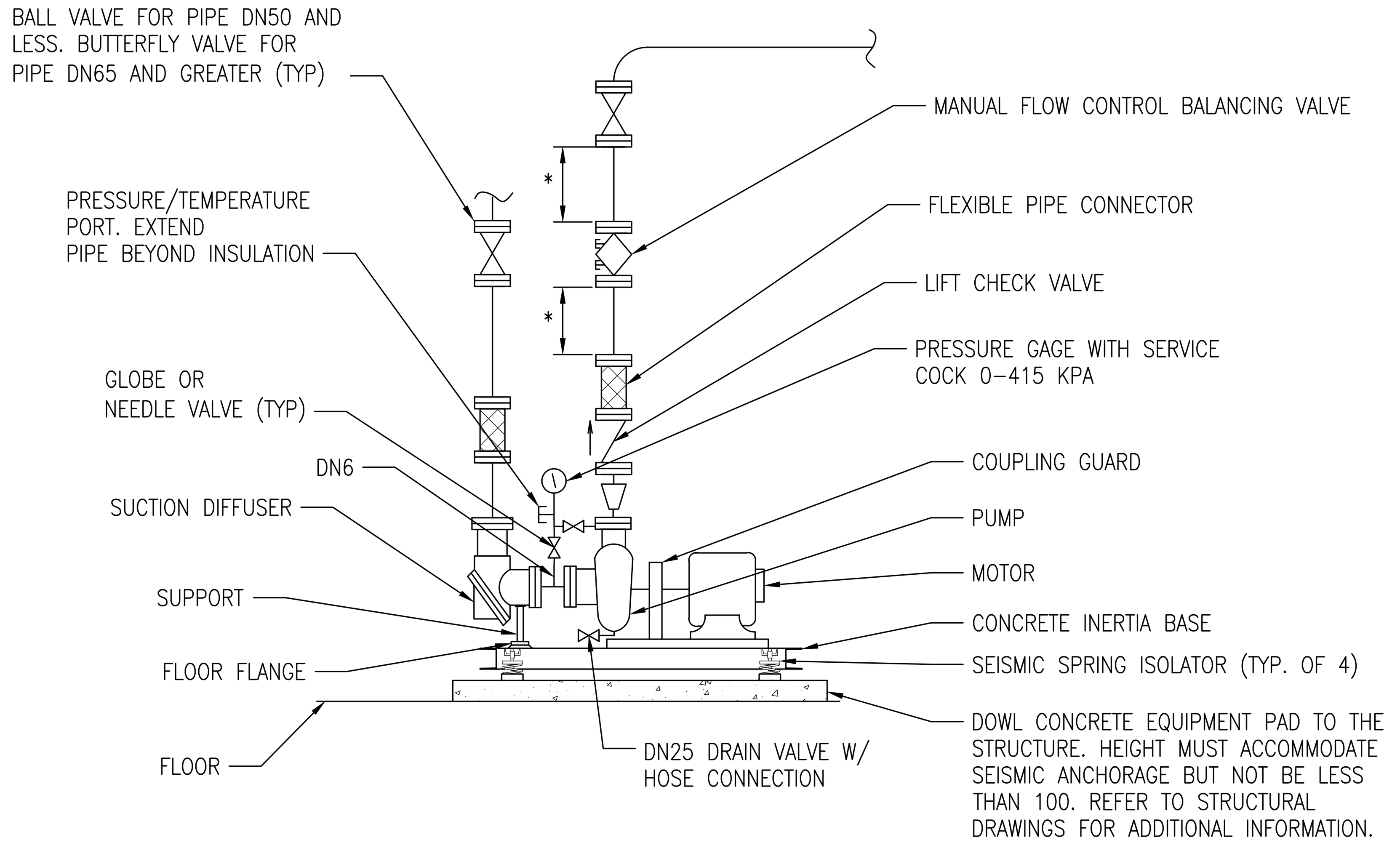


NOTES:
1. REFER TO FCU COIL PIPING FOR PIPING CONNECTIONS.
2. REFER TO THE FLOOR PLANS FOR DUCT SIZES FOR RETURN AND SUPPLY AIR DUCTS.

C4 FAN COIL UNIT INSTALATION DETAIL

SCALE: NOT TO SCALE M-102

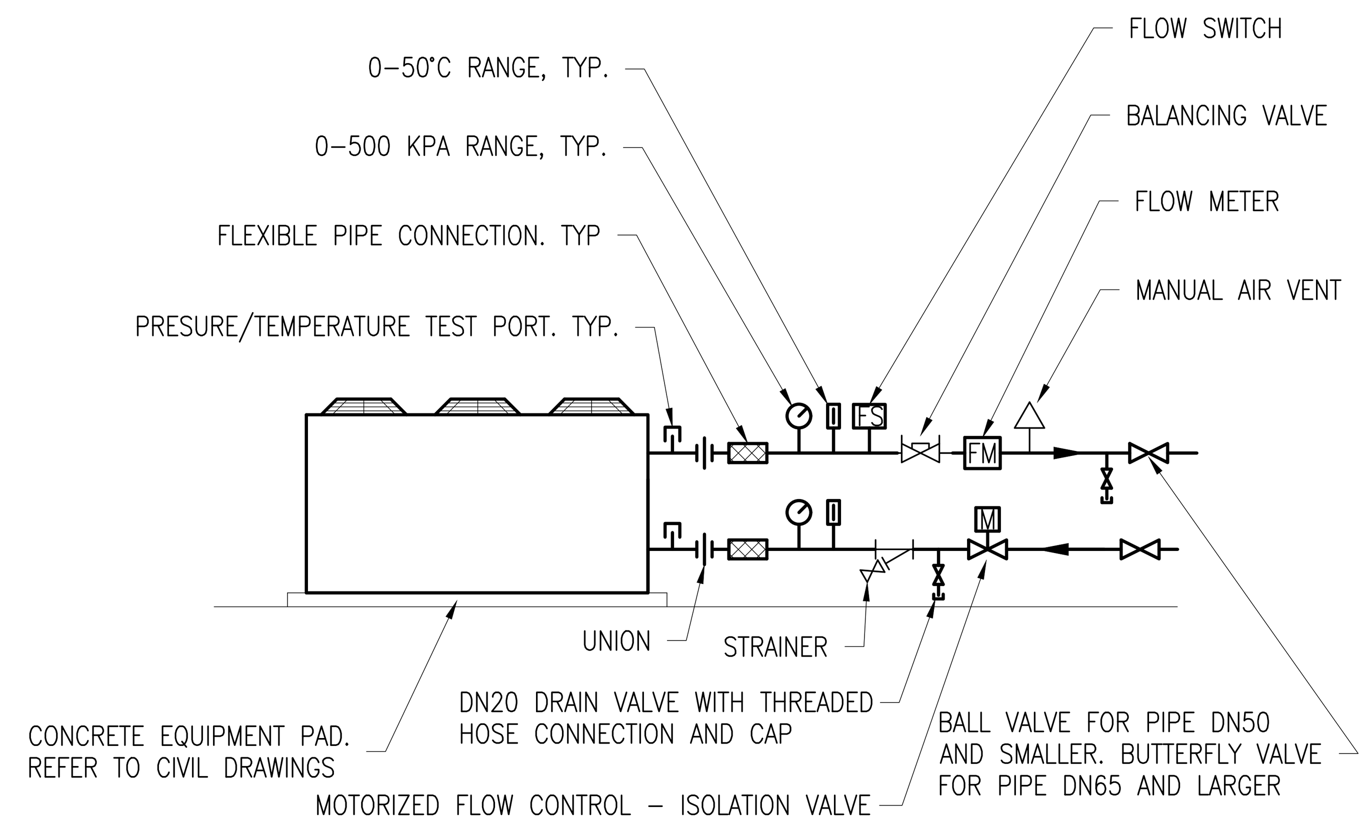
NOTES: 1. SUPPORT EQUIPMENT TO THE STRUCTURE, REFER TO SPEC SECTION 220548.20 "MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL" AND SECTION 230548.19 "SEISMIC BRACING FOR HVAC" FOR ADDITIONAL INFORMATION.



* STRAIGHT LENGTHS OF UNOBSTRUCTED PIPE WITHOUT INLINE APPURTENANCES SHALL BE INSTALLED UP/DOWNSTREAM OF FLOW CONTROL BALANCING VALVE A MINIMUM OF EIGHT PIPE DIAMETER OR PER MANUFACTURER'S INSTALLATION INSTRUCTIONS, WHICHEVER IS LONGEST.

A1 BASE MOUNTED CENTRIFUGAL PUMP PIPING DETAIL

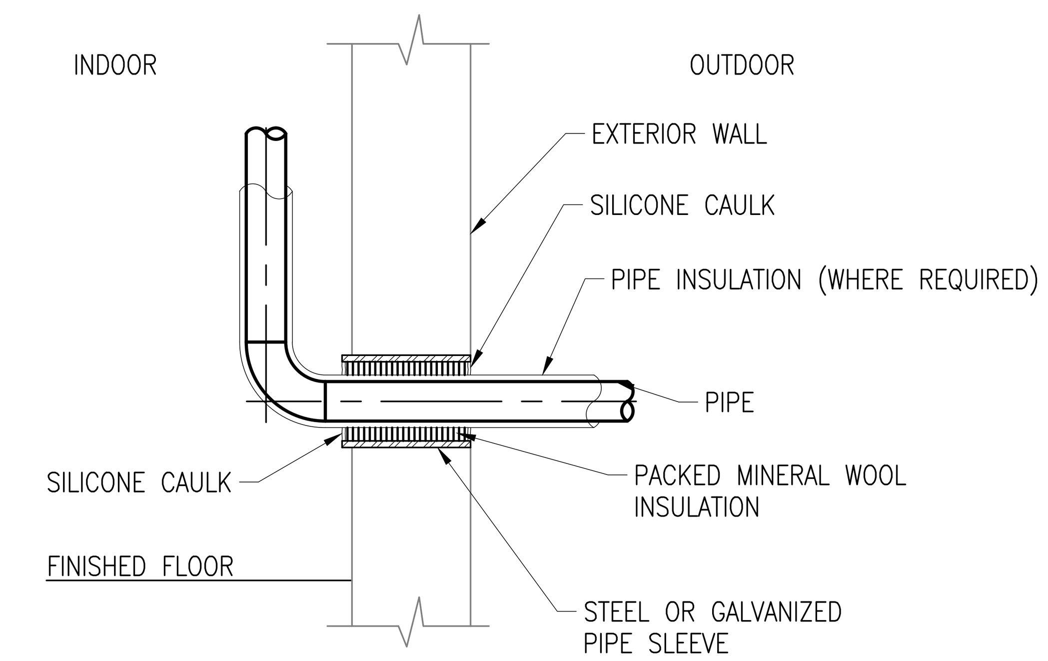
SCALE: NOT TO SCALE M-102



NOTE:
PROVIDE HEAT TRACE ON ALL EXTERIOR ABOVE GRADE PIPING. REFER TO DETAIL A4/M-501 FOR ADDITIONAL INFORMATION

A2 AIR COOLED CHILLER PIPING DETAIL

SCALE: NOT TO SCALE M-102



A4 PIPE WALL PENETRATION ABOVE GRADE DETAIL

SCALE: NOT TO SCALE M-102

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FOR COMMANDER NAVFAC	
ACTIVITY	D. EVANS VIA EMAIL
SATISFACTORY TO	DATE 01/15/2026
DES OLS	DRW OLS / CHK FISH
PM/DM	MJD/ATP
BRANCH MANAGER	SCE
DES PROD DIR	WBF
FIRE PROTECTION ENGINEER	DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

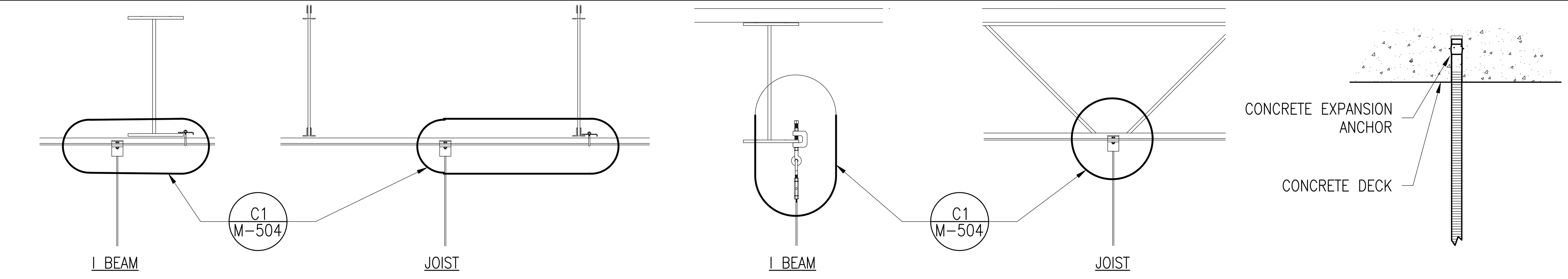
MECHANICAL DETAILS

SCALE:	AS NOTED
EPROJECT NO.:	1838720
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14157183
SHEET	121 OF 170

M-502

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\JOCNAVS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-502.dwg LAYOUT NAME: M-502 PLOTTED: Tuesday, January 20, 2026 - 3:11pm USER: jennifer.a.klein



JOIST OR I BEAM CENTERLINE PERPENDICULAR TO UNIT | JOIST OR I BEAM CENTERLINE PARALLEL TO UNIT | CONCRETE DECK

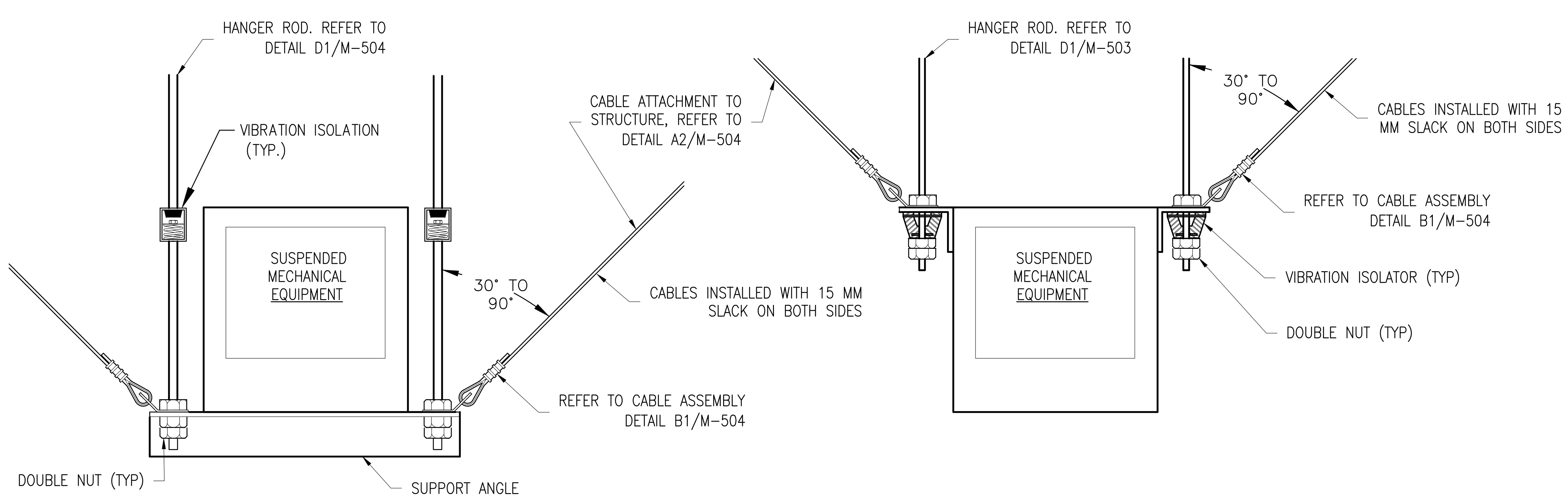
D1 HANGAR ROD ATTACHMENT DETAIL

SCALE: NOT TO SCALE

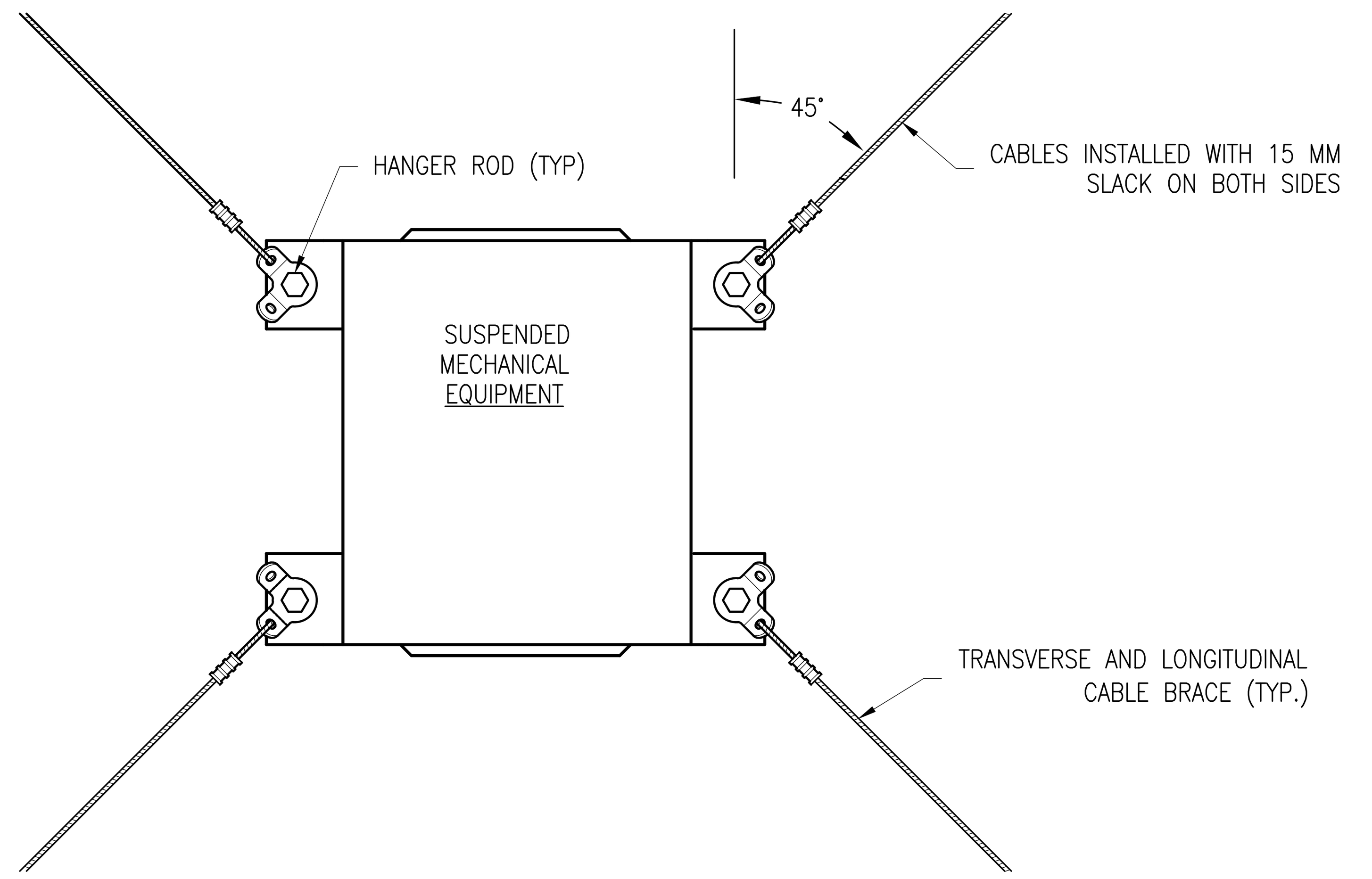
M-503

GENERAL SHEET NOTES

1. THE DETAILS ON THIS SHEET AND ON SHEET M-504 DO NOT PRECLUDE THE NEED TO DESIGN EQUIPMENT MOUNTINGS FOR FORCES REQUIRED BY OTHER CRITERIA SUCH AS FOR SEISMIC FORCES.
2. REFER TO SPECIFICATION SECTION 230548.19 "SEISMIC BRACING FOR HVAC" AND SECTION 220448.20 "MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL" FOR ADDITIONAL INFORMATION.



ELEVATIONS

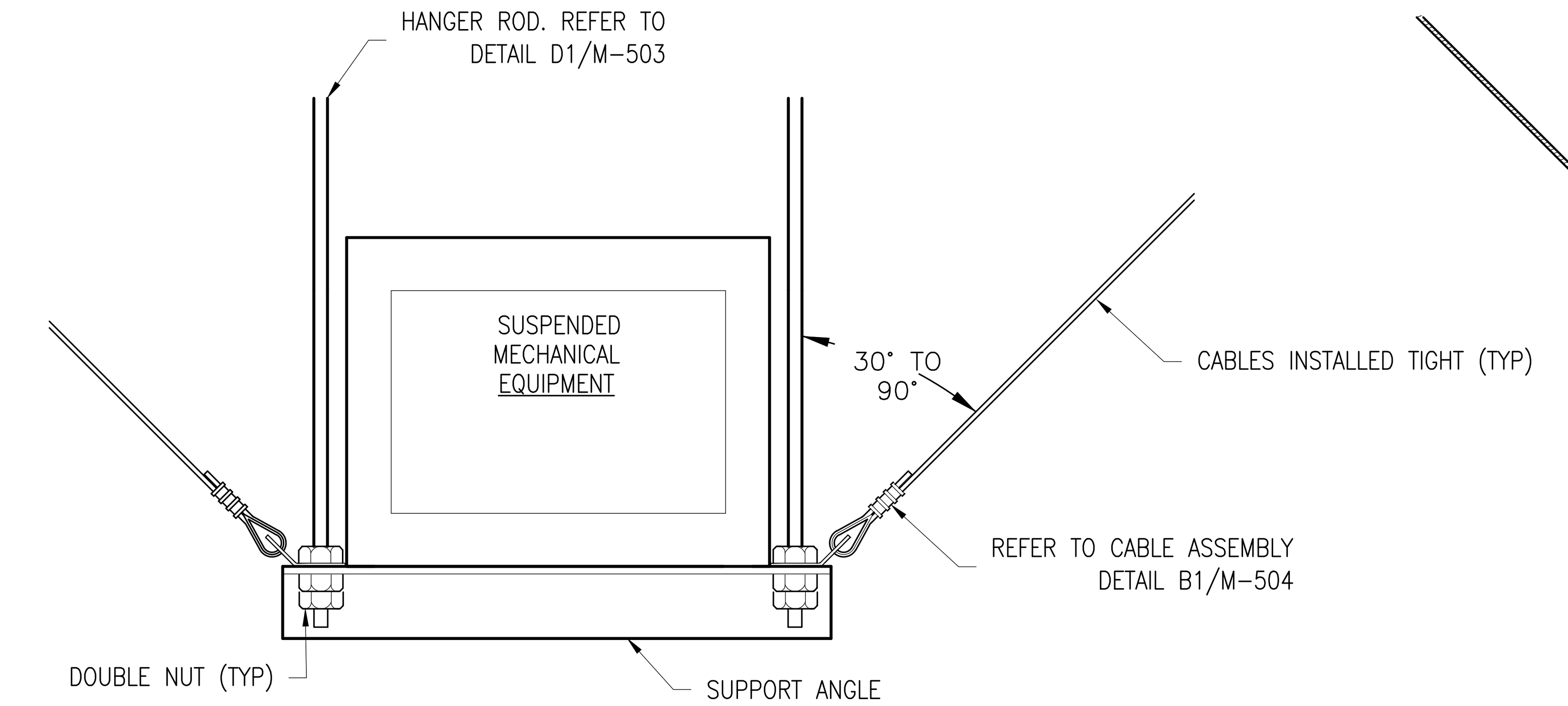


PLAN VIEW

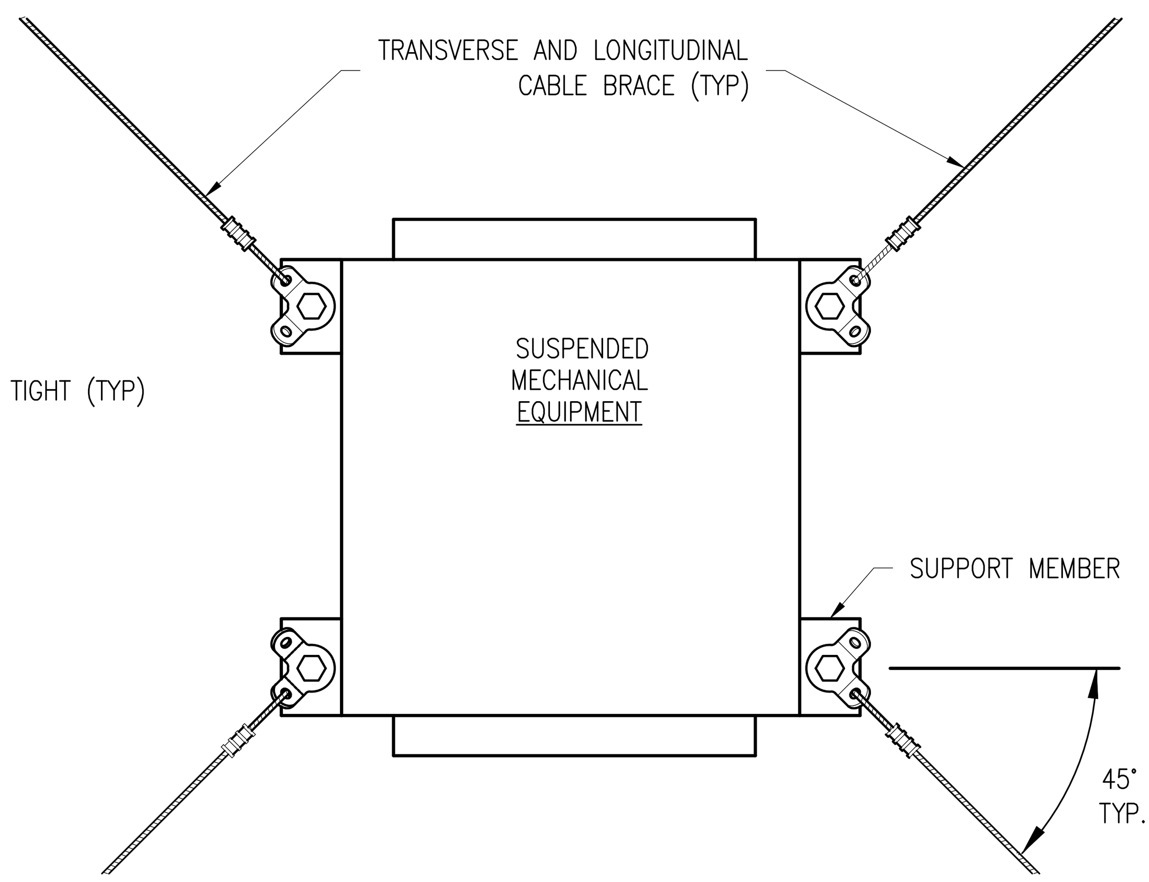
B1 BRACING DETAIL FOR SUSPENDED MECHANICAL EQUIPMENT WITH EXTERNAL VIBRATION ISOLATORS DETAIL

SCALE: NOT TO SCALE

M-503



ELEVATIONS

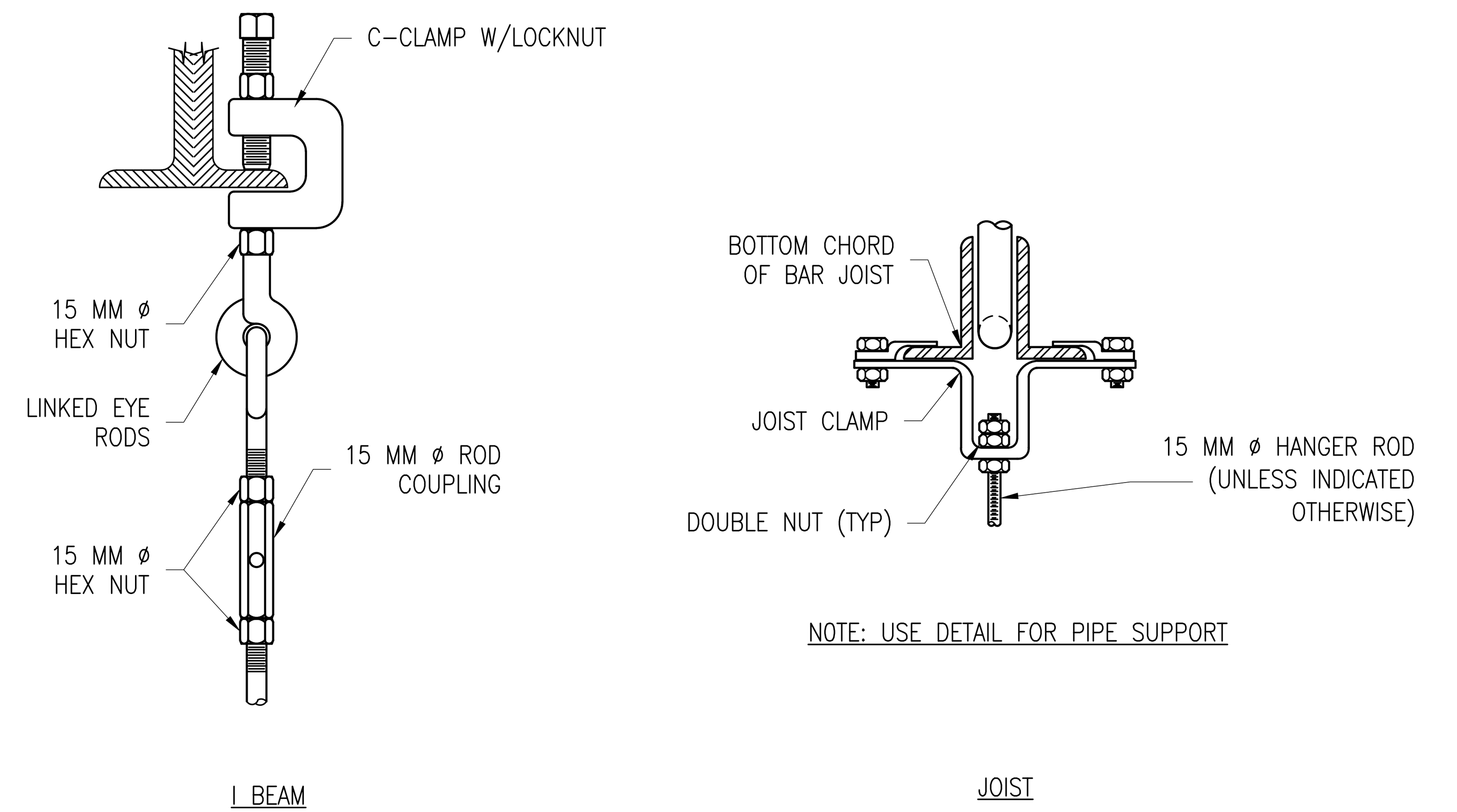


PLAN VIEW

A1 BRACING DETAIL FOR SUSPENDED MECHANICAL EQUIPMENT WITH INTERNAL VIBRATION ISOLATORS OR NO REQUIREMENT FOR VIBRATION ISOLATION DETAIL

SCALE: NOT TO SCALE

M-503



A4 CLOSEUP - PARALLEL CONNECTION DETAIL

SCALE: NOT TO SCALE

M-503

NAVFAC

APPROVED: Wm B Forbes — Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'

FOR COMMANDER NAVFAC

D. EVANS VIA EMAIL

SATISFACTORY TO — DATE 01/15/2026

DES OLS | DRW OLS | CHK FISH

PM/DM MJD/ATP

BRANCH MANAGER SCE

DES PROD DIR WBF

FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK, VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS

VARIOUS LOCATIONS

JOINT MARITIME FACILITY

MECHANICAL DETAILS

SCALE: AS NOTED

EPROJECT NO.: 1838720

CONSTR. CONTR. NO.

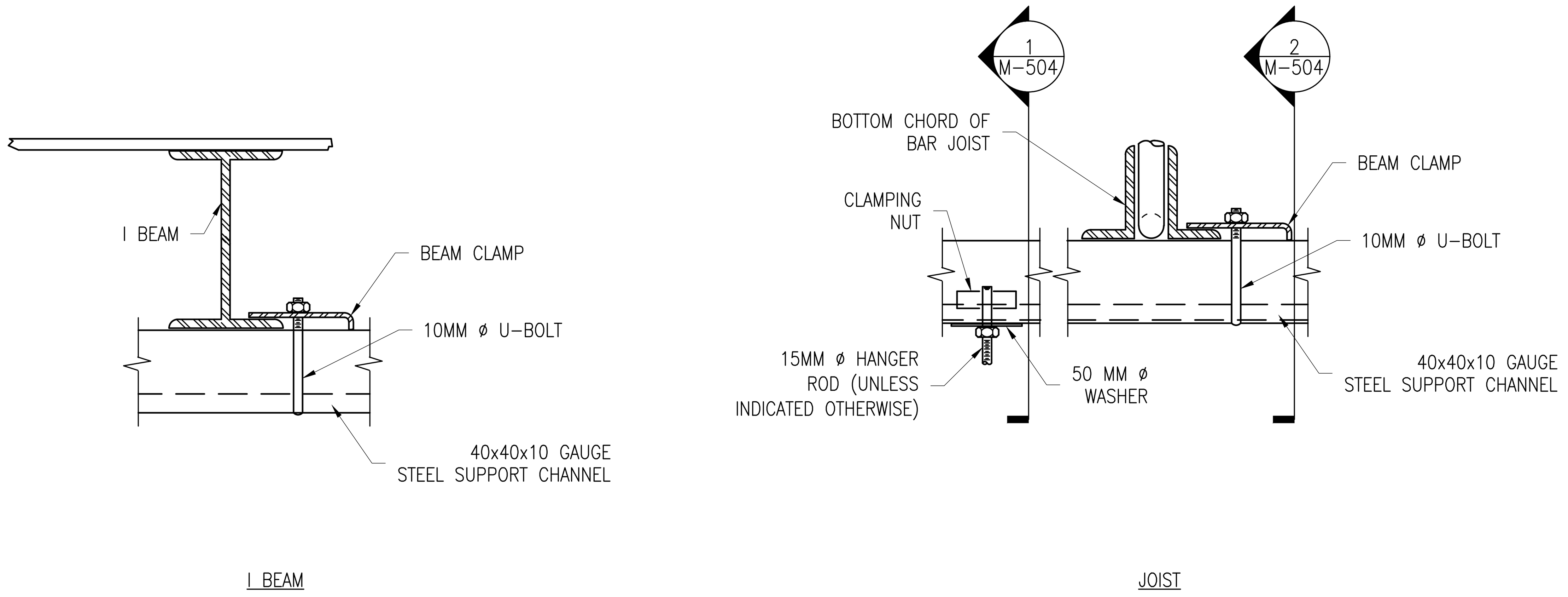
NAVFAC DRAWING NO. 14157184

SHEET 122 OF 170

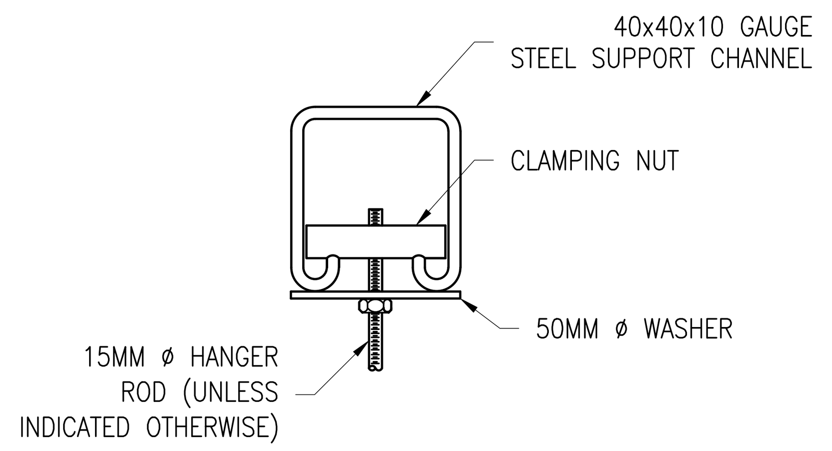
M-503

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

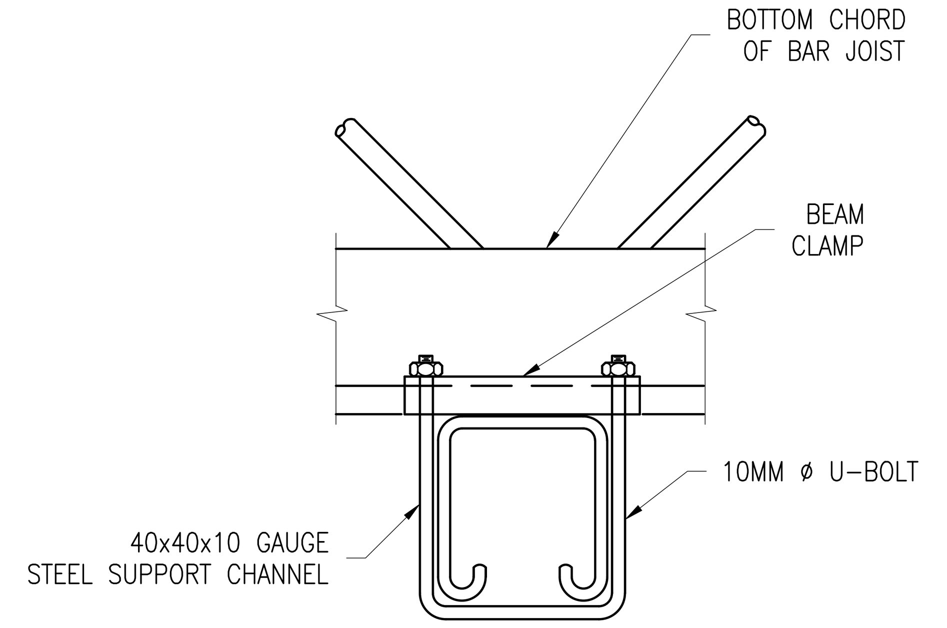
FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MUMF\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-503.dwg LAYOUT NAME: M-503 PLOTTED: Tuesday, January 20, 2026 - 3:11pm USER: jennifer.a.klein



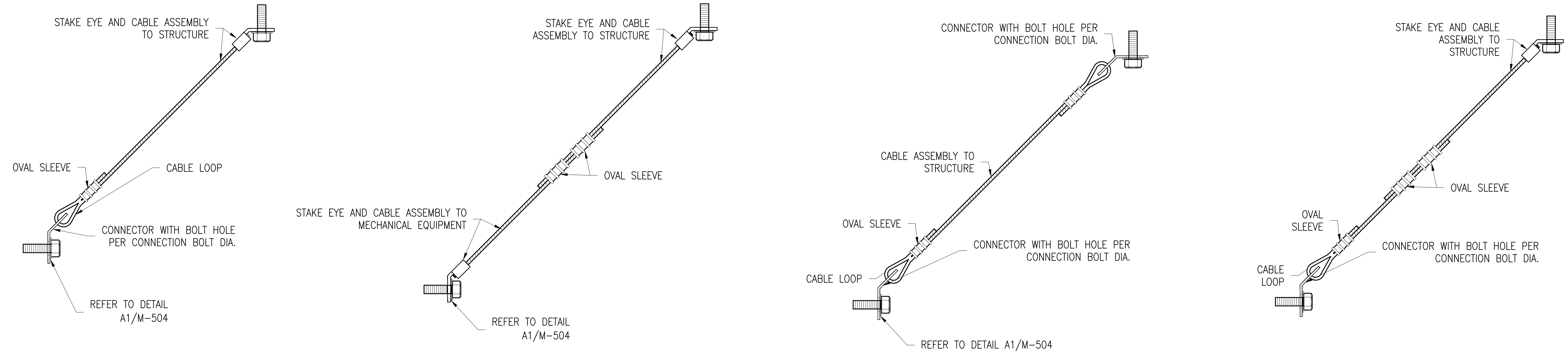
C1 CLOSEUP - PERPENDICULAR CONNECTION DETAIL
SCALE: NOT TO SCALE
M-503



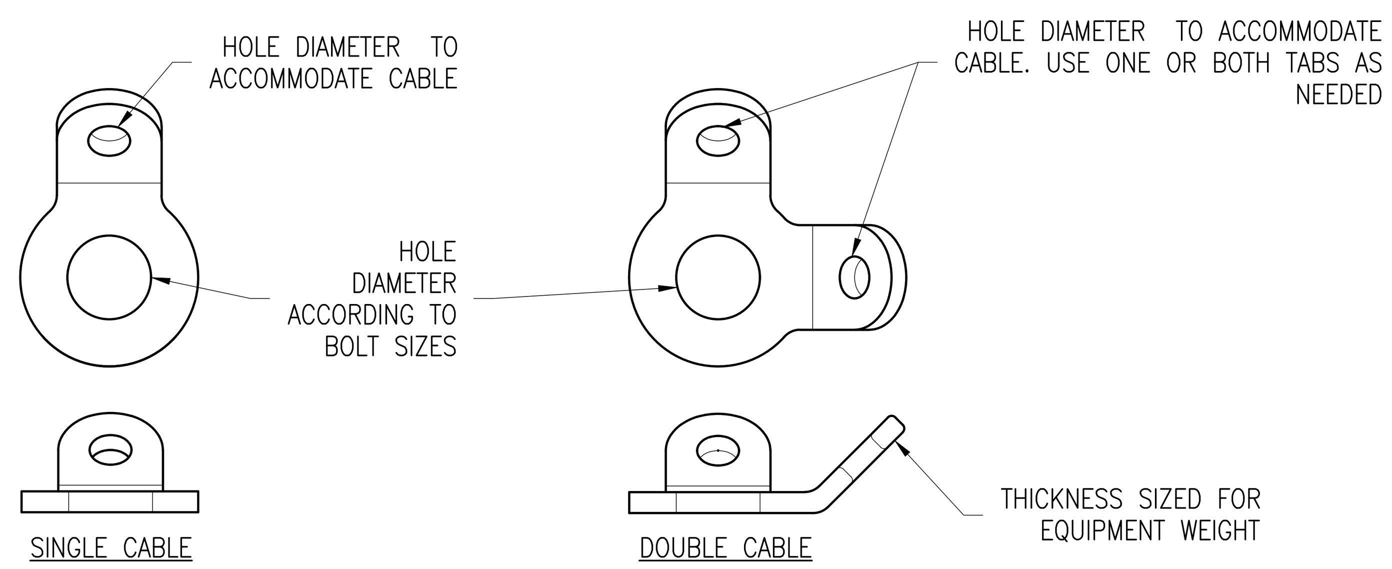
1 SECTION 1
SCALE: NOT TO SCALE
M-504



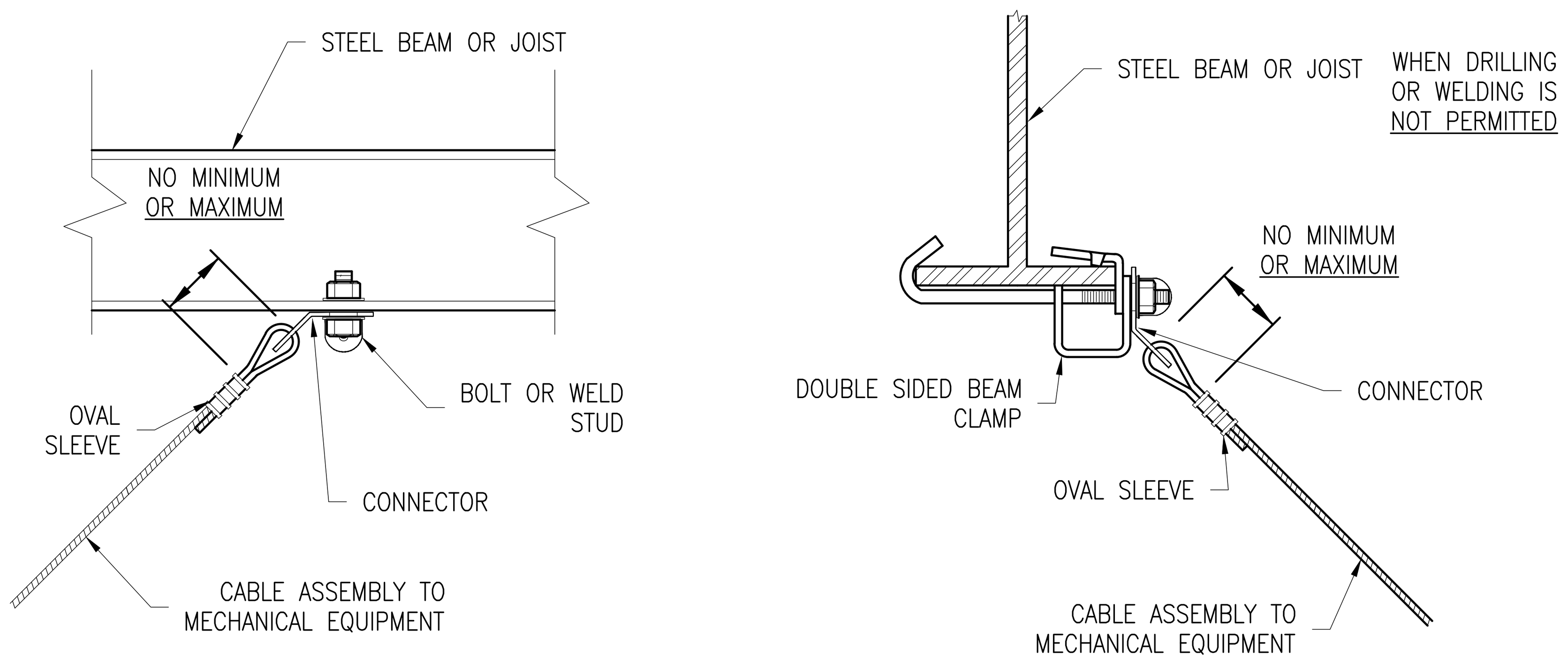
2 SECTION 2
SCALE: NOT TO SCALE
M-504



B1 CABLE BRACING ASSEMBLY DETAIL
SCALE: NOT TO SCALE
M-505



A1 BRACING CONNECTOR DETAIL
SCALE: NOT TO SCALE
M-504



A2 CABLE BRACING ATTACHMENT TO STRUCTURE DETAIL
SCALE: NOT TO SCALE
M-504

GENERAL SHEET NOTES

1. THE DETAILS ON THIS SHEET AND ON SHEET M-503 DO NOT PRECLUDE THE NEED TO DESIGN EQUIPMENT MOUNTINGS FOR FORCES REQUIRED BY OTHER CRITERIA SUCH AS FOR SEISMIC FORCES.
2. REFER TO SPECIFICATION SECTION 230548.19 "SEISMIC BRACING FOR HVAC" AND SECTION 220448.20 "MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL" FOR ADDITIONAL INFORMATION.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

SEAL

A/E INFO

APPROVED Wm B Forbes Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC
ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026
DES OLS DRW OLS CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
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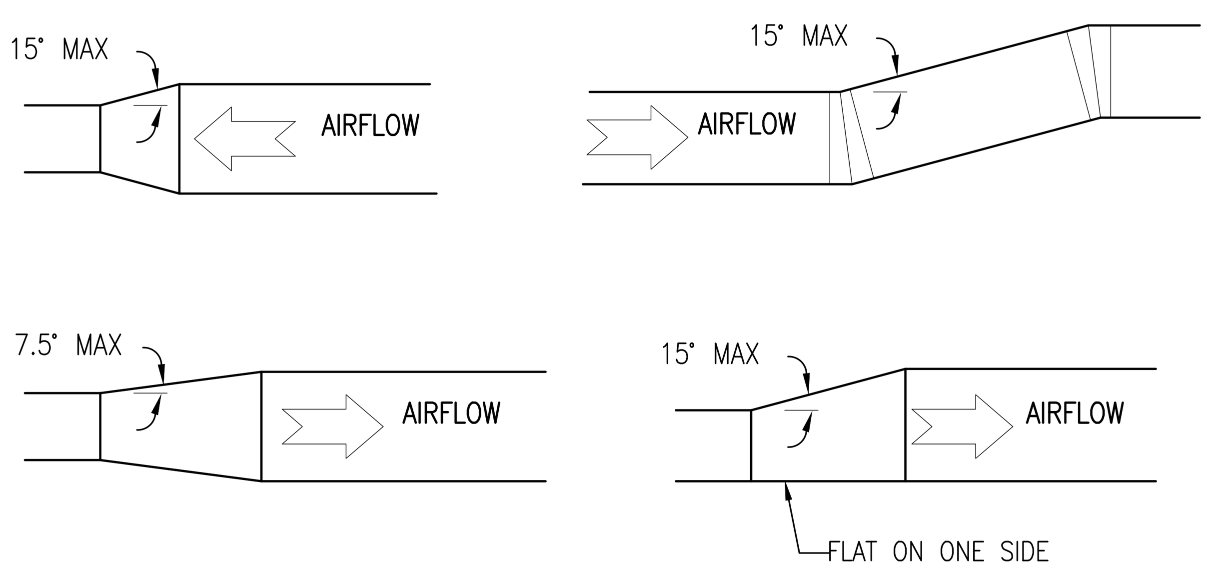
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
MECHANICAL DETAILS

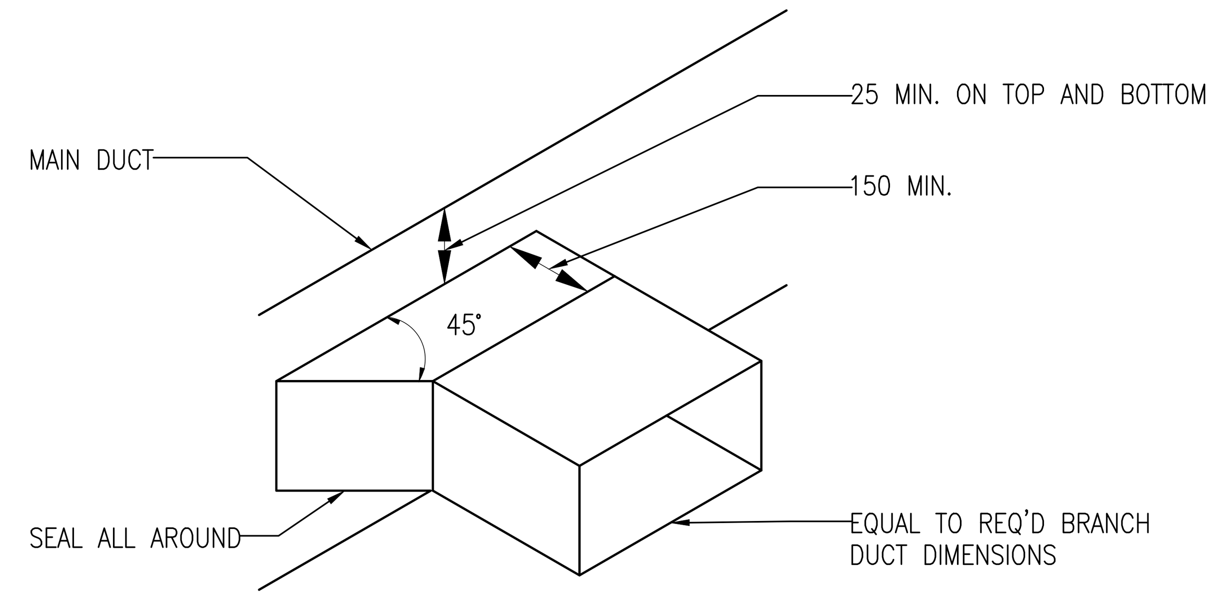
SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157185
SHEET 123 OF 170
M-504

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

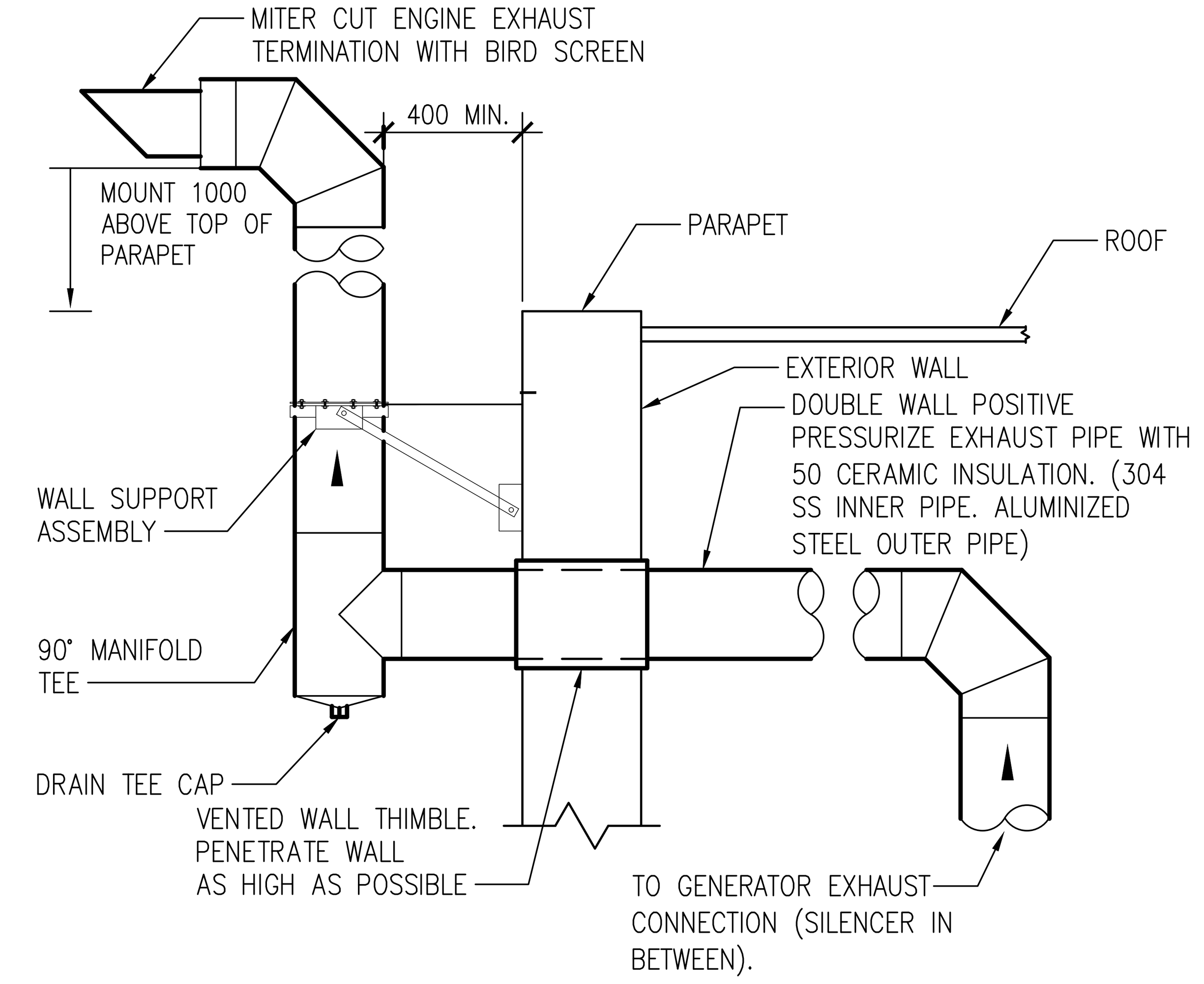
FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MUM\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-504.dwg LAYOUT NAME: M-504 PLOTTED: Tuesday, January 20, 2026 - 3:12pm USER: jennifer.a.klein



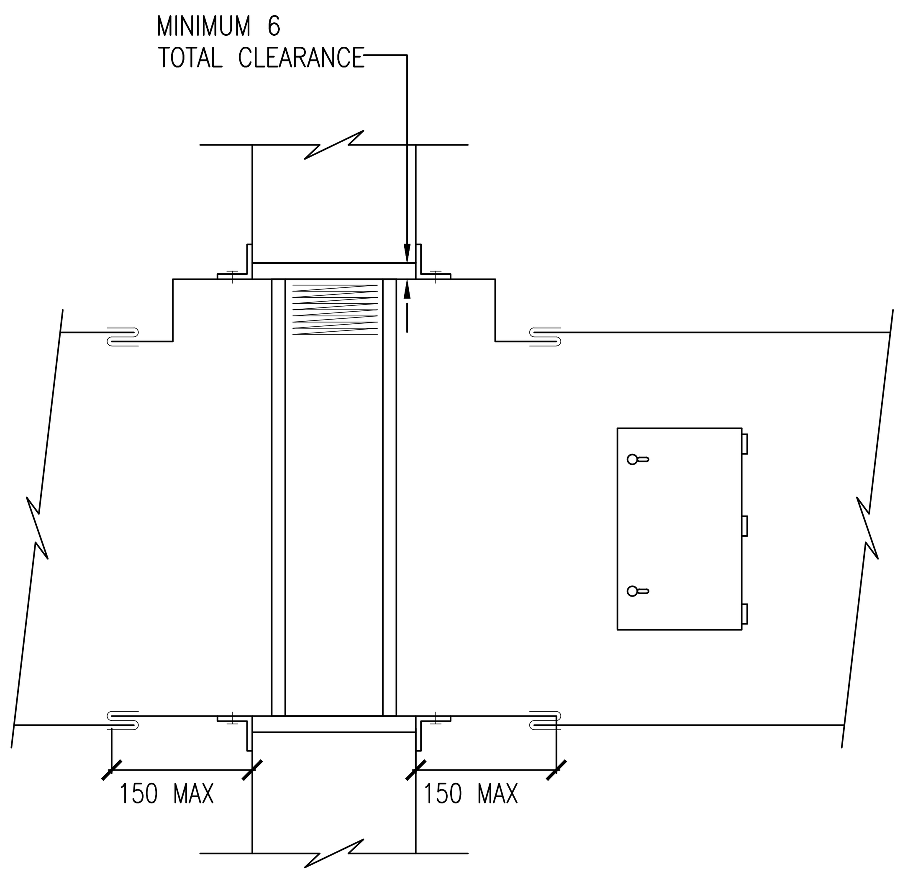
C1 LOW PRESSURE DUCT TRANSITIONS DETAIL
SCALE: NOT TO SCALE M-101



C2 LOW PRESSURE BRANCH TAKE-OFF FITTINGS DETAIL
SCALE: NOT TO SCALE M-101

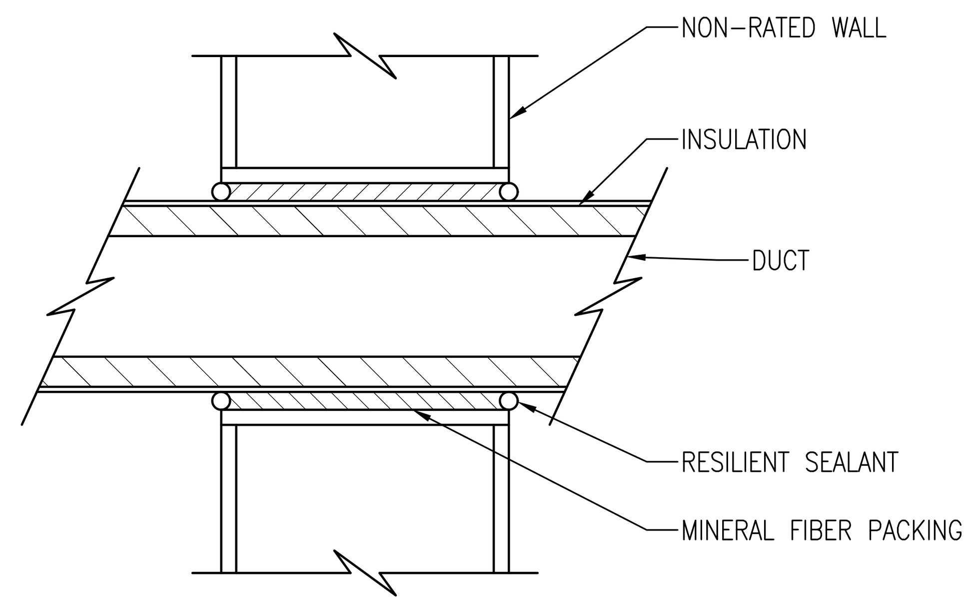


C4 EMERGENCY GENERATOR EXHAUST PIPING DETAIL
SCALE: NOT TO SCALE M-101



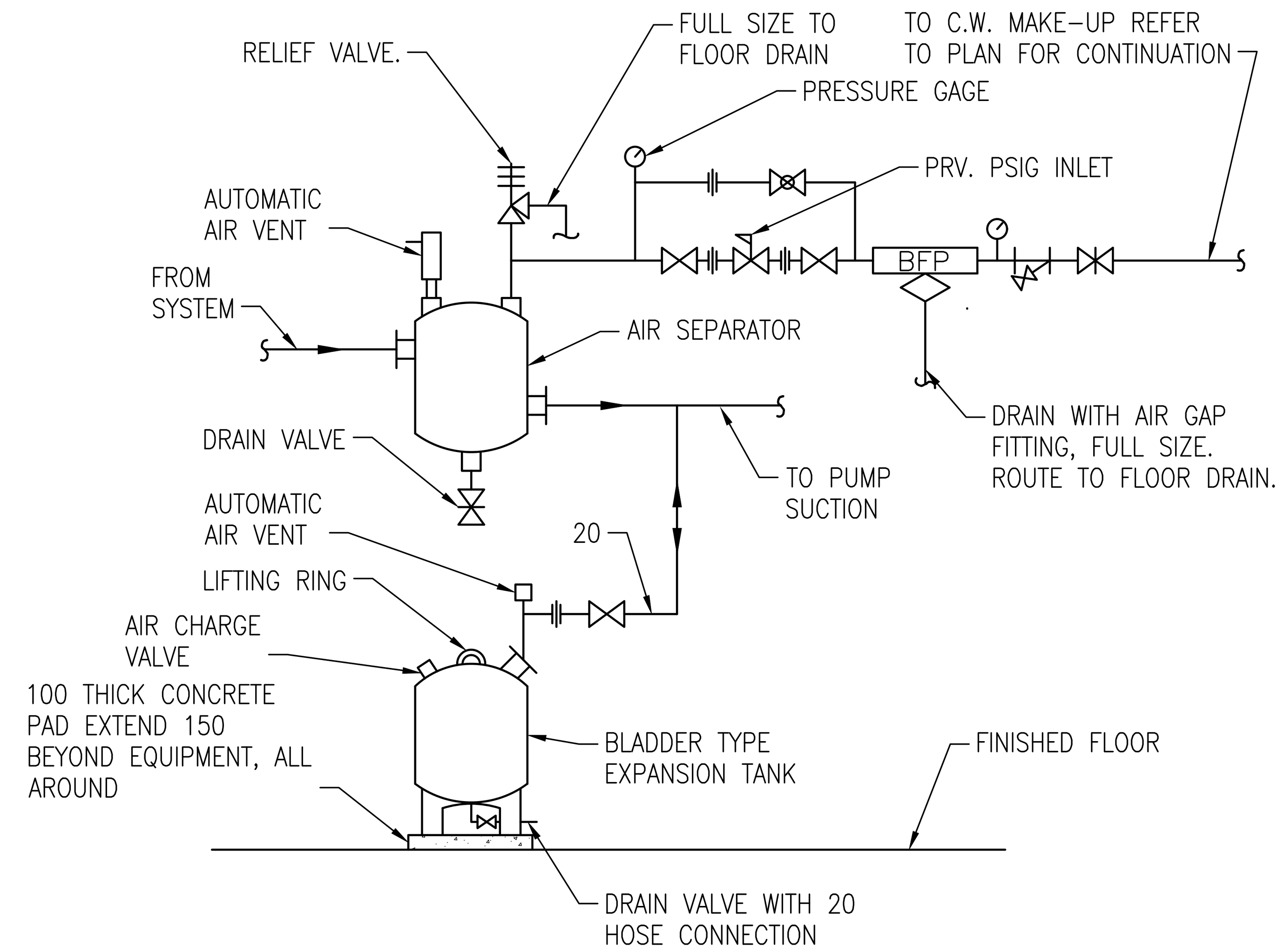
NOTES:
1. REFER TO FLOOR PLANS FOR DUCT DIMENSIONS
2. INSTALL FIRE DAMPER PER MANUFACTURER'S RECOMMENDATIONS TO MAINTAIN WALL RATING
3. INSTALL ACCESS DOOR SO THAT FIRE DAMPER FUSIBLE LINK CAN BE INSPECTED/MAINTAINED.
4. DUCT INSULATION NOT SHOWN FOR CLARITY.

A1 FIRE DAMPER INSTALLATION DETAIL
SCALE: NOT TO SCALE M-101

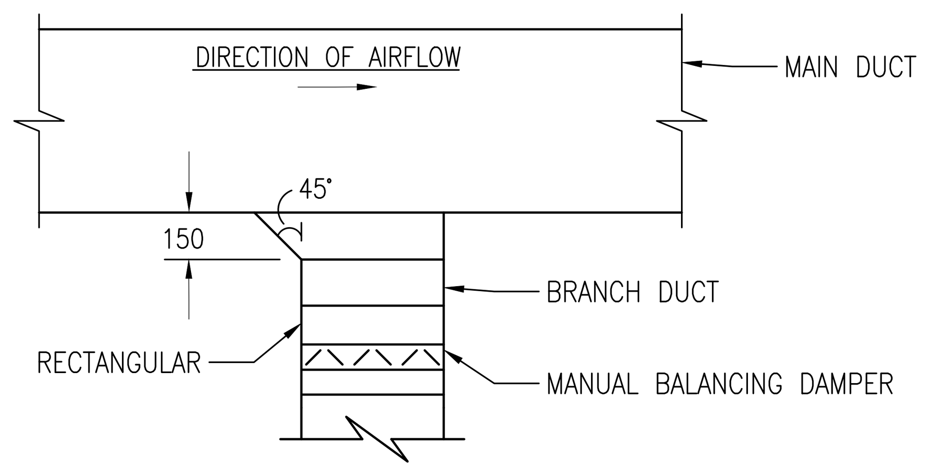


NOTES:
1. PROVIDE APPROXIMATELY 13 GAP BETWEEN INSULATION AND WALL.
2. SUPPORT DUCT ON BOTH SIDES OF WALL. CONTACT WITH WALL OR FRAMING IS NOT ALLOWED.
3. INSTALLATION OF UNINSULATED DUCT SIMILAR.

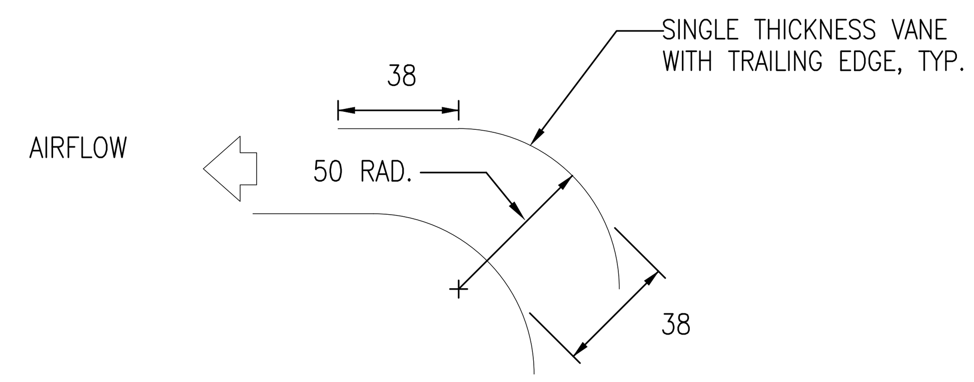
B2 DUCT PENETRATION, NON-RATED WALL - INTERIOR DETAIL
SCALE: NOT TO SCALE M-101



B4 AIR SEPARATOR & EXPANSION TANK PIPING DETAIL
SCALE: NOT TO SCALE M-102



A2 DUCT BRANCH TAKE-OFF DETAIL
SCALE: NOT TO SCALE M-101



NOTE: PROVIDE TURNING VANES ACROSS THE FULL WIDTH OF ALL SQUARE THROAT AND MITRED ELBOWS.

A4 TURNING VANES DETAIL
SCALE: NOT TO SCALE M-101

SYM	DESCRIPTION
DATE	APPR

NAVFAC

APPROVED: Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

ACTIVITY: D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES	OLS
DRW	OLS
CHK	FISH
PM/DM	MJD/ATP
BRANCH MANAGER	SCE
DES PROD DIR	WBF
FIRE PROTECTION ENGINEER	DSN

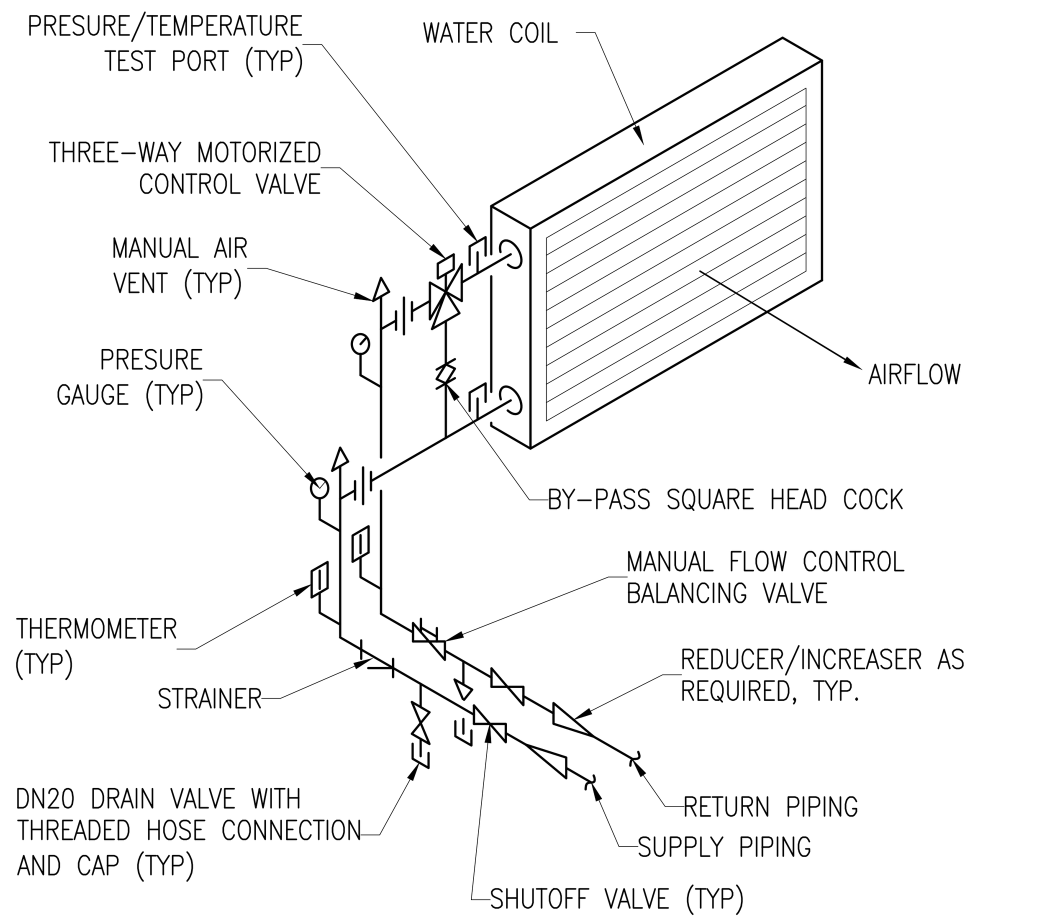
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK VA
VARIOUS LOCATIONS

JOINT MARITIME FACILITY
MECHANICAL DETAILS

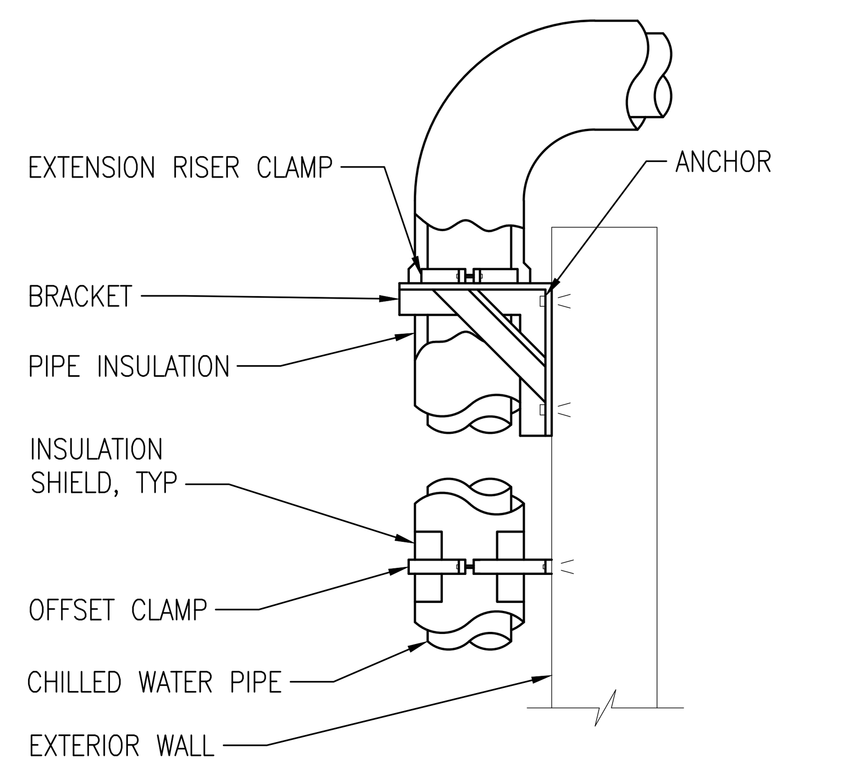
SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157186
SHEET 124 OF 170
M-505

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018



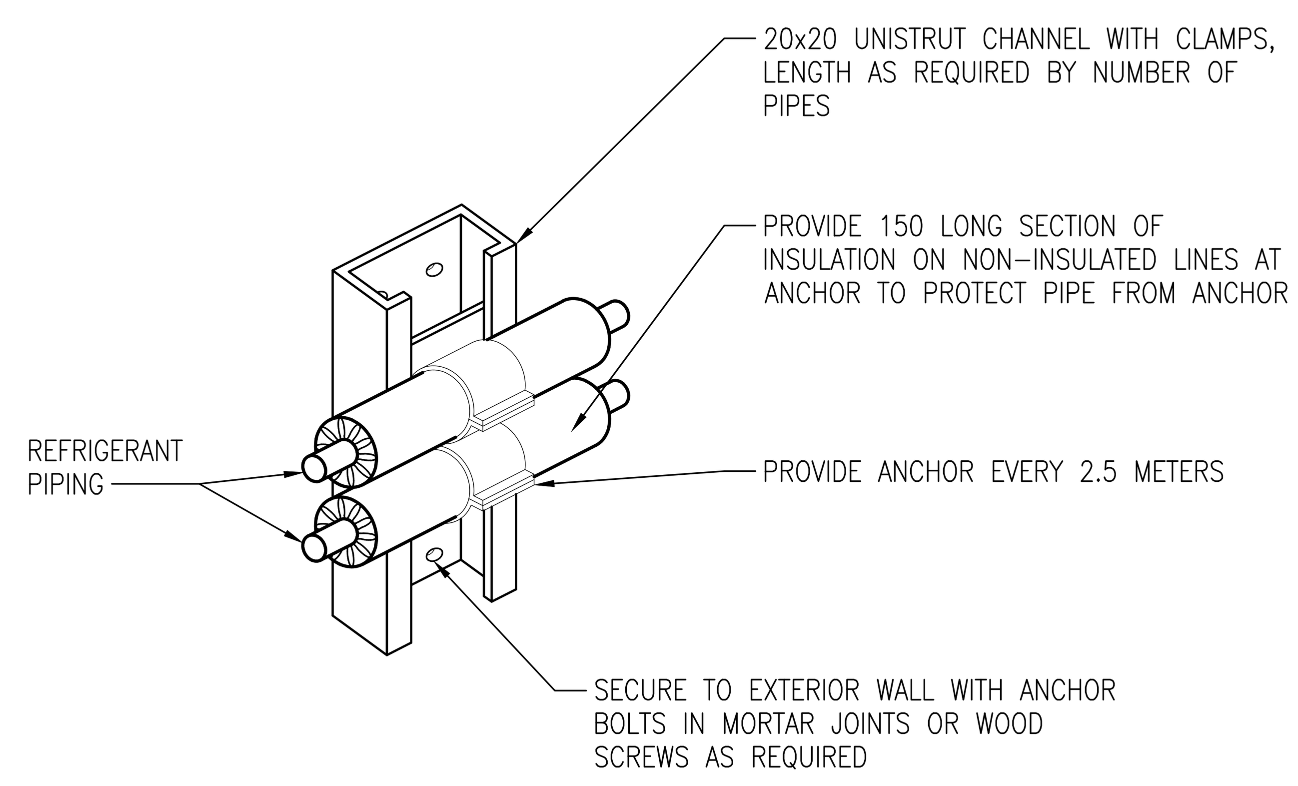
NOTES:
1. PIPE COIL COUNTER-FLOW AND ARRANGE PIPING TO PERMIT EASY REMOVAL OF COIL.
2. PROVIDE BALANCING VALVE MANUFACTURER'S RECOMMENDED STRAIGHT PIPING RUNS BOTH UPSTREAM AND DOWNSTREAM OF BALANCING VALVE.
3. SHUTOFF VALVE SHALL BE BALL VALVES FOR PIPE SIZES DN50 AND BELOW AND BUTTERFLY VALVES FOR PIPE SIZES DN65 AND ABOVE.
4. PRESSURE GAUGE AND THERMOMETER NOT REQUIRED FOR UNIT HEATERS.
5. PROVIDE DRAIN VALVES FOR AHU COOLING COILS AND HEATING COILS.

C1 WATER COIL PIPING DETAIL
SCALE: NOT TO SCALE — M-102



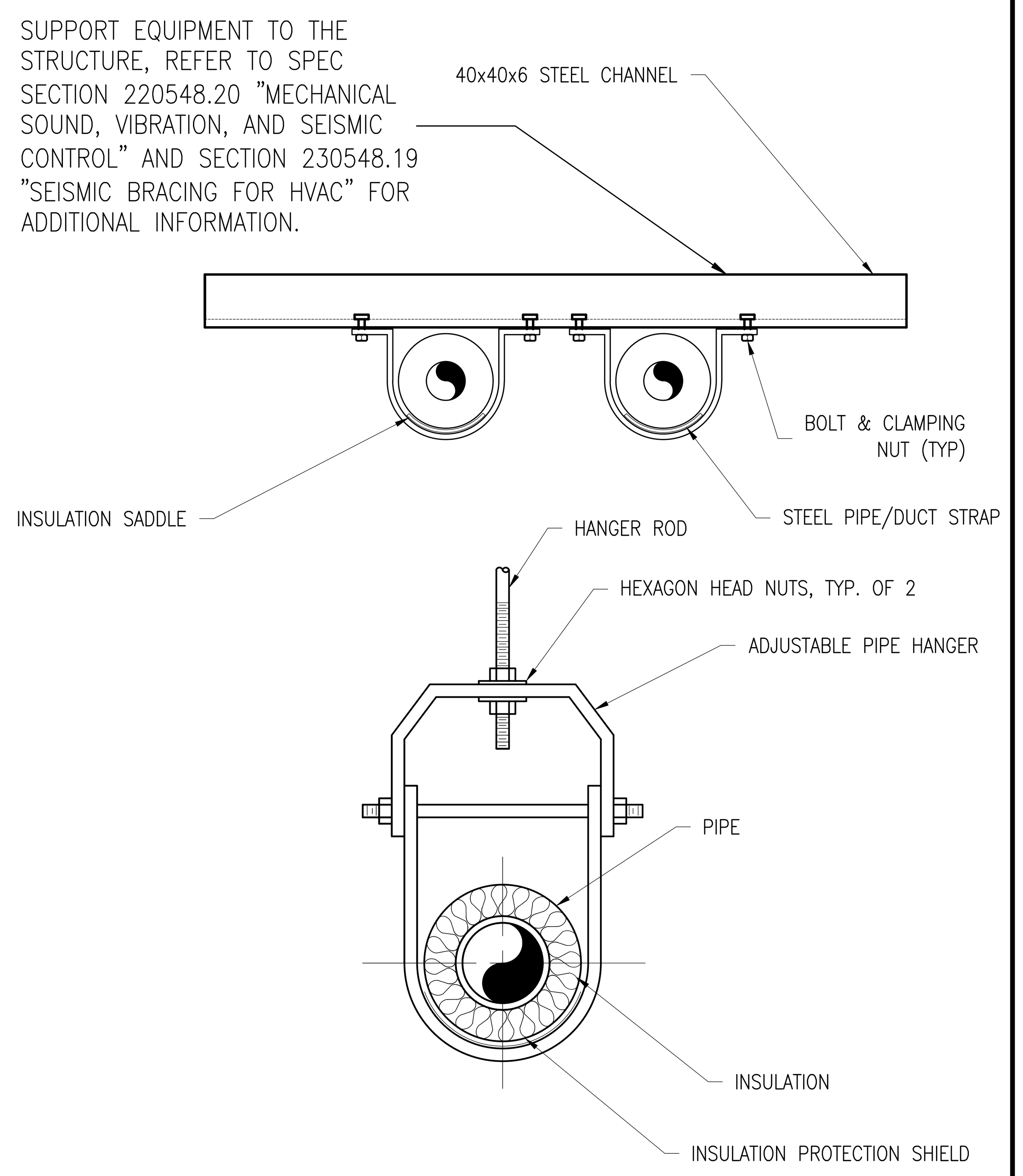
NOTES:
1. INSULATE OVER EXTENSION RISER CLAMP.
2. THE MAXIMUM DISTANCE BETWEEN EXTENSION RISER CLAMPS SHALL BE 3500 WITH OFFSET CLAMPS LOCATED MIDWAY. AT MINIMUM PROVIDE EXTENSION RISER CLAMPS AT THE HIGHEST AND LOWEST POINTS OF THE RISER.
3. BRACKETS CONSTRUCTED OF ? x ? x ? ANGLE
4. MATERIALS SHALL BE SUITABLE FOR OUTDOOR USE.

C2 CHILLED WATER RISER SUPPORT FROM WALL DETAIL
SCALE: NOT TO SCALE — M-102



NOTES: 1. MATERIALS SHALL BE SUITABLE FOR OUTDOOR USE.

C4 REFRIGERANT PIPE MOUNTING, EXTERIOR WALL DETAIL
SCALE: NOT TO SCALE — M-102



A1 PIPE SUPPORT DETAIL
SCALE: NOT TO SCALE — M-102

SYM	DESCRIPTION	DATE	APPR

NAVFAC

SEAL

A/E INFO

APPROVED: Wm B Forbes — Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

ACTIVITY: D. EVANS VIA EMAIL

SATISFACTORY TO — DATE 01/15/2026

DES OLS	DRW OLS	CHK FISH
PM/DM	MJD/ATP	
BRANCH MANAGER		SCE
DES PROD DIR		WBF
FIRE PROTECTION ENGINEER		DSN

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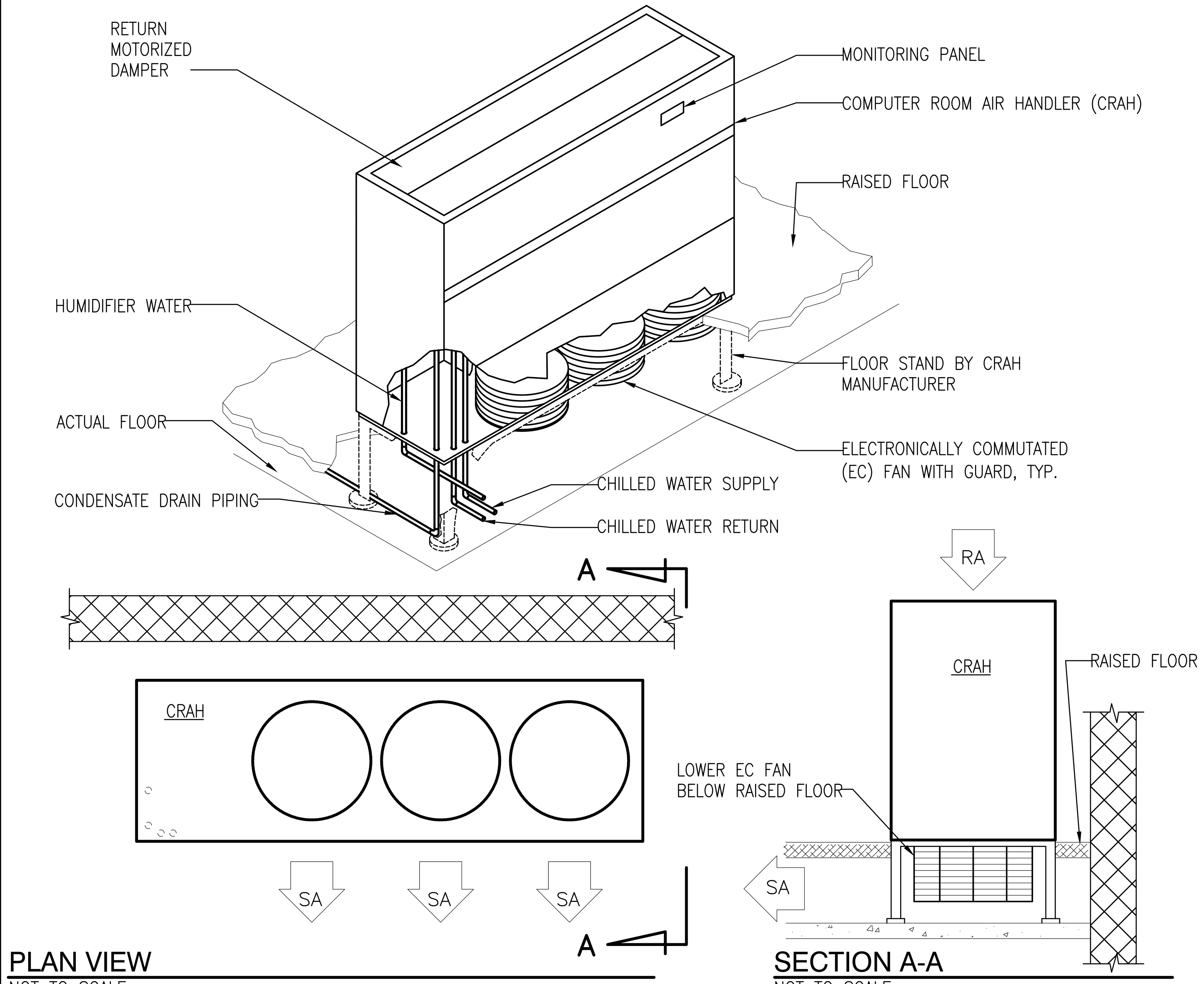
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK, VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

MECHANICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157187
SHEET 125 OF 170

M-506



PLAN VIEW
NOT TO SCALE

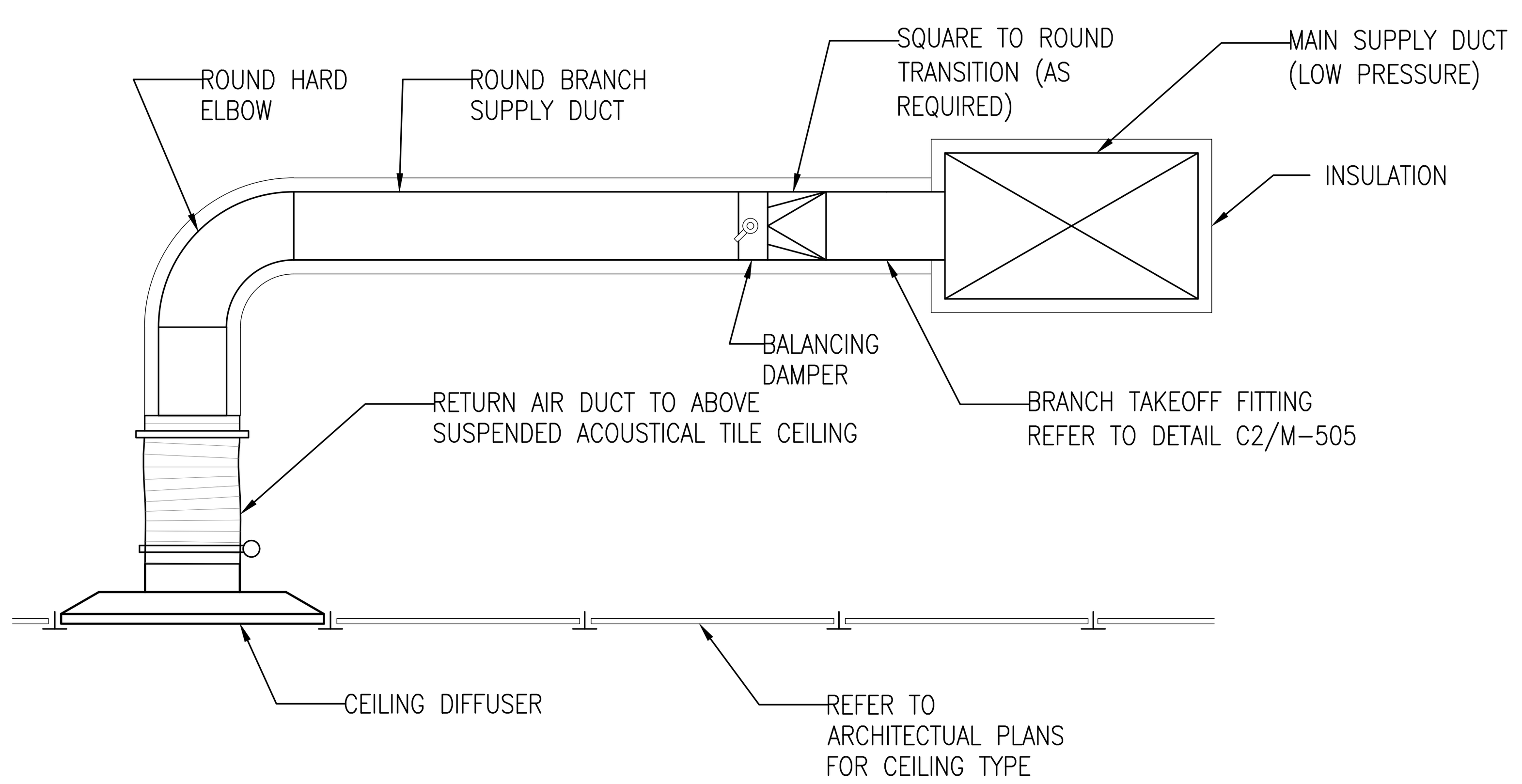
SECTION A-A
NOT TO SCALE

NOTES:
1. CRAH, EQUIPMENT STAND, AND ASSOCIATED APPURTENANCES SHALL BE SEISMICALLY RATED.
2. PROVIDE EQUIPMENT, EQUIPMENT STAND, PIPING AND ASSOCIATED APPURTENANCES WITH ANCHORS, VIBRATION MOUNTS AND SEISMIC COMPENSATION DEVICES AS DIRECTED BY THE CONTRACTOR TO ENSURE COMPLIANCE WITH SEISMIC REQUIREMENTS.
3. REFER TO FLOOR PLAN FOR PIPING CONTINUATION. REFER TO PIPING CHILLED WATER SYSTEM DIAGRAM FOR ADDITIONAL INFORMATION.

A1 COMPUTER ROOM AIR CONDITIONING UNIT DETAIL

SCALE: NOT TO SCALE / *SCALA: NON IN SCALA*

M-101



NOTES:
1. ROUND BRANCH SUPPLY DUCT DIAMETER, FLEXIBLE DUCT DIAMETER, AND BRANCH TAKEOFF HEIGHT AND WIDTH MUST BE THE SAME DIMENSION AS THE DIFFUSER NECK DIAMETER.

A3 SUPPLY AIR DIFFUSER, TYPICAL

SCALE: NOT TO SCALE

M-101

SYM | DESCRIPTION | DATE | APPR

NAVFAC

APPROVED Wm B Forbes Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC
ACTIVITY D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026
DES OLS DRW OLS CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
MECHANICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157188
SHEET 126 OF 170
M-507

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJM\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-507.dwg LAYOUT NAME: M-507 PLOTTED: Tuesday, January 20, 2026 - 3:12pm USER: jennifer.a.klein

GRILLE, REGISTER, AND DIFFUSER SCHEDULE

MARK	SERVICE	TYPE	AIRFLOW (LPS RANGE)	MAX. NC	MAX. SP (PA)	NECK SIZE (MM)	MODULE SIZE	THROW IN METERS (AT 0.25 MPS TERMINAL VELOCITY)	NOTES
S1	SUPPLY	LAY-IN	0–66	14	13.69	150ø	600X600	2.1	1
S2	SUPPLY	LAY-IN	66–116	17	13.94	200ø	600X600	2.4	1
S3	SUPPLY	SURFACE MOUNT	123–165	20	13.94	300X300	300X300	5.5	2
S4	SUPPLY	SURFACE MOUNT	0–75	20	16.17	150X150	150X150	2.8	2
S5	SUPPLY	LAY-IN	104–179	18	16.92	250ø	600X600	3	1
S6	SUPPLY	SURFACE MOUNT	80–132	21	29.11	230X230	230X230	5.2	2
S7	SUPPLY	SURFACE MOUNT	132–185	22	18.9	300ø	500X500	3.9	2
R1	RETURN	LAY-IN	0–87	22	12.44	200X200	600X600	–	3
R2	RETURN	LAY-IN	153–229	25	12.44	300X300	600X600	–	3
R3	RETURN	LAY-IN	153–304	28	19.41	400X400	600X600	–	3
R4	RETURN	SURFACE MOUNT	35–87	22	19.41	200X200	–	–	4
R5	RETURN	SURFACE MOUNT	185–389	27	19.41	450X450	–	–	4
R6	RETURN	SURFACE MOUNT	0–45	24	19.41	150X150	–	–	4
R7	RETURN	LAY-IN	231–530	24	14.93	350X350	600X600	–	3
R8	RETURN	SURFACE MOUNT	231–530	24	14.93	350X350	–	–	4
R9	RETURN	SURFACE MOUNT	125–147	<20	24.9	250X250	–	–	4

NOTES:

1. LAY-IN LOUVERED CEILING DIFFUSER, ALUMINUM CONSTRUCTION, SQUARE FACE, INTEGRAL ROUND NECK, ADJUSTABLE DISCHARGE PATTERN.
2. SURFACE MOUNTED CEILING DIFFUSER, ALUMINUM CONSTRUCTION, SQUARE FACE, INTEGRAL SQUARE NECK, ADJUSTABLE DISCHARGE PATTERN.
3. LAY-IN CEILING RETURN GRILLE, ALUMINUM CONSTRUCTION, SQUARE FACE.
4. SURFACE MOUNTED RETURN GRILLE, ALUMINUM CONSTRUCTION, SQUARE FACE.

FAN SCHEDULE

MARK	LOCATION	SERVICE	TYPE	AIR FLOW (LPS)	ESP (PA)	DRIVE TYPE	ELECTRICAL DATA		WEIGHT (KG)	NOTES
							V/PH/HZ	POWER (KW)		
EF-1	UPS-B 119	EXHAUST	CABINET	209	57	DIRECT	120/1/60	0.08	23	3
EF-2	UPS-A 117	EXHAUST	CABINET	270	73	DIRECT	120/1/60	0.08	23	3
EF-3	OPS EQUIPMENT 116	EXHAUST	INLINE	1062	81	DIRECT	120/1/60	0.75	75	1
EF-4	OPS EQUIPMENT 116	EXHAUST	INLINE	1062	81	DIRECT	120/1/60	0.75	75	1
EF-5	ABOVE CEILING	EXHAUST	INLINE	47	73	DIRECT	120/1/60	0.07	–	2
EF-6	ABOVE CEILING	EXHAUST	INLINE	47	72	DIRECT	120/1/60	0.07	–	2
EF-7	ABOVE CEILING	EXHAUST	INLINE	47	80	DIRECT	120/1/60	0.07	–	2
EF-8	ABOVE CEILING	EXHAUST	INLINE	47	64	DIRECT	120/1/60	0.07	–	2

NOTES:

1. PROVIDE INLET GUARD
2. RADON MITIGATION FAN, PLUG TYPE. SEE ELECTRICAL DRAWINGS FOR RECEPTACLE LOCATION.
3. EXPLOSION PROOF FAN

DESIGN CONDITIONS

SPACE TYPE	COOLING		HEATING
	DRY BULB AT 1% OCCURRENCE	HUMIDITY RATIO AT 1% OCCURRENCE	DRY BULB AT 1% OCCURRENCE
	°C DB/WB	°C DB/WB	°C DB
OFFICE AREAS	25.0/18.0	29.0/22.0	20.0
ASSEMBLY AREAS	25.0/18.0	29.0/22.0	20.0
EQUIPMENT ROOMS	--		5.0
OUTDOOR CONDITIONS	32.0/21.0		4.0

LOUVER SCHEDULE

MARK	TYPE	WIDTH (mm)	HEIGHT (mm)	DEPTH (mm)	AIRFLOW (LPS)	MIN. NET FREE AREA (m^2)	MAX. AIR PRESSURE DROP (PA)	NOTES
L-1	EXHAUST	2500	2500	300	20000	3.77	35	1,3,4,5
L-2	INTAKE	3500	3500	300	20970	6.25	25	2,3,4,5
L-3	EXHAUST	1000	600	100	1960	0.29	62	1,3,4,5
L-4	INTAKE	1500	600	125	1960	0.46	30	2,3,4,5
L-5	EXHAUST	1300	550	125	2635	0.49	35	1,3,4,5
L-6	INTAKE	1500	550	150	1700	0.21	37	2,3,4,5
L-7	EXHAUST	460	460	125	575	0.07	25	1,3,4,5

NOTES:
1. DRAINABLE BLADE TYPE.
2. WIND DRIVEN RAIN TYPE, AMCA-500-L (WIND DRIVEN RAIN)
3. THE BEGINNING POINT OF WATER PENETRATION IS ABOVE 6.35 METERS PER SECOND FREE AREA VELOCITY.
4. FLANGED FRAME, HORIZONTAL BLADE, 150 mm DEEP, 25 mm SQUARE MESH BIRD SCREEN, 70% KYNAR PVDF/100% FLUOROPOLYMER FEVE FINISH, AMCA-500-L (AIR AND WATER).
5. REFER TO ARCHITECTURAL SHEETS FOR ADDITIONAL INFORMATION.

ELECTRIC UNIT HEATER SCHEDULE

MARK	SERVING ROOM	AIRFLOW (LPS)	ENTERING AIR TEMP (°C)	CAPACITY (KW)	AMPS	V/HZ/PH
EUH-1	113 FIRE PUMP	300	7	2.0	9.7	208/1/60
EUH-2	114 GENERATOR FUEL ROOM	300	7	2.0	9.7	208/1/60
EUH-3	114 GENERATOR FUEL ROOM	300	7	2.0	9.7	208/1/60
EUH-4	115 GENERATOR ROOM	300	7	2.0	9.7	208/1/60
EUH-5	115 GENERATOR ROOM	300	7	2.0	9.7	208/1/60
EUH-6	118 ELECTRICAL	300	7	2.0	9.7	208/1/60
EUH-7	118 ELECTRICAL	300	7	2.0	9.7	208/1/60
EUH-8	112 MECHANICAL	300	7	2.0	9.7	208/1/60
EUH-9	112 MECHANICAL	300	7	2.0	9.7	208/1/60
EUH-10	100 SECURE VESTIBULE	300	7	1.5	7.2	208/1/60
EUH-11	111 CORRIDOR	300	7	1.5	7.2	208/1/60

NOTES:
1. HORIZONTAL DISCHARGE UNIT COMPLETE WITH MOTOR, FAN, HIGH LIMIT TEMPERATURE CONTROL WITH AUTOMATIC RESET, STAINLESS STEEL SHEATHED TUBULAR HEATING ELEMENT, BUILT IN THERMAL OVERLOAD, FACTORY INSTALLED CONTROL CIRCUIT AND CONTACTOR, WALL MOUNT THERMOSTAT, AND WALL BRACKET OR SUSPENSION KIT.
2. MOUNT BOTTOM OF UNIT 2,7M AFF.

DUCTLESS SPLIT-SYSTEM HEAT PUMP UNIT SCHEDULE

MARK	LOCATION	TYPE	COOLING DATA				HEATING DATA			ELECTRICAL DATA					SEER	NOTES
			AIRFLOW (LPS)	TOTAL CAPACITY (KW)	SENSIBLE CAPACITY (KW)	EAT DB/WB (°C)	AIRFLOW (LPS)	TOTAL CAPACITY (KW)	EAT DB/WB (°C)	VOLTS	PHASE	HZ	MCA	MOP		
DDIU-1 /DDOU-1	TELECOM 110	WALL MOUNTED	135	2,5	2,2	26,7/19,4	151	3,2	21/16	208	1	60	9	15	18,0	1,2,3

NOTES:
1. COMPLETE WITH INDOOR UNIT, INVERTER DUTY VARIABLE SPEED COMPRESSOR, ANTI-CORROSION OUTDOOR COIL COATING, UL OR ETL LISTING.
2. PROVIDE LOW AMBIENT CONTROL FOR COOLING OPERATION AT HEATING OUTDOOR DESIGN CONDITION.
3. PROVIDE RIGID CONDENSATE PIPING WITH A DIAMETER NO LESS THEN THE CONNECTION DIAMETER AND AS RECOMMENDED BY THE MANUFACTURER.

SYM | DESCRIPTION | DATE | APPR

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A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

ACTIVITY D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES OLS | DRW OLS | CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

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JOINT MARITIME FACILITY

MECHANICAL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157189
SHEET 127 OF 170

M-601

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-601.dwg LAYOUT NAME: M-601 PLOTTED: Tuesday, January 20, 2026 – 3:12pm USER: jennifer.a.klein

AIR COOLED CHILLER SCHEDULE

MARK	AMBIENT TEMPERATURE (°C)	FLUID	FOULING FACTOR ($m^2°C/W$)	EVAPORATOR FLOWRATE (LPS)	EVAPORATOR CAPACITY (KW)	EVAPORATOR EWT (°C)	EVAPORATOR LWT (°C)	EVAPORATOR PRESSURE DROP (kPA)	IPLV	ELECTRICAL V/PH/HZ	ELECTRICAL MCA (AMPS)	ELECTRICAL MOP (AMPS)
ACCH-1,2,3	21.1	30% PROPYLENE GLYCOL	0.0176	5.2	107	11.1	5.6	21.1	16.7	480/3/60	69	100

NOTES:
1. HIGH EFFICIENCY AIR COOLED UNIT COMPLETE WITH 2 INDEPENDENT VARIABLE CAPACITY REFRIGERANT CIRCUITS WITH 2 HERMETICALLY SEALED SCROLL COMPRESSORS PER CIRCUIT WITH A FULL OPERATING CHARGE OF REFRIGERANT.
2. FEATURES AND ACCESSORIES INCLUDE CORROSION RESISTANT STEEL UNIT PANELS AND FRAME, STAINLESS STEEL BRAZED PLATE HEAT EXCHANGER WITH BLANKET HEATER AND INSULATION, CONDENSER COIL WITH INTEGRAL SUBCOOLING CIRCUIT, EC FANS, OIL MANAGEMENT SYSTEM, LOW NOISE COMPRESSOR, AUTO-ADAPTIVE MICROPROCESSOR CONTROL, LOW AMBIENT CONTROLS DOWN TO 0°C OPERATION, AND NEOPRENE PADS.
3. PROVIDE SINGLE POINT ELECTRICAL CONNECTION, ELECTRONIC SOLID STATE SOFT STARTER, AND NON-FUSED DISCONNECT SWITCH.
4. UNIT MANUFACTURED FOR OUTDOOR LOCATIONS COMPLETE WITH ANTI-CORROSION EPOXY COIL COATING.
5. PROVIDE UNIT WITH FEATURES AND ACCESSORIES REQUIRED FOR BACNET DDC INTEGRATION.
8. PROVIDE A SEISMICALLY RATED UNIT WITH VIBRATION ISOLATORS.

AIR SEPARATOR SCHEDULE

MARK	ROOM LOCATION	SYSTEM	LPS	SIZE MM
AS-1	112 MECHANICAL	CHILLED WATER	8.1	75

NOTES:
1. TANGENTIAL TYPE, HEAVY DUTY CAST IRON OR CARBON STEEL CONSTRUCTION COMPLETE WITH PERFORATED BAFFLE, THREADED OR FLANGED CONNECTIONS, 175°C MAX OPERATING TEMPERATURE, 10 BAR MAX WORKING PRESSURE, AIR VENT, REMOVABLE STRAINER AND STAINLESS STEEL INTERNALS.

EXPANSION TANK SCHEDULE

MARK	ROOM LOCATION	SYSTEM	MINIMUM TANK VOLUME (L)	ACCEPTANCE VOLUME (L)	FILL PRESSURE (kPA)
ET-1	112 MECHANICAL	BUILDING CHILLED WATER	37,8	37,8	190

NOTES:
1. BLADDER TYPE EXPANSION TANK IN VERTICAL FREE STANDING CONFIGURATION.
2. TANK DESIGN PRESSURE IS 860 kPA.

PUMP SCHEDULE

MARK	SERVICE	LOCATION	TYPE	FLUID	FLOWRATE LPS	E.S.P. KPA	MIN. EFFICIENCY %	ELECTRICAL DATA KW	ELECTRICAL DATA V/PH/HZ
CHWP-1	BUILDING CHILLED WATER	112 MECHANICAL	END SUCTION	30% PROPYLENE GLYCOL	13.8	225	70	4.4	480/3/60
CHWP-2	BUILDING CHILLED WATER	112 MECHANICAL	END SUCTION	30% PROPYLENE GLYCOL	13.8	225	70	4.4	480/3/60

NOTES:
1. PROVIDE A VARIABLE SPEED DRIVE FROM PUMP MANUFACTURER FOR EACH PUMP.
2. PROVIDE PREMIUM EFFICIENCY MOTORS.

VIBRATION ISOLATION SCHEDULE

EQUIPMENT TYPE	ISOLATOR TYPE	MIN DEFLECTION (mm)
AIR HANDLER	4	19
CHILLER	4	25.0
FAN	3	6.4
HIGH WALL	1	6.4
SPLIT HEAT PUMP OUTSIDE UNIT	4	6.4

ISOLATOR TYPES:
1. PAD, RUBBER, OR GLASS FIBER
2. RUBBER FLOOR ISOLATOR OR HANGER
3. SPRING FLOOR ISOLATOR OR HANGER
4. RESTRAINED SPRING ISOLATOR

APPROVED: Wm B forbes — Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

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SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157190
SHEET 128 OF 170
M-602

FAN COIL UNIT SCHEDULE

MARK	SERVING ROOM	SUPPLY FAN			COOLING COIL DATA										ELECTRIC HEATING COIL DATA				ELECTRICAL DATA		NOTES
		FLOW RATE (LPS)	ESP (PA)	POWER (kW)	TOTAL CAPACITY (KW)	SENSIBLE CAPACITY (KW)	EAT		LAT		EWT (°C)	LWT (°C)	LPS	WATER P.D. (KPA)	TOTAL CAPACITY (KW)	EAT (°C)	LAT (°C)	NONIMAL CURRENT (A)	FLA	V/PH/HZ	
							°C DB	°C WB	°C DB	°C WB											
FCU-1	MULTIPLE	404	63	0.3	4.0	3.7	24.0	17.9	15.4	14.7	5.6	11.1	0.19	3.0	2.6	21.2	27.4	7.3	2.4	208/3/60	1-6
FCU-2	MULTIPLE	656	81	0.45	8.2	8.1	24.0	16.4	13.1	12.2	5.6	11.1	0.40	3.2	7.9	21.2	31.9	21.8	2.4	208/3/60	1-6
FCU-3	MULTIPLE	223	140	0.2	2.8	2.8	24.0	17.0	14.1	13.3	5.6	11.1	0.13	1.6	3.8	21.2	35.2	10.7	2.4	208/3/60	1-6
FCU-4	117 UPS A	1080	50	1.7	15.9	15.9	25.1	16.8	13.0	12.2	5.6	11.1	0.77	8.1	1.3	20.1	21.1	3.7	11.2	208/3/60	1-6
FCU-5	117 UPS A	1080	50	1.7	15.9	15.9	25.1	16.8	13.0	12.2	5.6	11.1	0.77	8.1	1.3	20.1	21.1	3.7	11.2	208/3/60	1-6
FCU-6	119 UPS B	1100	50	1.8	16.2	16.2	25.1	16.8	13.0	12.2	5.6	11.1	0.78	8.3	1.5	20.1	21.3	4.2	11.2	208/3/60	1-6
FCU-7	119 UPS B	1100	50	1.8	16.2	16.2	25.1	16.8	13.0	12.2	5.6	11.1	0.78	8.3	1.5	20.1	21.3	4.2	11.2	208/3/60	1-6
FCU-8	MULTIPLE	632	176	0.9	4.0	4.0	29.6	19.4	15.1	14.3	5.6	11.1	0.19	13.2	1.6	12.9	18.8	4.5	4.6	208/3/60	1-6

NOTES:
1. UNIT COMPLETE WITH CHILLED WATER COOLING COIL, ELECTRIC HEATING COIL, 50MM THICK PLEATED MERV 11 AIR FILTER, BACNET CONTROL INTERFACE, AND CONDENSATE PUMP.
2. PROVIDE DUAL SLOPED STAINLESS STEEL DRAIN PAN UNDER THE COOLING COIL. PIPE COOLING COIL CONDENSATE TO NEAREST FLOOR DRAIN.
3. PROVIDE SINGLE POINT ELECTRICAL CONNECTION AND ELECTRIC DISCONNECT.
4. COOLING COIL FLOW BASED ON GLYCOL SOLUTION
5. PROVIDE 2-WAY MODULATING CONTROL VALVES FROM UNIT MANUFACTURER.
6. FAN MOTOR POWER IS APPROXIMATE, PROVIDE FAN MOTOR COMBINATION THAT CAN MEET THE FLOW RATE AT EXTERNAL STATIC PRESSURE INCLUDING INTERNAL STATIC PRESSURE.

BUFFER TANK SCHEDULE

MARK	ROOM LOCATION	SYSTEM	TANK VOLUME (L)	TANK DIAMETER (MM)	PIPING CONNECTION SIZE (MM)
BT-1	112 MECHANICAL	CHILLED WATER	1500	1100	100

NOTES:
1. DESIGN PRESSURE 860 kPA
2. CONSTRUCTED IN ACCORDANCE WITH ASME SECTION VIII, DIVISION I, LATEST EDITION
3. STAINLESS STEEL CONSTRUCTION
3. PROVIDE WITH R12.5 INSULATING JACKET

DEDICATED OUTDOOR AIR SYSTEM (DOAS) AIR HANDLING UNIT SCHEDULE

MARK	ROOM LOCATION	SUPPLY FAN			EXHAUST FAN			ELECTRIC PREHEAT COIL			COOLING COIL												ELECTRIC REHEAT COIL			STATIC ENTHALPY CORE										
		AIRFLOW (LPS)	ESP (PA)	MOTOR (KW)	AIRFLOW (LPS)	ESP (PA)	MOTOR (KW)	TOTAL CAPACITY (KW)	EAT (°C)	LAT (°C)	TOTAL CAPACITY (KW)	LATENT CAPACITY (KW)	EAT			LAT			EWT (°C)	LWT (°C)	WATER FLOW (LPS)	WATER P.D. (KPA)	TOTAL CAPACITY (KW)	EAT (°C)	LAT (°C)	LATENT CAPACITY (kW)	SUPPLY AIR						EXHAUST AIR			MIN. LATENT EFFICIENCY (%)
																											EAT			LAT			EAT			
													°C DB	°C WB	°C DP	°C DB	°C WB	°C DP									°C DB	°C WB	°C DP	°C DB	°C WB	°C DP	°C DB	°C WB	°C DP	
DOAS-1	115 MECHANICAL	1,238	306	2.2	799	151	1.2	34.9	-10.6	12.8	22.1	9.9	20.9	17.5	15.6	12.8	12.2	11.8	5.6	11.1	1.8	50.3	18.3	12.8	25	5.0	18.3	17.8	17.6	20.9	17.5	15.6	25.0	17.0	12.2	55

NOTES:
1. 480V/3PH/60HZ UNIT WITH SUPPLY AIR AND RETURN AIR PLUG FANS. FAN MOTORS MUST BE RATED FOR USE WITH VARIABLE FREQUENCY DRIVES.
2. COMPLETE UNIT INCLUDES BUT NOT LIMITED TO: FANS, VARIABLE FREQUENCY DRIVES, PREHEAT COIL, CHILLED WATER COIL, REHEAT COIL, AIR FILTERS, TOTAL ENERGY HEAT EXCHANGER, ACCESS SECTIONS, MOTORIZED DAMPERS, BACNET CONTROL INTERFACE.
3. FILTER AIR PRESSURE DROP SHALL BE MINIMUM 124.5 PA IN THE SUBMITTAL TO SIMULATE DIRTY FILTER.
4. REFER TO CONTROL DRAWING AND SEQUENCE OF OPERATION FOR ADDITIONAL INFORMATION.
5. PROVIDE COILS WITH ANTI-CORROSION EPOXY COATING.
6. PROVIDE MINIMUM CLASS 2 OPPOSED BLADE DAMPERS AT THE OUTDOOR AIR AND EXHAUST AIR.
7. FAN MOTOR POWER IS APPROXIMATE, PROVIDE FAN MOTOR COMBINATION THAT CAN MEET THE FLOW RATE AT EXTERNAL STATIC PRESSURE INCLUDING INTERNAL STATIC PRESSURE.

COMPUTER ROOM AIR HANDLING (CRAH) UNIT SCHEDULE

MARK	SERVING ROOM	SUPPLY FANS			COOLING COIL DATA										ELECTRICAL DATA				ELECTRIC HEATER	NOTES
		SUPPLY AIR (LPS)	ESP (PA)	KW	TOTAL CAPACITY (KW)	SENSIBLE CAPACITY (KW)	EAT		LAT		EWT °C	LWT °C	LPS	WATER P.D. (KPA)	FLA	MCA	MOCP	V/HZ/PH	KW	
							°C DB	°C WB	°C DB	°C WB										
CRAH-1,2,3,4	116	2750	750	9	44	44	26.0	15.6	12.7	10.3	5.6	11.1	2.1	72	56	69	75	480/60/3	30	1-10

NOTES:
1. UNIT COMPLETE WITH CHILLED WATER COOLING COIL, UNDER-FLOOR VARIABLE SPEED SUPPLY FAN, TOP RETURN, 50MM THICK PLEATED MERV 11 AIR FILTER, BACNET CONTROL INTERFACE, CONDENSATE PUMP AND FLOOR STAND (BY UNIT MANUFACTURER). FLOOR STAND HEIGHT TO MATCH HEIGHT OF RAISED FLOOR.
2. MINIMUM 3 STAGE ELECTRIC REHEAT.
3. N+1 CONFIGURATION (3 RUN AT ONE TIME)
4. PROVIDE FACTORY CONTROL MODULE, WIRING, HARDWARE AND SOFTWARE SO THAT UNITS MAY COMMUNICATE CONTROL WITH EACH OTHER. REFER TO CONTROL DRAWING AND SEQUENCE OF OPERATION FOR ADDITIONAL INFORMATION.
5. PROVIDE DUAL SLOPED STAINLESS STEEL DRAIN PAN UNDER COOLING COIL. ROUTE CONDENSATE DRAIN TO NEAREST FLOOR DRAIN.
6. PROVIDE SINGLE POINT ELECTRICAL CONNECTION AND ELECTRIC DISCONNECT.
7. UNITS MUST BE DELIVERED FROM THE FACTORY WITH SEISMIC BRACING AND SHALL BE INSTALLED IN COMPLIANCE WITH MANUFACTURER'S SEISMIC INSTALLATION INSTRUCTIONS UNLESS OTHERWISE INDICATED.
8. PROVIDE CABLE TYPE LEAK DETECTION SYSTEM. SYSTEM SHALL BE COMPLETE WITH CONTROL MODULE THAT WILL CONSTANTLY MONITOR AT MINIMUM STATUS, POINTS OF LEAKS, INTERNAL FAULTS, AND POWER FAILURES. PROVIDE MINIMUM LENGTH 9M PLENUM RATED CABLE TO MONITOR AS MUCH UNDER RAISED FLOOR AREA AS POSSIBLE.
9. PROVIDE WITH 2-WAY MODULATING CONTROL VALVE FROM UNIT MANUFACTURER.
10. SUPPLY FANS POWER KW IS AN ESTIMATE FOR APPROXIMATING CIRCUIT SIZING. SELECT UNIT THAT MEETS SUPPLY AIR FLOW RATE AT EXTERNAL STATIC PRESSURE INCLUDING INTERNAL STATIC PRESSURES.

SYM | DESCRIPTION | DATE | APPR

NAVFAC

SEAL

A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
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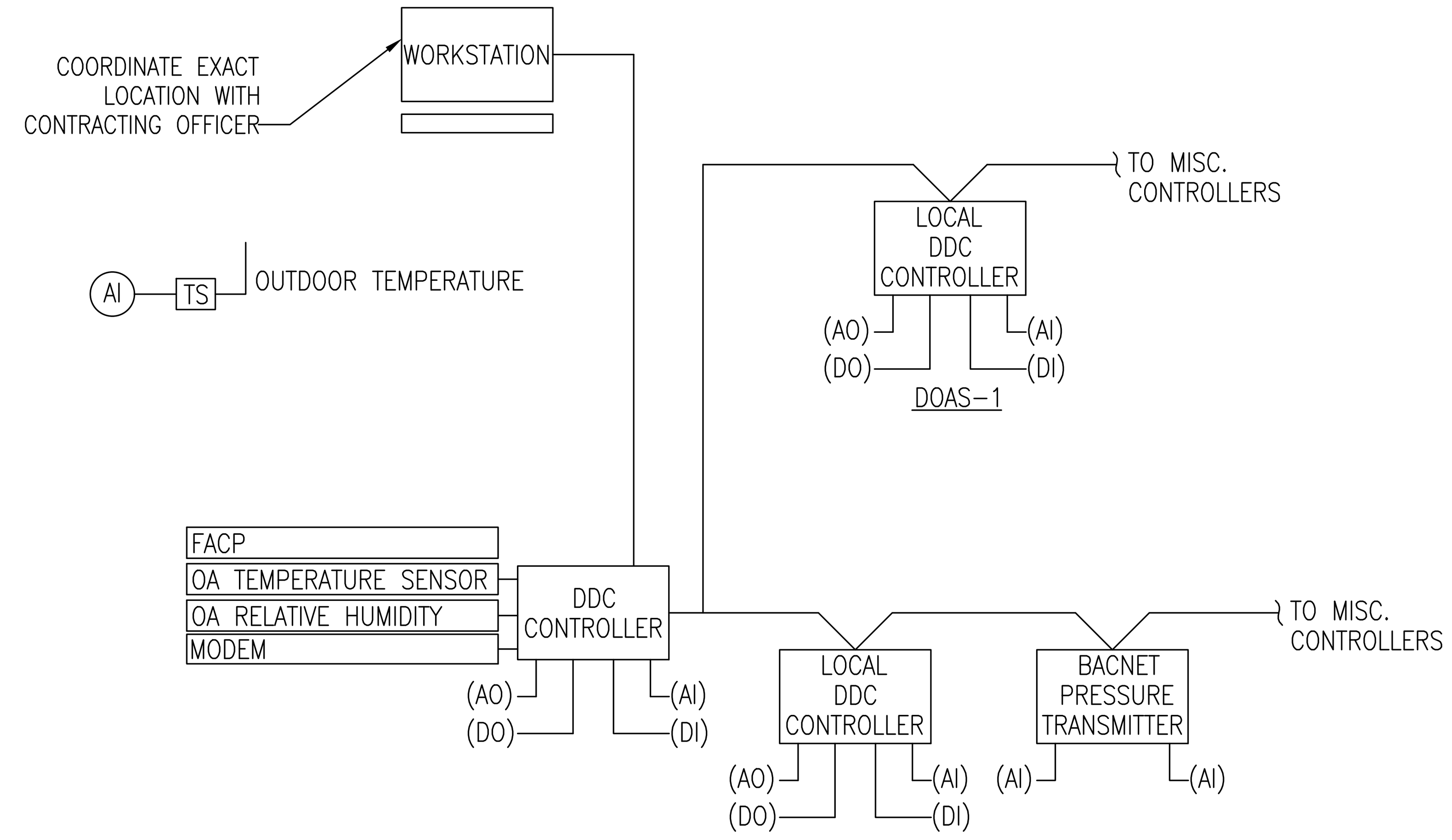
NAVFAC DRAWING NO. 14157191
SHEET 129 OF 170
M-603
NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-603.dwg LAYOUT NAME: M-603 PLOTTED: Tuesday, January 20, 2026 - 3:13pm USER: jennifer.a.klein

MECHANICAL CONTROL LEGEND

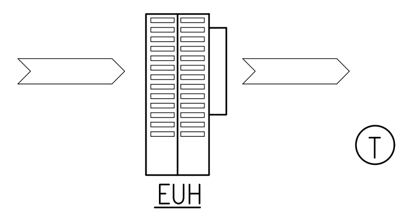
EH	ELECTRIC HEATING COIL
SD	SMOKE DETECTOR
MD	MOTORIZED DAMPER
TS	TEMPERATURE SENSOR
CAV	CLEAN AGENT VENTILATION SWITCH
CO_2	CARBON DIOXIDE CONCENTRATION SENSOR
DPS	DIFFERENTIAL PRESSURE SENSOR
HGD	HYDROGEN GAS DETECTOR
HOA	HAND–OFF–AUTOMATIC SWITCH
MS	MOTOR STARTER
CSR	CURRENT SENSOR RELAY
H	HUMIDISTAT
T	SPACE THERMOSTAT
T	SPACE TEMPERATURE SENSOR
H	SPACE HUMIDITY SENSOR
FZ	FREEZESTAT
SP	STATIC PRESSURE SENSOR
W	WATER SENSING SWITCH
FACP	FIRE ALARM CONTROL PANEL
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
ON/OFF	ON/OFF SWITCH
OS	OCCUPANCY SENSOR
OO	OCCUPANCY OVERRIDE
HD	HYDROGEN DETECTOR

AMS	AIRFLOW MEASURING STATION
AF	AIRFLOW
CO	CARBON MONOXIDE SENSOR
CS	CURRENT SWITCH
LS	LIMIT SWITCH
NO2	NITROGEN DIOXIDE SENSOR
OCC	OCCUPANCY SENSOR
	THREE WAY CONTROL VALVE
	TWO WAY CONTROL VALVE
AO	ANALOG OUTPUT
AI	ANALOG INPUT
DO	DIGITAL OUTPUT
DI	DIGITAL INPUT
	STATUS LIGHT
	DISCONNECT
FS	FLOW SWITCH
FM	FLOW METER
VFD	VARIABLE FREQUENCY DRIVE
MD	MOTORIZED DAMPER
SCR	SILICON CONTROLLED RECTIFIER
	AND GATE RELAY



DDC SYSTEM ARCHITECTURE

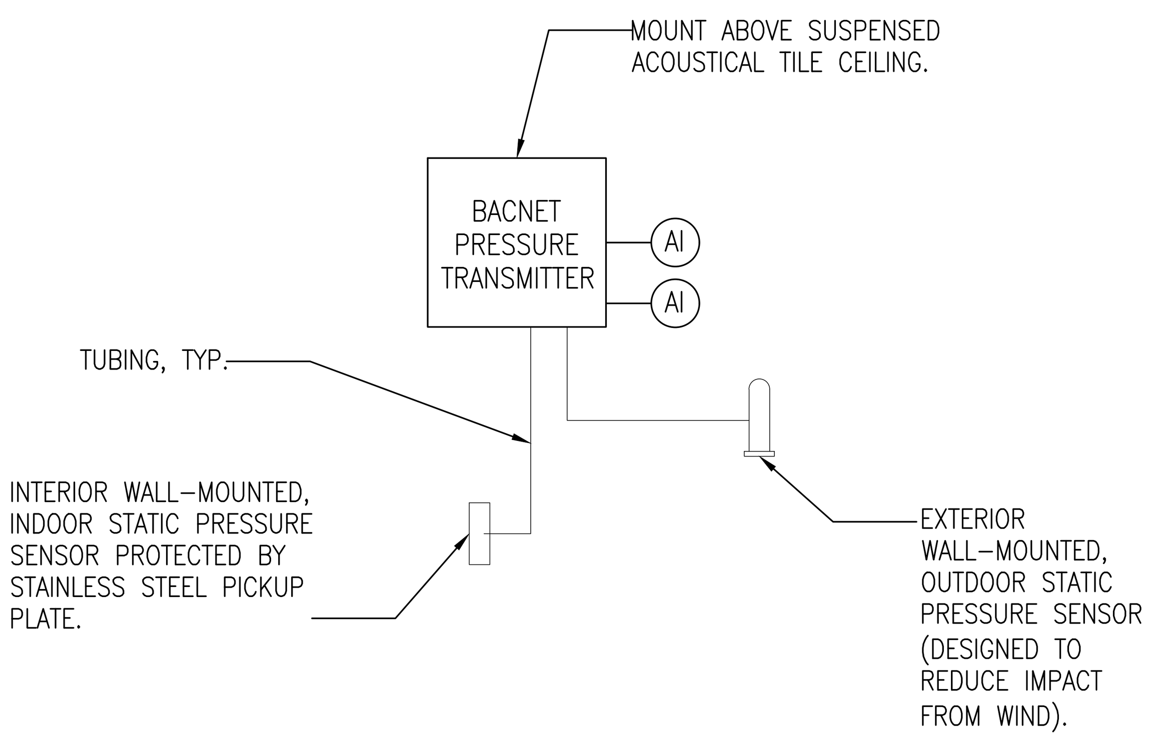
NOT TO SCALE



EUH CONTROL DIAGRAM, TYPICAL

NOT TO SCALE

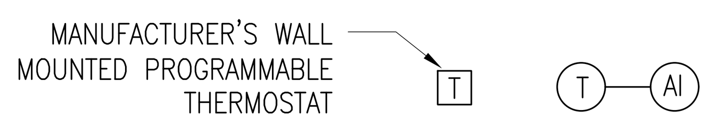
LOCAL DDC CONTROLLER	
INPUTS	
ANALOG	INTERNAL BUILDING PRESSURE (AHU–1A)
	OUTDOOR PRESSURE (NORTH)
OUTPUTS	
ANALOG	



NOTES:
1. THE PRESSURE TRANSMITTER SHALL HAVE BUILT–IN DIGITAL DAMPENING OPTIONS TO MINIMIZE UNWANTED FLUCTUATIONS CAUSED BY AMBIENT CONDITIONS.
2. HIDE TUBING IN WALLS OR ABOVE CEILING.

BUILDING AND OUTDOOR PRESSURE CONTROL DIAGRAM

NOT TO SCALE



LOCAL DDC CONTROLLER	
INPUTS	
ANALOG	TEMPERTAURE SENSOR
OUTPUTS	
ANALOG	

DDIH/DDOU CONTROL DIAGRAM, TYPICAL

NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

SEAL

A/E INFO

APPROVED Wm B forbes Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

ACTIVITY D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

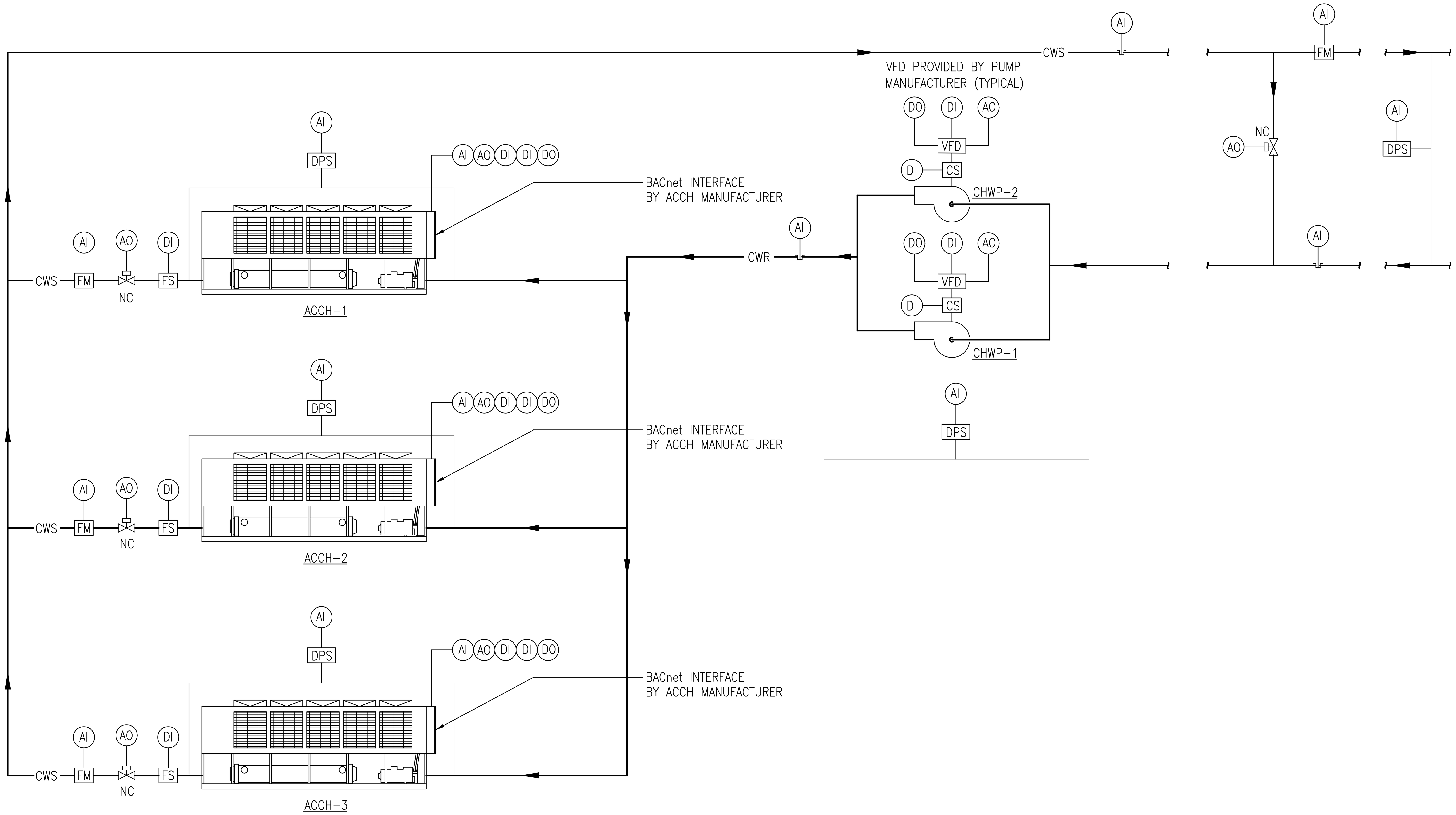
DES RER | DRW RER | CHK FISH
PM/DM MJD/ATP
BRANCH MANAGER SCE
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
MECHANICAL CONTROLS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157192
SHEET 130 OF 170
M-701

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\12_MECH\P1360_1783264_M-701.dwg LAYOUT NAME: M-701 PLOTTED: Tuesday, January 20, 2026 – 3:13pm USER: jennifer.a.klein



CHILLED WATER PLANT AND LOOP			
	INPUTS		OUTPUTS
ANALOG	ACCH-1,2,3 CHILLER FAULT CODE	ANALOG	ACCH-1,2,3 SETPOINT RESET
	ACCH-1,2,3 FLOW RATE		ACCH-1,2,3 FLOW CONTROL – ISOLATION
	ACCH-1,2,3 DIFFERENTIAL PRESSURE		ACCH-1,2,3 DEMAND LIMIT
			CHWP-1,2 SPEED
	CWS PLANT TEMP		FLOW CONTROL – BYPASS
	CWR PLANT TEMP		
	CWR LOOP TEMP		
	CWS LOOP FLOW RATE		
	CHWP DIFFERENTIAL PRESSURE		
	LOOP DIFFERENTIAL PRESSURE		
DIGITAL	ACCH-1,2,3 STATUS	DIGITAL	ACCH-1,2,3 ENABLE/DISABLE
	ACCH-1,2,3 ALARM		CHWP-1,2 ENABLE/DISABLE
	ACCH-1,2,3 FLOW SWITCH STATUS		
	CHWP-1,2 STATUS		
	CHWP-1,2 ALARM		

CHILLED WATER PLANT AND LOOP CONTROL DIAGRAM

NOT TO SCALE

SYM DESCRIPTION DATE APPR

NAVFAC

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DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC

NORFOLK, VA

PLANNING, DESIGN AND CONSTRUCTION

VARIOUS LOCATIONS

VARIOUS LOCATIONS

JOINT MARITIME FACILITY

CHILLED WATER CONTROL

SCALE: AS NOTED

EPROJECT NO.: 1838720

CONSTR. CONTR. NO.

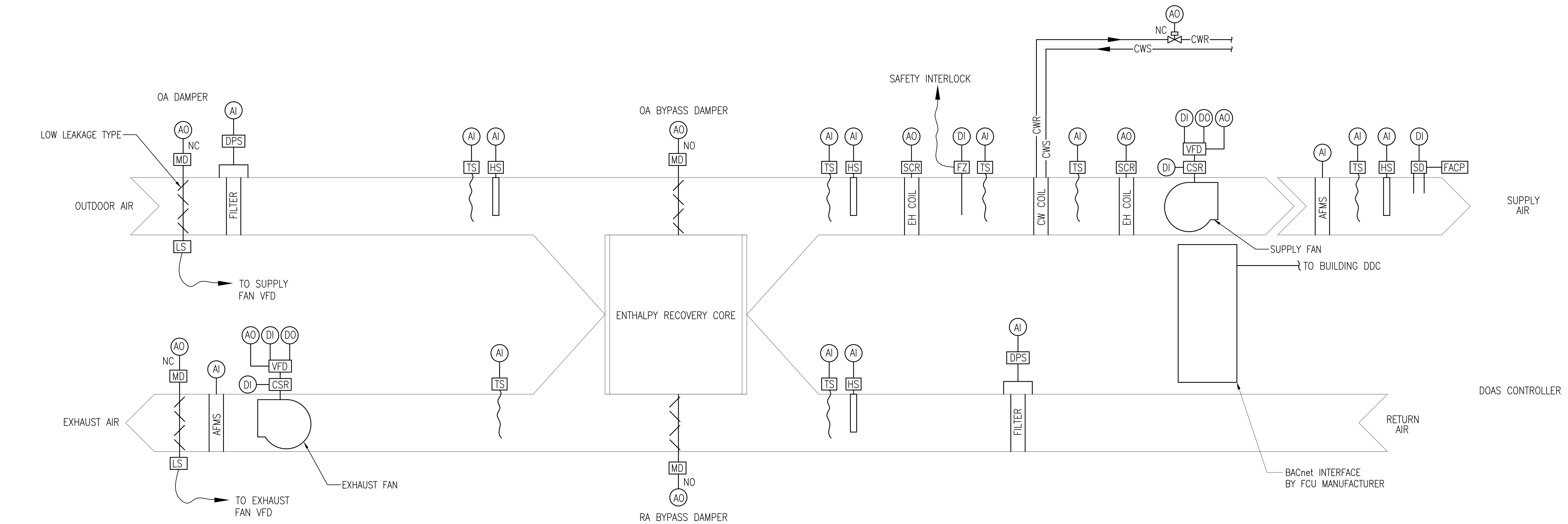
NAVFAC DRAWING NO. 14157193

SHEET 131 OF 170

M-702

DOAS-1					
	INPUTS		OUTPUTS		CALCULATED
ANALOG	OA FILTER DIFFERENTIAL PRESSURE	ANALOG	COOLING COIL CONTROL VALVE	ANALOG	OA ENTHALPY UPSTREAM OF CORE
	RA FILTER DIFFERENTIAL PRESSURE		HEATING COIL MODULATION		OA ENTHALPY DOWNSTREAM OF CORE
			PREHEAT COIL MODULATION		RA ENTHALPY UPSTREAM OF CORE
	OA DRY BULB UPSTREAM OF CORE				SA ENTHALPY DOWNSTREAM OF FAN
	OA DRY BULB DOWNSTREAM OF CORE		CORE OA BYPASS DAMPER		
	OA DRY BULB DOWNSTREAM OF PREHEAT COIL		CORE RA BYPASS DAMPER		OA DEW POINT UPSTREAM OF CORE
	OA DRY BULB DOWNSTREAM OF COOLING COIL		OUTDOOR AIR DAMPER POSITION		OA DEW POINT DOWNSTREAM OF CORE
	OA DRY BULB DOWNSTREAM OF REHEAT COIL		EXHAUST AIR DAMPER POSITION		SA DEW POINT DOWNSTREAM FAN
	RA DRY BULB UPSTREAM OF CORE		EXHAUST FAN VFD SPEED		
			SUPPLY FAN VFD SPEED		
	OA HUMIDITY UPSTREAM OF CORE				
	OA HUMIDITY DOWNSTREAM OF CORE				
	RA HUMIDITY UPSTREAM OF CORE				
	RA HUMIDITY DOWNSTREAM OF CORE				
	EA FLOW DOWNSTREAM EXHAUST FAN				
DIGITAL	FREEZESTAT	DIGITAL	EXHAUST FAN ENABLE	DIGITAL	
			SUPPLY FAN ENABLE		
	EXHAUST FAN STATUS				
	EXHAUST FAN VFD FAULT		SUPPLY SMOKE DETECTOR TO FACP		
	SUPPLY FAN STATUS				
	SUPPLY FAN VFD FAULT				

SUPPLY AIR					
	INPUTS		OUTPUTS		CALCULATED
ANALOG	SA FLOW DOWNSTREAM SUPPLY FAN	ANALOG		ANALOG	
	SA DRY BULB DOWNSTREAM OF SUPPLY FAN				
	SA HUMIDITY DOWNSTREAM OF COOLING COIL				
DIGITAL	SUPPLY AIR SMOKE DETECTOR	DIGITAL		DIGITAL	



NOTES
1. ALL CONTROL ACCESSORIES, SOFTWARE AND HARDWARE POINTS INDICATED ARE A SET REQUIRED TO MEET THE SEQUENCE OF OPERATION AND MONITORING.

DOAS-1 CONTROL DIAGRAM

NOT TO SCALE

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D. EVANS VIA EMAIL
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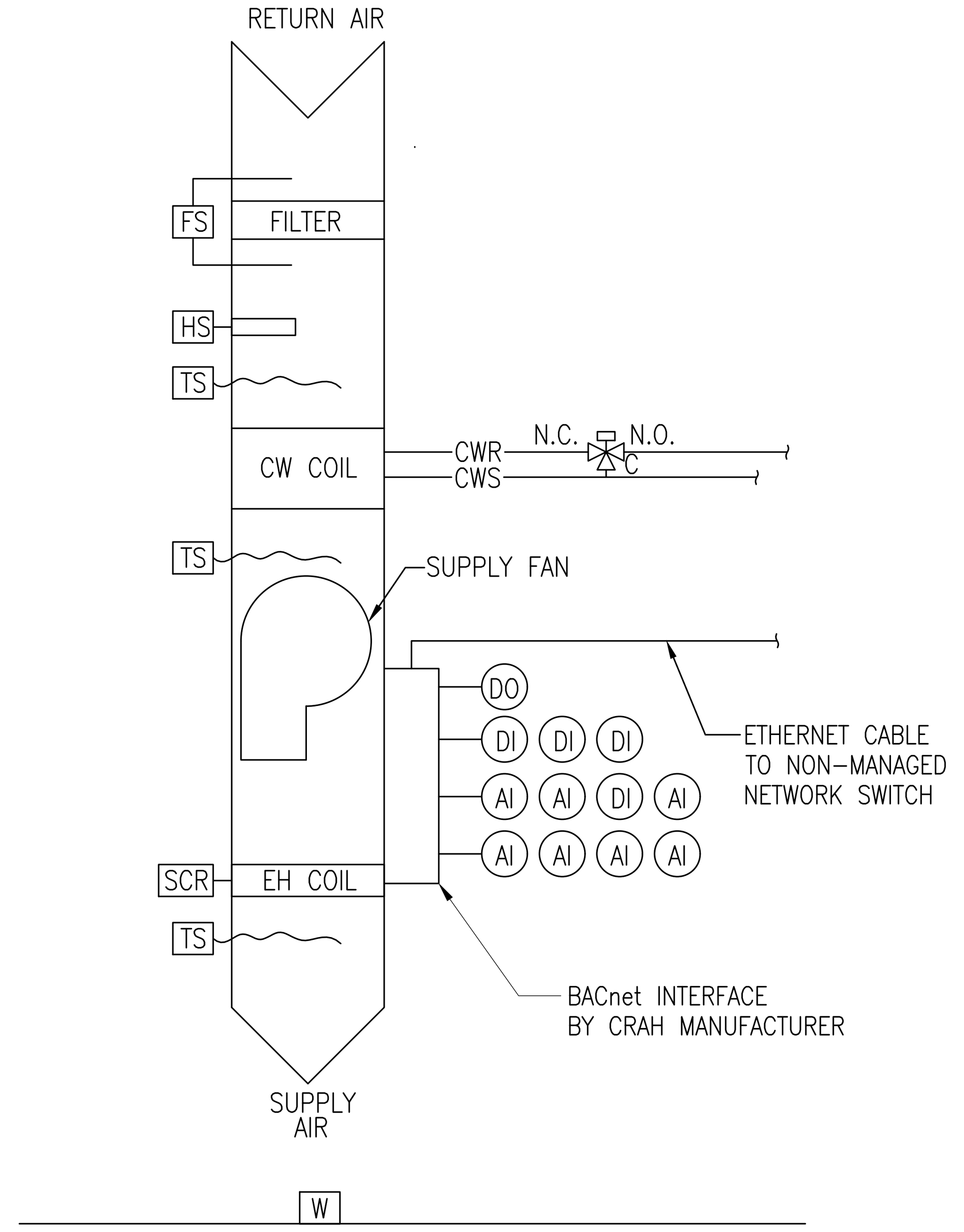
JOINT MARITIME FACILITY
MECHANICAL CONTROLS

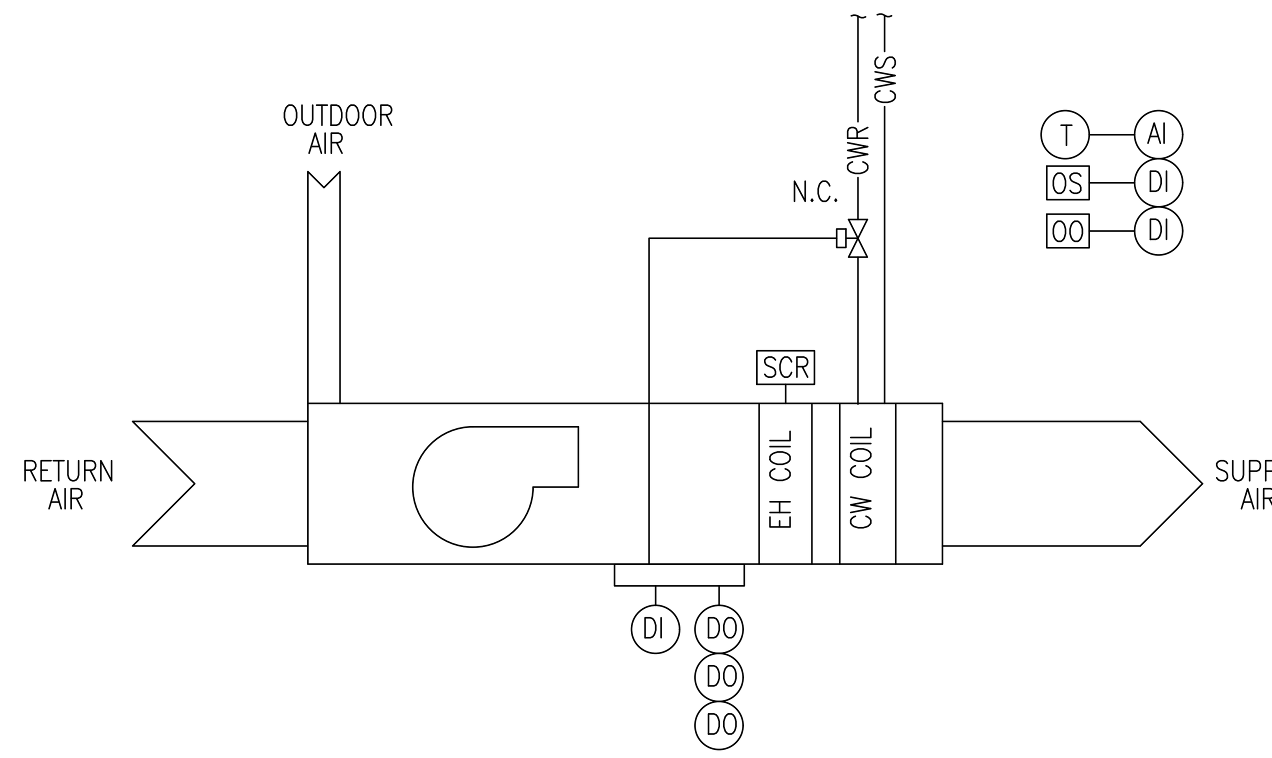
SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157194
SHEET 132 OF 170
M-703



LOCAL DDC CONTROLLER			
INPUTS		OUTPUTS	
ANALOG	RETURN AIR TEMPERATURE	ANALOG	SA TEMPERATURE
	CW COIL LEAVING TEMPERATURE		
	SUPPLY AIR TEMPERATURE		
	RETURN AIR RELATIVE HUMIDITY		
	TEMPERATURE SETPOINT		
	RELATIVE HUMIDITY SETPOINT		
	FAN SPEED		
DIGITAL	FAN STATUS	DIGITAL	CRAH ENABLE/DISABLE
	ELECTRIC HEATING COIL STATUS		
	LOW TEMPERATURE ALARM		
	HIGH TEMPERATURE ALARM		
	FAN FAULT ALARM		
	LOSS OF POWER ALARM		
	LOSS OF AIRFLOW ALARM		
	FILTER STATUS		
	CONDENSATE PUMP STATUS		
	DEHUMIDIFICATION STATUS		
	EMERGENCY POWER OFF ALARM		
	RAISED FLOOR WATER DETECTION		

CRAH-1 CONTROL DIAGRAM

NOT TO SCALE



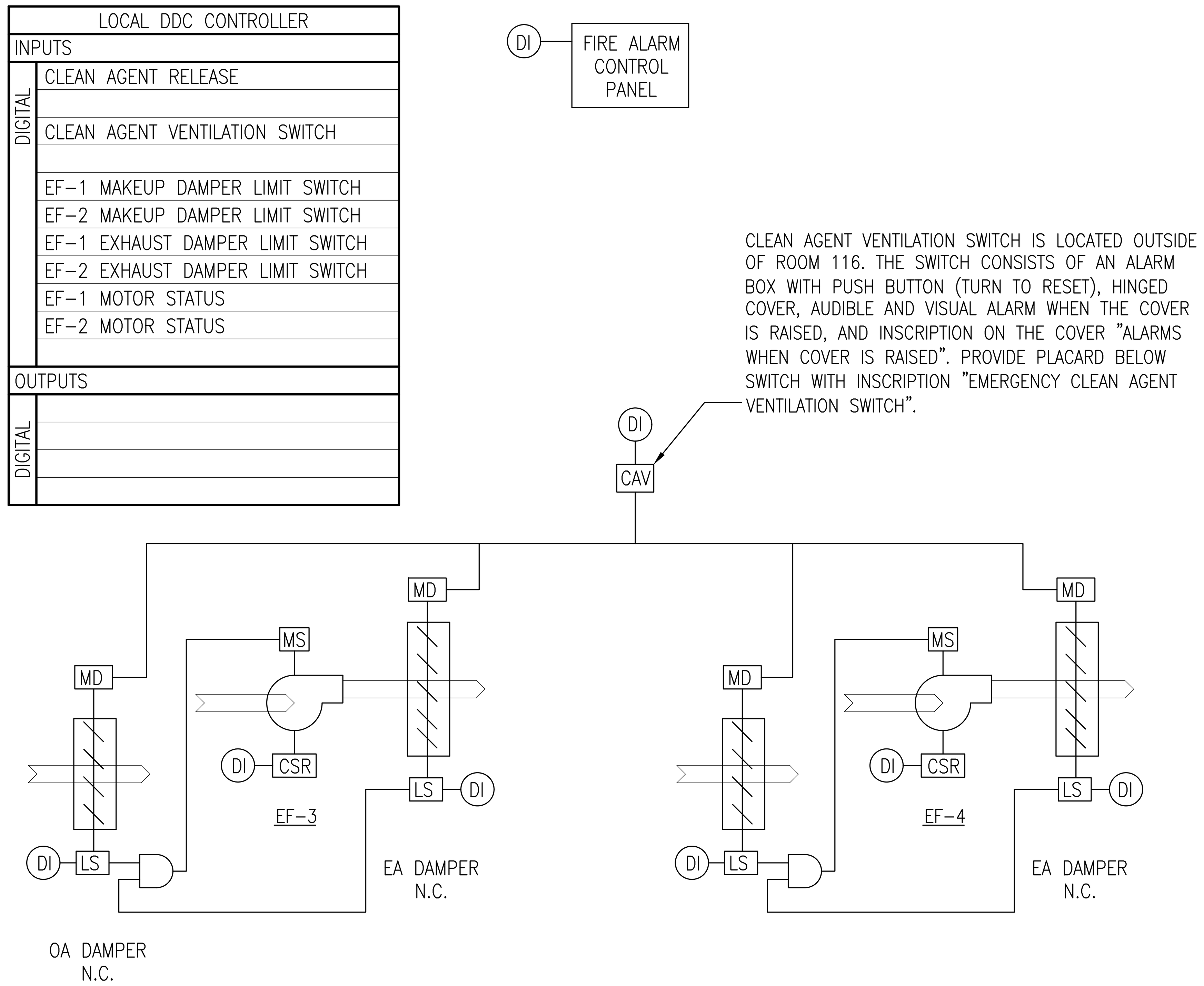
FCU	
INPUTS	
ANALOG	SPACE TEMPERATURE
DIGITAL	OCCUPANCY SENSOR
	OCCUPANCY OVERRIDE
	FAN STATUS
OUTPUTS	
ANALOG	
DIGITAL	ENABLE/DISABLE
	COOLING CALL
	HEATING CALL

FAN COIL CONTROL DIAGRAM , TYPICAL

NOT TO SCALE

LOCAL DDC CONTROLLER	
INPUTS	
DIGITAL	CLEAN AGENT RELEASE
	CLEAN AGENT VENTILATION SWITCH
	EF-1 MAKEUP DAMPER LIMIT SWITCH
	EF-2 MAKEUP DAMPER LIMIT SWITCH
	EF-1 EXHAUST DAMPER LIMIT SWITCH
	EF-2 EXHAUST DAMPER LIMIT SWITCH
	EF-1 MOTOR STATUS
	EF-2 MOTOR STATUS
OUTPUTS	
DIGITAL	

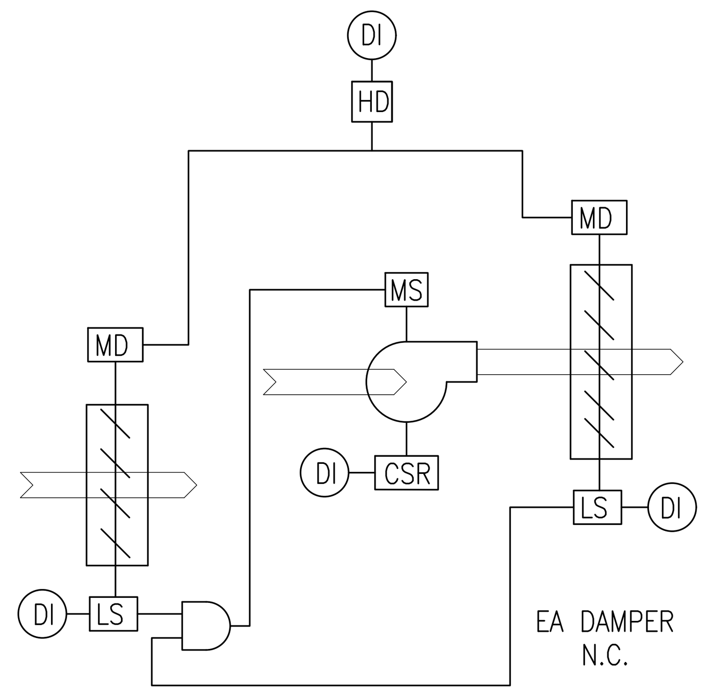
CLEAN AGENT VENTILATION SWITCH IS LOCATED OUTSIDE OF ROOM 116. THE SWITCH CONSISTS OF AN ALARM BOX WITH PUSH BUTTON (TURN TO RESET), HINGED COVER, AUDIBLE AND VISUAL ALARM WHEN THE COVER IS RAISED, AND INSCRIPTION ON THE COVER "ALARMS WHEN COVER IS RAISED". PROVIDE PLACARD BELOW SWITCH WITH INSCRIPTION "EMERGENCY CLEAN AGENT VENTILATION SWITCH".



EF-3 AND EF-4 OPS ROOM CONTROL DIAGRAM

NOT TO SCALE

LOCAL DDC CONTROLLER	
INPUTS	
DIGITAL	HYDROGEN DETECTION
	EF-1 MAKEUP DAMPER LIMIT SWITCH
	EF-1 EXHAUST DAMPER LIMIT SWITCH
	EF-1 MOTOR STATUS
OUTPUTS	
DIGITAL	



EF-1 AND EF-2 UPS ROOM CONTROL DIAGRAM

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JOINT MARITIME FACILITY
MECHANICAL CONTROLS

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157195
SHEET 133 OF 170
M-704

SEQUENCES OF OPERATION

DOAS-1 OUTSIDE AIR

SAFETIES

FREEZE SAFETY: DISABLE THE UNIT FAN(S), CLOSE OUTSIDE AIR DAMPERS WHENEVER A FREEZE PROTECTION DEVICE (FREEZE STAT) IS ACTIVATED. THIS IS A HARDWIRED SEQUENCE WITH THE DDC HARDWARE MONITORING ACTIVATION.

SMOKE DETECTOR: SHUT DOWN THE UNIT WHENEVER ANY UNIT SMOKE DETECTOR OR ASSOCIATED FIRE ALARM RELAY ACTIVATES. THIS IS A HARDWIRED SEQUENCE WITH THE DDC HARDWARE MONITORING ACTIVATION.

CLEAN AGENT: SHUT DOWN THE UNIT WHEN THE CLEAN AGENT SIGNAL IS ACTIVE. THIS IS A HARDWIRED SEQUENCE WITH THE DDC HARDWARE MONITORING ACTIVATION.

SAFETY RESET: WHEN A HARDWARE SAFETY ACTIVATES, THE UNIT SHALL REMAIN SHUTDOWN UNTIL BOTH THE SAFETY RESET INPUT (AS SHOWN ON THE POINTS SCHEDULE) IS ACTIVE, AND NO LATCHING HARDWARE SAFETIES ARE ACTIVE. USE OF THE SAFETY RESET WILL ALSO RE-ENABLE FAN OR ENTHAPY CORE BYPASS COMMANDS THAT HAVE BEEN REMOVED DUE TO FAILURE OF PROOF.

FILTER MONITORING: ALARM DDC WHEN EITHER FILTER SWITCH IS ACTIVATED FOR MORE THAN 5 CONTINUOUS MINUTES (ADJ.)

SUPPLY FAN

ENABLE/DISABLE: ENABLE SUPPLY FAN TO RUN CONTINUOUSLY DURING OCCUPIED MODE. DISABLE SUPPLY FAN DURING UNOCCUPIED MODE.

PROOF: IF THE FAN IS NOT PROVEN ON WITHIN 30 SECONDS (ADJ.) AFTER BEING COMMANDED TO START, REMOVE THE FAN RUN COMMAND AND GENERATE AN ALARM. AFTER A FAILURE OF PROOF, THE FAN CAN BE RE-ENABLED VIA THE SAFETY RESET COMMAND.

SPEED CONTROL: ADJUST THE FAN SPEED TO MAINTAIN THE VOLUMETRIC FLOW RATE AS SCHEDULED

EXHAUST FAN

ENABLE/DISABLE: ENABLE THE EXHAUST FAN TO RUN DURING OCCUPIED MODE. DISABLE THE EXHAUST FAN DURING UNOCCUPIED MODE OR SUPPLY FAN FAILURE.

PROOF: IF THE FAN IS NOT PROVEN ON WITHIN 30 SECONDS (ADJ.) AFTER BEING COMMANDED TO START, REMOVE THE FAN RUN COMMAND AND GENERATE AN ALARM. AFTER A FAILURE OF PROOF, THE FAN CAN BE RE-ENABLED VIA THE SAFETY RESET COMMAND.

SPEED CONTROL: MODULATE THE EXHAUST FAN SPEED TO MAINTAIN THE BUILDING STATIC PRESSURE SET POINT WHEN THE SUPPLY FAN IS PROVEN ON.

BUILDING STATIC PRESSURE SET POINT: THE BUILDING STATIC PRESSURE SET POINT IS A FIXED VALUE: 5 Pa

PRE-HEATING CONTROL

PREHEAT TEMPERATURE SET POINT: 10°C (ADJ.).

PRE-HEATING COIL: MODULATE THE PRE-HEAT COIL TO MAINTAIN THE PRE-HEAT COIL LEAVING TEMPERATURE AT PREHEAT TEMPERATURE SET POINT.

COOLING CONTROL

COOLING TEMPERATURE SET POINT: 13°C (ADJ.)

CHILLED WATER COIL VALVE: MODULATE THE CHILLED WATER COIL VALVE TO MAINTAIN CHILLED WATER COIL LEAVING AIR TEMPERATURE AT COOLING TEMPERATURE SET POINT, WHEN THE SUPPLY FAN IS PROVEN ON.

RE-HEATING CONTROL

REHEAT TEMPERATURE SET POINT: 23°C (ADJ.).

RE-HEATING COIL: MODULATE THE RE-HEAT COIL TO MAINTAIN COIL LEAVING AIR TEMPERATURE AT REHEAT TEMPERATURE SET POINT, WHEN THE SUPPLY FAN IS PROVEN ON AND THE OUTSIDE AIR TEMPERATURE IS BELOW THE REHEAT TEMPERATURE SET POINT MINUS 3°C (ADJ.).

ENTHALPY CORE BYPASS CONTROL

ENERGY RECOVERY ENABLE LIMIT: 3°C (ADJ.).

FROST PROTECTION LIMIT: OUTSIDE AIR TEMPERATURE DEWPOINT UPSTREAM CORE PLUS 3°C (ADJ.).

CORE DEHUMIDIFICATION MODE: WHEN THE OUTSIDE AIR DEW POINT IS HIGHER THAN 14°C (ADJ.) AND THE RETURN AIR DEW POINT IS LOWER THAN THE OUTSIDE AIR DEW POINT MINUS 1°C (ADJ.)

CORE ENERGY RECOVERY MODE: WHEN THE CORE IS NOT IN DEHUMIDIFICATION MODE, THE ABSOLUTE VALUE OF RETURN AIR TEMPERATURE MINUS OUTSIDE AIR TEMPERATURE IS GREATER THAN THE ENERGY TRANSFER ENABLE LIMIT, AND WHEN THE CORE IS -EITHER- ADDING ENTHALPY TO THE OUTSIDE AIR WHEN THE OUTSIDE AIR TEMPERATURE IS LOWER THAN THE COOLING TEMPERATURE SET POINT MINUS 1°C (ADJ.) -OR- REMOVING ENTHALPY FROM THE OUTSIDE AIR WHEN THE OUTSIDE AIR TEMPERATURE IS GREATER THAN THE COOLING TEMPERATURE SET POINT PLUS 1°C (ADJ.).

CORE BYPASS MODE: WHEN THE CORE IS NOT IN DEHUMIDIFICATION MODE AND IS NOT IN ENERGY TRANSFER MODE.

CORE FROST PROTECTION MODE: WHEN THE EXHAUST AIR TEMPERATURE DOWNSTREAM THE CORE IS BELOW THE FROST PROTECTION LIMIT PLUS 1°C (ADJ.),

WHEN THE CORE IS EITHER IN DEHUMIDIFICATION MODE OR ENERGY TRANSFER MODE BOTH AIRSTREAMS ARE FLOWING THROUGH THE CORE AND NO AIR IS BYPASSED. WHEN THE CORE IS IN BYPASS MODE BOTH AIRSTREAMS ARE FLOWING AROUND THE CORE AND NO AIR IS FLOWING THROUGH THE CORE. WHEN THE CORE IS IN FROST PROTECTION MODE THE RETURN AIR IS FLOWING THROUGH THE CORE AND THE OUTSIDE AIR DAMPER MODULATES AIR THROUGH AND AROUND THE CORE TO MAINTAIN EXHAUST AIR TEMPERATURE AT THE FROST PROTECTION LIMIT.

CHILLED WATER SYSTEM - ACCH-1,2,3 AND CHWP-1,2

CHILLED WATER SYSTEM ENABLE: ENABLE WHEN -EITHER- OUTSIDE AIR TEMPERATURE RISES ABOVE THE HIGH LIMIT 80°F (ADJ.) WITH A 10°F DEAD BAND (ADJ.) -OR- OUTSIDE AIR TEMPERATURE RISES ABOVE 55°F WITH A 5°F DEAD BAND (ADJ.), AND COOL REQUESTS FROM LOADS SERVED EXCEEDS 20% (ADJ.) WITH A 10% DEAD BAND (ADJ.) OF ALL CONNECTED COOLING LOADS. OTHERWISE, THE SYSTEM IS DISABLED. ONCE ENABLED, DO NOT REMOVE THE SYSTEM ENABLE FOR A MINIMUM OF 1 HOUR (ADJ.)

SYSTEM MINIMUM FLOW: SUM OF FACTORY MINIMUM FLOWS OF CHILLERS ENABLED.

SYSTEM MAXIMUM FLOW: SUM OF FACTORY MAXIMUM FLOWS OF CHILLERS ENABLED.

PUMPS

PUMP ROTATION: SELECT THE PUMP WITH THE LEAST RUNTIME AS LEAD WHEN THE SYSTEM IS STARTED. IF THE LEAD PUMP RUNS CONTINUOUSLY FOR MORE THAN 180 HOURS (ADJ.) THE NEXT AVAILABLE PUMP IS SELECTED AS LEAD.

PUMP START/STOP: COMMAND THE LEAD PRIMARY PUMP TO START WHENEVER THE SYSTEM IS ENABLED. COMMAND THE PUMP TO STOP AFTER A 5 MINUTES (ADJ.) DELAY WHENEVER THE SYSTEM IS DISABLED.

PUMP PROOF: IF THE LEAD PRIMARY PUMP IS NOT PROVEN ON WITHIN 30 SECONDS (ADJ.) AFTER BEING COMMANDED TO START, REMOVE THE PUMP RUN COMMAND AND GENERATE AN ALARM. WHEN THE LEAD PUMP FAILS TO PROVE, REPORT THE LEAD PUMP FAILED, AND START THE NEXT AVAILABLE PUMP AS LEAD.

PUMP SPEED CONTROL: MODULATE THE PUMP SPEED TO MAINTAIN THE SYSTEM DIFFERENTIAL PRESSURE AT SET POINT 10 PSI (ADJ.), SUBJECT TO ALSO MAINTAINING FLOW TO THE CHILLER BETWEEN THE CHILLER MANUFACTURER MINIMUM FLOW AND MANUFACTURER MAXIMUM FLOW.

MINIMUM FLOW BYPASS: MODULATE MINIMUM FLOW BYPASS WHEN LOADS ARE BELOW SYSTEM MINIMUM FLOW. OTHERWISE NO FLOW THROUGH BYPASS.

CHILLERS

CHILLER PLANT LOGIC BY CHILLER MANUFACTURER. LOGIC CAN RESIDE IN CHILLER PACKAGED CONTROLLERS OR SEPARATE CHILLER MANUFACTURER PLANT CONTROLLER. CHILLER PLANT CONTROLLER CAN BE USED TO SEQUENCE PUMPS USING CHILLER MANUFACTURER SEQUENCES.

CHILLER ROTATION: CHILLERS ARE CONFIGURED LEAD/LAG/STANDBY WHERE ONLY TWO MAX OF THREE CHILLERS ARE TO RUN AT AN INSTANCE. IF ANY CHILLER RUNS CONTINUOUSLY FOR MORE THAN 180 HOURS (ADJ.) THEN PLANT LOGIC ROTATES THE LEAD/LAG/STANDBY CONFIGURATION OF THE CHILLERS.

CHILLER STAGING: CHILLER PLANT LOGIC CONTROLS LEAD/LAG SEQUENCING OF CHILLERS TO SATISFY LOAD. CHILLED WATER FLOWS FOR EACH CHILLER ARE DEMAND CONTROLLED TO CHILLER MANUFACTURER SPECIFICATIONS OF MIN/MAX FLOW RATES AND MAX RATE OF CHANGE OF FLOW RATE.

CHILLERS ENABLE: CHILLER PLANT LOGIC ENABLES THE LEAD/LAG CHILLERS TO RUN WHEN THE SYSTEM IS ENABLED FOR COOLING, THE CHILLER HAS BEEN CALLED, AND ITS RESPECTIVE FLOW RATE HAS BEEN PROVED.

CHILLER DISABLE: REMOVE THE CHILLER ENABLE WHEN EITHER THE CHILLED WATER SYSTEM IS DISABLED, CHILLER CALL IS REMOVED, OR ITS RESPECTIVE FLOW PROOF IS LOST.

CHILLED WATER SUPPLY TEMPERATURE SET POINT: 9°F (ADJ.)

CHILLED WATER RESET SCHEDULE BASED ON OUTSIDE TEMPERATURE.

TEMPERATURE CONTROL: THE RESPECTIVE FACTORY CONTROLLER WILL SEQUENCE ITS CHILLER TO MAINTAIN THE CHILLED WATER SUPPLY TEMPERATURE SET POINT.

ALARM: SEND AN ALARM WHEN A FACTORY CONTROLLER ALARM IS ACTIVE.

FLOW RATE IS MEASURED USING FLOW METERS WITH DIFFERENTIAL PRESSURE SENSORS AS BACKUP.

FCU - ALL

ZONE OVERRIDE SWITCH: CHANGE TO OCCUPIED MODE FOR 4 HOURS (ADJ.) WHENEVER THE LOCAL ZONE OVERRIDE SWITCH IS ACTIVATED. RESUME THE ORIGINAL SCHEDULED MODE WHENEVER THE LOCAL ZONE OVERRIDE SWITCH IS RESET OR THE OVERRIDE PERIOD HAS EXPIRED.

OCCUPANCY SENSOR: CHANGE TO OCCUPIED MODE FOR 1 HOUR (ADJ.) WHENEVER THE SENSOR DETECTS OCCUPANCY. RESUME THE ORIGINAL SCHEDULED MODE WHEN THE OVERRIDE PERIOD HAS EXPIRED.

UNOCCUPIED HEATING SET POINT: 14°C (ADJ.) FOR HEATING.

OCCUPIED HEATING SET POINT: 20°C (ADJ.) FOR HEATING.

OCCUPIED COOLING SET POINT: 26°C (ADJ.) FOR COOLING.

UNOCCUPIED COOLING SET POINT: 32°C (ADJ.) FOR COOLING.

COOLING MODE: WHENEVER THE ZONE TEMPERATURE IS ABOVE THE EFFECTIVE COOLING SET POINT WITH A 1°C (ADJ.) DEADBAND.

HEATING MODE: WHENEVER THE ZONE TEMPERATURE IS BELOW THE EFFECTIVE HEATING SET POINT WITH A 1°C (ADJ.) DEADBAND.

CHILLED WATER COIL: MODULATE THE CHILLED WATER COIL VALVE DURING COOLING MODE TO MAINTAIN EFFECTIVE SET POINT WHEN THE FAN IS PROVEN ON AND THE HEATING COIL IS NOT ENERGIZED, OTHERWISE COMMAND THE CHILLED WATER COIL VALVE CLOSED.

ELECTRIC HEATING COIL: MODULATE THE HEATING COIL DURING HEATING MODE TO MAINTAIN EFFECTIVE SETPONT WHEN THE FAN IS PROVEN ON, AND THE CHILLED WATER VALVE IS CLOSED. OTHERWISE COMMAND THE HEATING COIL OFF.

SMOKE DAMPERS: SHUT DOWN THE UNIT AND CLOSE ALL UNIT SMOKE DAMPERS WHENEVER ANY UNIT SMOKE DETECTOR ACTIVATES. THIS IS A HARDWIRED SEQUENCE WITH THE DDC HARDWARE MONITORING ACTIVATION.

FAN START/STOP: FAN RUNS CONTINUOUSLY WHEN ENABLED

FAN PROOF: IF THE FAN IS NOT PROVEN ON WITHIN 30 SECONDS (ADJ.) AFTER BEING COMMANDED TO START, SEND ALARM.

FCU UNITS 4&5 AND FCU UNITS 6&7 ARE REDUNDANT PAIRS IN A LEAD/STANDBY CONFIGURATION. SELECT THE FCU WITH THE LEAST RUNTIME AS LEAD. IF THE LEAD FCU RUNS FROM MORE THAN 180 HOURS THE NEXT FCU IS SELECTED AS LEAD.

DDIU-1/DDOU-1

ALWAYS ON. KEEP COOLING AND HEATING SET POINT 5.7°C APART. AUTOMATIC CHANGEOVER

COOLING SET POINT: 26°C (ADJ.)

HEATING SET POINT: 20°C (ADJ.)

COOLING MODE: ACTIVATE WHENEVER THE ZONE TEMPERATURE IS ABOVE THE COOLING SET POINT.

HEATING MODE: ACTIVATE WHENEVER THE ZONE TEMPERATURE IS BELOW THE HEATING SET POINT.

FAN CONTROL: RUN WHEN COMPRESSOR IS RUNNING AND 3 MINUTES (ADJ.) AFTER COMPRESSOR STOPS.

EUH-1

ALWAYS ON.

HEATING SET POINT: 10°C (ADJ.)

HEATING MODE: ACTIVATE WHENEVER THE ZONE TEMPERATURE IS BELOW THE HEATING SET POINT.

FAN CONTROL: RUN CONTINUOUSLY IN HEATING MODE

EF-3, 4 (CLEAN AGENT ALARM/VENTILATION)

WHEN A SIGNAL IS RECEIVED FROM THE FIRE ALARM CONTROL PANEL THAT THE CLEAN AGENT SERVING ROOM 116 HAS RELEASED, AN ALARM IS SENT TO THE DDC AND THE DOAS . WHEN THE CLEAN AGENT RELEASE ALARM IS CLEARED AT THE DDC AND THE CLEAN AGENT VENTILATION BUTTON IS NOT PRESSED, THE DAMPER SHALL OPEN.

WHEN THE CLEAN AGENT VENTILATION BUTTON IS PRESSED, IT ILLUMINATES, DAMPERS OPEN AND WHEN DAMPER LIMIT SWITCHES ARE ACTIVATED THE FANS ENERGIZE. THIS IS THROUGH HARDWIRE. WHEN THE EMERGENCY FIRE PROTECTION VENTILATION BUTTON IS RESET, BY TURNING, THE REVERSE SHALL OCCUR.

EF-1, 2 (HYDROGEN VENTILATION)

WHEN A SIGNAL IS RECEIVED FROM THE HYDROGEN SENSOR, HYDROGEN HAS BEEN DETECTED ABOVE THRESHOLD AND THE FAN AND DAMPERS WILL ENERGIZE. THE FAN WILL RUN AND DAMPER REMAIN OPEN UNTIL MANUALLY RESET.

RD-1,2,3,4 (RADON FANS)

RADON FANS RUN CONTINOUS

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VARIOUS LOCATIONS
VARIOUS LOCATIONS
JOINT MARITIME FACILITY
MECHANICAL CONTROLS
SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157196
SHEET 134 OF 170
M-705
NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

ELECTRICAL LEGEND - INTERIOR

LIGHTING -

- LED TYPE LUMINAIRE AS INDICATED. SEE LUMINAIRE SCHEDULE ON SHEET EL601
- LED TYPE EXIT SIGN - LUMINAIRE TYPE G
- LED TYPE SELF-CONTAINED EMERGENCY LIGHTING UNIT - LUMINAIRE TYPE E. SINGLE HEAD REMOTE TYPE AS INDICATED.
- A — LUMINAIRE TYPE SYMBOL - SEE LUMIARE SCHEDULE ON SHEET EL601
- S_{3D} — SINGLE POLE WALL SWITCH - 20 AMP, 120V, SUBSCRIPT "3" INDICATES THREE WAY SWITCH, SUBSCRIPT "D" INDICATES DIMMING
- O — OCCUPANCY SENSOR FOR LIGHTING CONTROL
- P — EXTERIOR-MOUNTED PHOTOCELL FOR LIGHTING CIRCUIT CONTROL
- TS — ASTRONOMICAL TIMESWITCH FOR LIGHTING CIRCUIT CONTROL

POWER AND EQUIPMENT -

- J — JUNCTION BOX AS INDICATED
- E — EQUIPMENT CONNECTION AS INDICATED
- 5 — MOTOR CONNECTION, kW AS INDICATED
- DISCONNECT SWITCH - XX AMP, 208V, 3 POLE UON IN IP20 ENCLOSURE
- VFD — VARIABLE FREQUENCY DRIVE CONNECTION
- S_M — MANUAL MOTOR STARTER SWITCH
- GFI / WC / WP — DUPLEX RECEPTACLE - 20 AMP, 120V, P + N + GND, MOUNT AT 460 TO CENTER UON, SUBSCRIPT GFI INDICATES GROUND FAULT INTERRUPTING TYPE; SUBSCRIPT 'WC' INDICATES FOR CONNECTION TO WATER COOLER; SUBSCRIPT 'WP' INDICATES WEATHERPROOF WITH HINGED, IN-USE COVER (GROUND FAULT INTERRUPTING TYPE)
- TWIST LOCK L5-20
- 3PH POWER TRANSFORMER AS NOTED
- ELECTRICAL PANELBOARD - 480Y/277V, 3PH, 4W, SURFACE-MOUNTED UON
- ELECTRICAL PANELBOARD - 208Y/120V, 3PH, 4W, SURFACE-MOUNTED UON
- (OR) — BRANCH CIRCUIT OR FEEDER WIRING IN CONDUIT. NO TICK MARKS INDICATE 2 x #12 + 1 x #12 GND IN 16 mm CONDUIT. TICK MARKS, WHEN SHOWN, INDICATE NUMBER OF #12 AWG CONDUCTORS IF OTHER THAN THREE. INDICATES GROUND. CONDUIT LARGER THAN 16mm AND WIRING LARGER THAN #12 AWG AS INDICATED.
- INDICATES A CONDUIT RUN CONCEALED IN CEILING, WALL FLOOR, ABOVE SUSPENDED CEILING, OR EXPOSED IN WAREHOUSE AREA
- CONDUIT TURNED UP
- CONDUIT TURNED DOWN
- PA-1 — HOMERUN TO PANELBOARD - SUBSCRIPT INDICATES PANELBOARD CIRCUIT NO.

TELECOMMUNICATIONS -

- $\blacktriangle_A$ — UNCLASSIFIED TELECOMMUNICATIONS WORKSTATION OUTLET - CATEGORY 6, RJ45 TYPE
- $\blacktriangle_B$ — CLASSIFIED TELECOMMUNICATIONS WORKSTATION OUTLET - DUPLEX SC, MM FOC TYPE
- WALL-MOUNTED HANDSET OUTLET AT 1350 AFF
- OVERHEAD OR UNDERFLOOR TELECOMMUNICATIONS CABLE TRAY
- TELECOMMUNICATIONS EQUIPMENT CABINET

ELECTRONIC SECURITY -

- BMS — INTRUSION DETECTION SYSTEM 'HIGH SECURITY' BALANCED MAGNETIC SWITCH
- INTRUSION DETECTION SYSTEM PASSIVE INFRARED OCCUPANCY SENSOR FOR MOTION DETECTION
- PCU — ESS PROCESSING CONTROL UNIT
- CK — ACS CARD READER WITH KEYPAD
- CR — ACS CARD READER
- VSS CAMERA
- RAD — REMOTE ARM/DISARM KEYPAD
- DS — ELECTRICALLY-OPERATED DOOR STRIKE
- EL — ELECTRICALLY-OPERATED DOOR LOCK
- A — LOCAL DOOR OPEN ALARM ANNUNCIATOR
- PIR FOR DOOR EXIT FUNCTION

ELECTRICAL LEGEND - EXTERIOR

- MANHOLE OR HANDHOLE AS INDICATED
- EXISTING MANHOLE OR HANDHOLE AS INDICATED
- EXISTING PAD-MOUNTED TRANSFORMER
- PAD-MOUNTED TRANSFORMER
- DUCTBANK SECTION LOOKING IN DIRECTION OF ARROWS
- SPARE DUCT
- HEAVY LINE INDICATES BOTTOM OF DUCT
- 2 — CABLE DESIGNATION - SEE EXTERIOR CONDUIT AND CABLING SCHEDULE ON SHEET E-601
- —UP— UNDERGROUND PRIMARY CIRCUIT IN CONCRETE-ENCASED DUCTBANK
- —US— UNDERGROUND SECONDARY CIRCUIT IN CONCRETE-ENCASED DUCTBANK
- —UC— UNDERGROUND COMMUNICATIONS CABLING IN CONCRETE-ENCASED DUCTBANK
- – – USL– – EXISTING UNDERGROUND SITE LIGHTING CIRCUIT
- —USL— REMOVE EXISTING UNDERGROUND SITE LIGHTING CIRCUIT
- EXISTING OVERHEAD WOODEN UTILITY POLE
- – – 12.47 – – EXISTING OVERHEAD 12.47kV, 3-WIRE PRIMARY
- —OC— OVERHEAD TELECOMMUNICATIONS AS INDICATED
- – – ST – – EXISTING OVERHEAD SECONDARY TRIPLEX
- —ST— REMOVE EXISTING OVERHEAD SECONDARY TRIPLEX
- —OC— EXISTING OVERHEAD TELECOMMUNICATIONS
- —OC— REMOVE EXISTING OVERHEAD TELECOMMUNICATIONS
- EXISTING SITE LIGHTING POLE WITH LUMINAIRE
- REMOVE EXISTING SITE LIGHTING POLE WITH LUMINAIRE
- SITE LIGHTING POLE WITH LUMINAIRE
- EXISTING POLE GUY
- REMOVE EXISTING POLE GUY
- POLE GUY AS INDICATED
- —VSS— UNDERGROUND VIDEO SURVEILLANCE SYSTEM DUCTBANK

ELECTRICAL LEGEND - GROUNDING

- LIGHTNING PROTECTION SYSTEM AIR TERMINAL
- —LRC— LIGHTNING PROTECTION SYSTEM ROOF CONDUCTOR
- —LDC— LIGHTNING PROTECTION SYSTEM DOWN CONDUCTOR
- G — GROUND ROD
- G_T — GROUND ROD WITH TEST WELL
- G — INTERIOR SIGNAL REFERENCE GROUND LUGS FLUSH WITH CONCRETE FLOOR
- PBB — TELECOMMUNICATIONS PRIMARY BONDING BUSBAR
- SBB — TELECOMMUNICATIONS SECONDARY BONDING BUSBAR
- MEGB — MAIN ELECTRICAL GROUND BUS
- SEGB — SECONDARY ELECTRICAL GROUND BUS
- —GC— UNDERGROUND ELECTRICAL GROUNDING COUNTERPOISE
- —EGC— EQUIPMENT GROUNDING CONDUCTOR

ABBREVIATIONS

A	AMPERES
ACS	ACCESS CONTROL SYSTEM
ACT	ACOUSTICAL CEILING TILE
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AWG	AMERICAN WIRE GAUGE
C	CONDUIT
CAC	COMMON ACCESS CARD
CAT	CATEGORY
CCT	CORROLATED COLOR TEMPERATURE
COND	CONDUIT
CL	CENTERLINE
CU	COPPER
DWG	DRAWING
EC	EMPTY CONDUIT
ESS	ELECTRONIC SECURITY SYSTEM
FLA	FULL LOAD AMPS
FOC	FIBER OPTIC CABLING
GC	GROUNDING COUNTERPOISE
GFI	GROUND FAULT INTERRUPTER
GND	GROUND
GRS	GALVANIZED RIGID STEEL
HMI	HUMAN MACHINE INTERFACE
HT	HEIGHT
Hz	HERTZ
IDS	INTRUSION DETECTION SYSTEM
IMC	INTERMEDIATE METAL CONDUIT
J	JUNCTION BOX
kVA	KILO VOLT AMPERE
kW	KILO WATT
LED	LIGHT EMITTING DIODE
LPS	LIGHTNING PROTECTION SYSTEM
LRA	LOCKED ROTOR AMPS
LSI	LONG TIME, SHORT TIME, INSTANTANEOUS
LSIG	LONG TIME, SHORT TIME, INSTANTANEOUS, GROUND FAULT
M	METER
MCB	MAIN CIRCUIT BREAKER
mm	MILLIMETER
MM	MULTIMODE
MIN	MINIMUM
MLO	MAIN LUGS ONLY
MTG	MOUNTING
NTS	NOT TO SCALE
OSP	OUTSIDE PLANT
PBB	PRIMARY BONDING BUSBAR
PCU	PROCESSING CONTROL UNIT FOR ESS
PH	PHASE
PIR	PASSIVE INFRARED
PNL	PANEL
PVC	POLYVINYL CHLORIDE
RMC	RIGID METAL CONDUIT
SBB	SECONDARY BONDING BUSBAR
SE	SERVICE ENTRANCE
SPD	SURGE PROTECTION DEVICE
SQ	SQUARE
TEL	TELEPHONE
TYP	TYPICAL
V	VOLTS
VAC	VOLTS ALTERNATING CURRENT
VFD	VARIABLE FREQUENCY DRIVE
VRLA	VALVE-REGULATED LEAD-ACID
VSS	VIDEO SURVEILLANCE SYSTEM
UON	UNLESS OTHERWISE NOTED
UPS	UNINTERRUPTIBLE POWER SUPPLY
W	WIRE
WHD	WATT HOUR METER
WP	WEATHERPROOF EQUIPMENT

ROOM SCHEDULE

ROOM NO. - NAME	ROOM NO. - NAME	ROOM NO. - NAME
100 - SECURE VESTIBULE	107 - BUNK	114 - GENERATOR FUEL ROOM
101 - OPEN OFFICE	108 - BUNK	115 - GENERATOR ROOM
102 - UNSECURE CORRIDOR	109 - STAGING AREA	116 - OPS EQUIPMENT
103 - SHOWER	110 - COMMERCIAL TELECOM	117 - UPS A
104 - TOILET	111 - CORRIDOR	118 - ELECTRICAL
105 - BREAK	112 - MECHANICAL	119 - UPS B
106 - JANITOR	113 - FIRE PUMP	

SINGLE CONDUCTOR QUANTITY

CONDUIT SIZE IN mm

(2) 3 x 6 + 1 x 10 GND - 63

NUMBER OF PARALLEL CIRCUITS IF APPLICABLE

CONDUCTOR SIZE IN AWG

CONDUIT AND WIRE SIZE DESCRIPTION

SYM | DESCRIPTION | DATE | APPR

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DES CAB | DRW CAB | CHK SEK

PM/DM MJD/ATP

BRANCH MANAGER SEK

DES PROD DIR WBF

FIRE PROTECTION ENGINEER DSN

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PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS

VARIOUS LOCATIONS

JOINT MARITIME FACILITY

ELECTRICAL LEGEND AND ABBREVIATIONS

SCALE: NONE

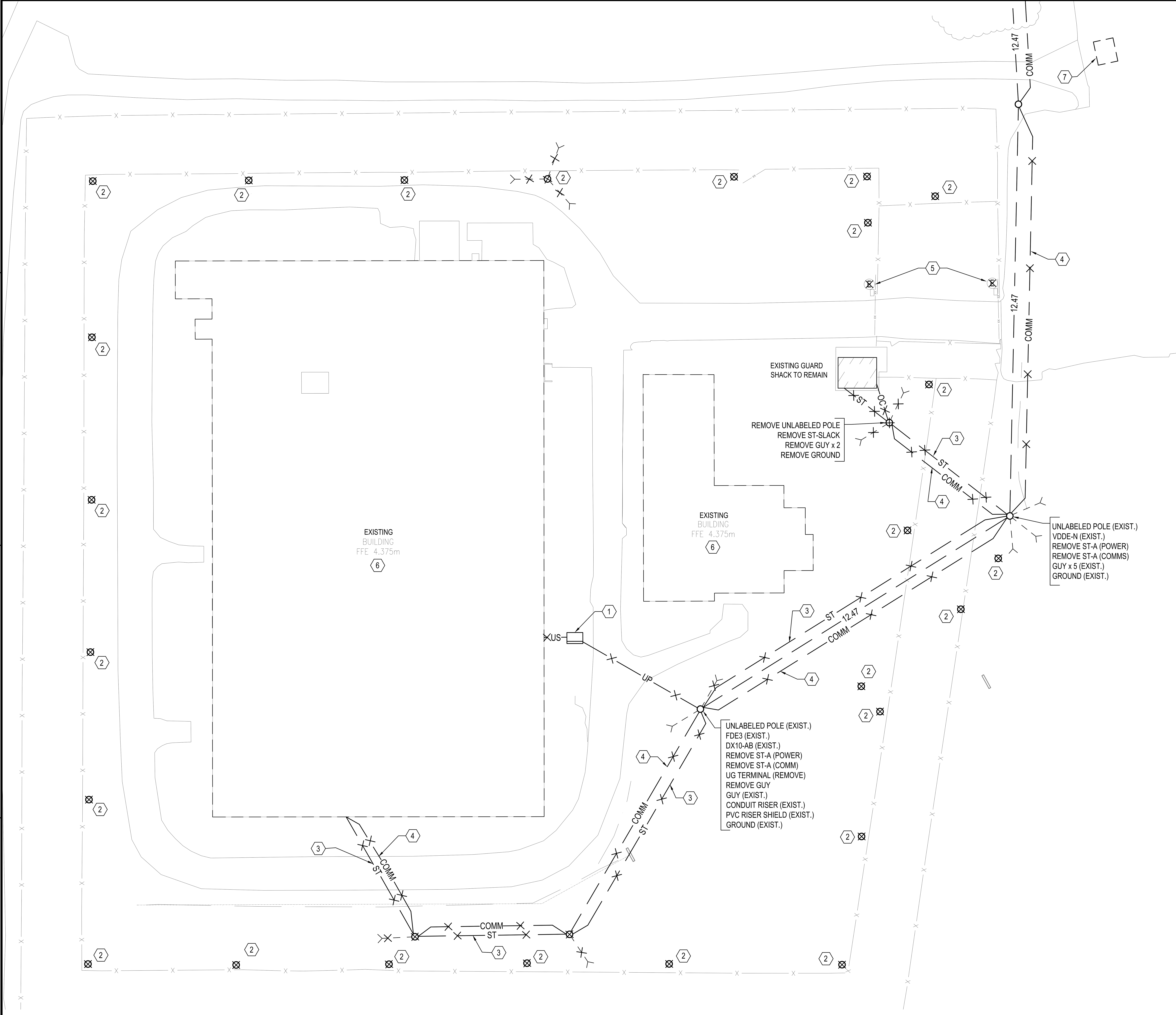
EPROJECT NO.: 1838720

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 14157197

SHEET 135 OF 170

E-001



ELECTRICAL DEMOLITION SITE PLAN

SCALE: 1:300

PLAN NORTH

GENERAL SHEET NOTES

A. COORDINATE ALL WORK SHOWN WITH CIVIL DEMOLITION PLAN ON SHEET CD101 AND SEPARATE DEMOLITION DRAWINGS IN FRONT OF DRAWING SET.

B. COMPLETELY REMOVE ALL UNDERGROUND ELECTRICAL CONDUITS, DUCTBANKS, AND STRUCTURES WHERE THEY WILL INTERFERE WITH NEW WORK.

C. COMPLETELY REMOVE ALL ELECTRICAL CIRCUITS, FEEDERS AND WIRING.

SHEET KEYNOTES

1. REMOVE EXISTING 300 kVA PAD-MOUNTED TRANSFORMER WITH UNDERGROUND PRIMARY CIRCUIT FROM POLE AND SECONDARY SERVICE TO BUILDING.
2. REMOVE EXISTING 6m TALL WOOD POLE.
3. REMOVE EXISTING OVERHEAD SECONDARY ELECTRICAL SERVICE.
4. REMOVE EXISTING OVERHEAD COMMUNICATIONS SERVICE.
5. REMOVE EXISTING VEHICLE GATE CONTROLLER CONNECTIONS AND CIRCUITING FOR POWER AND CONTROL BACK TO GUARD SHACK.
6. ENTIRE FACILITY DEMOLISHED PER DEMOLITION DRAWINGS.
7. EXISTING UNDERGROUND MANHOLE FOR SITE TELECOMMUNICATIONS. SEE NEW WORK SITE PLAN ON ES102.

GRAPHIC SCALE

1:300 0 10 20 30m

SYM	DESCRIPTION	DATE	APPR

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SATISFACTORY TO DATE 01/15/2026

DES CAB DRW CAB CHK SEK

PM/DM MJD/ATP

BRANCH MANAGER SEK

DES PROD DIR WBF

FIRE PROTECTION ENGINEER DSN

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PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

ELECTRICAL DEMOLITION SITE PLAN

SCALE: AS NOTED

EPROJECT NO.: 1838720

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 14157198

SHEET 136 OF 170

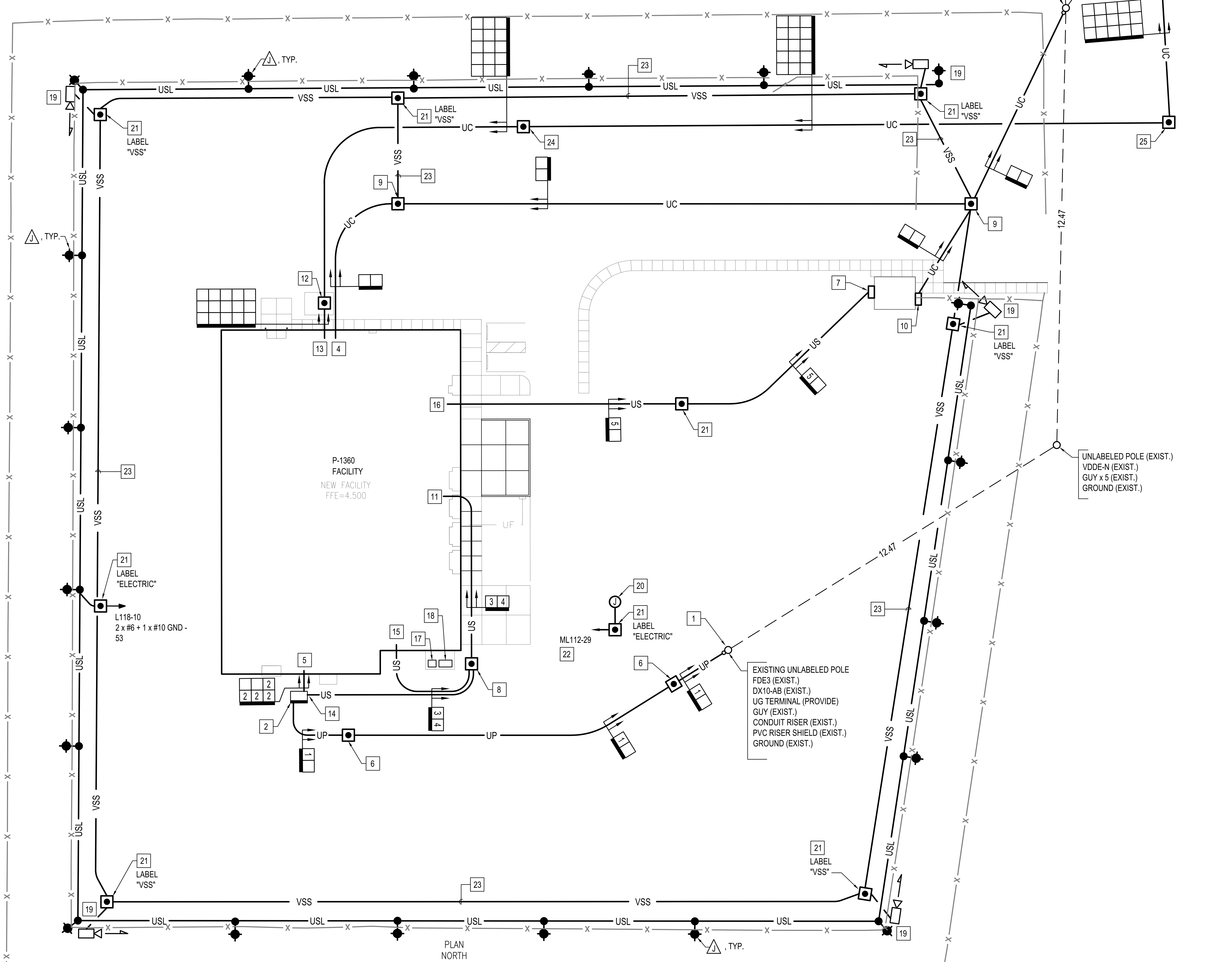
ES101

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\13_ELEC\P1360_1783264_ES101.dwg LAYOUT NAME: ES101 - Electrical Demolition Site Plan PLOTTED: Tuesday, January 20, 2026 - 12:15pm USER: eric.herlan

EXTERIOR CONDUIT AND CABLING SCHEDULE

IDENTIFICATION	CONDUIT/ DUCT SIZE	CABLE DESCRIPTION	REMARKS
1	100	3 x #4/0 (8kV)	PRIMARY ELECTRICAL SERVICE FROM EXISTING OVERHEAD CIRCUIT
2	100	4 x 350 kCMIL	SECONDARY ELECTRICAL SERVICE ENTRANCE CONDUCTORS
3	100	3 x #1/0 + 1 x #6 GND	SECONDARY FIRE PUMP FEEDER
4	100	3 x #1/0 + 1 x #6 GND	PRIMARY FIRE PUMP FEEDER
5	100	4 x #6 + 1 x #10 GND	ELECTRICAL SERVICE TO EXISTING GATE HOUSE



ELECTRICAL SITE PLAN

SCALE: 1:300

GENERAL SHEET NOTES

A. PROVIDE ALL ELECTRICAL INSTALLATION IN ACCORDANCE WITH THE CANADIAN ELECTRICAL CODE (CAN/CSA-C22.3), AS WELL AS ADDITIONAL REQUIREMENTS LISTED IN THE SPECIFICATIONS.

B. SEE CIVIL UTILITY PLAN ON SHEET CU101 FOR COORDINATION WITH OTHER UTILITIES.

SHEET KEYNOTES

1. TRANSITION EXISTING OVERHEAD 12.47 kV, 3 PHASE PRIMARY CIRCUIT TO UNDERGROUND AT EXISTING OVERHEAD POLE.
2. 750 kVA, 12.47 kV DELTA - 480Y/277V, 3 PHASE, 4 WIRE PAD-MOUNTED SERVICE TRANSFORMER.
3. OVERHEAD TELECOMMUNICATIONS CONNECTION POINT.
4. TURN UP AND TERMINATE TELECOMMUNICATIONS SERVICE CONDUITS IN TELECOMM ROOM 110. SEE ENLARGED ROOM 110 TELECOMMUNICATIONS PLAN ON SHEET TN401.
5. TURN UP AND TERMINATE ELECTRICAL SERVICE IN ELECTRICAL ROOM 118. SEE ELECTRICAL POWER PLAN ON SHEET EP101.
6. PRIMARY VOLTAGE ELECTRICAL MANHOLE TYPE 1 (UG-1).
7. PROVIDE WEATHERPROOF EXTERIOR PULL BOX TO FACILITATE ROUTING OF PANELBOARD FEEDER AND GATE CONTROLLER CIRCUITS INTO GATE HOUSE.
8. SECONDARY ELECTRICAL HAND HOLE TYPE 2 (UG-4).
9. TELECOMMUNICATIONS HAND HOLE TYPE 2 (UG-4).
10. PROVIDE WEATHERPROOF EXTERIOR PULL BOX TO FACILITATE ROUTING OF TELECOMMUNICATIONS CABLING (PROVIDED BY OTHERS) INTO GATE HOUSE.
11. TURN UP AND TERMINATE CIRCUITS (PRIMARY AND SECONDARY) FOR FIRE PUMP AT FIRE PUMP CONTROLLER IN FIRE PUMP ROOM 113. SEE ELECTRICAL POWER PLAN ON SHEET EP101.
12. TELECOMMUNICATIONS MANHOLE UC1A. SEE OUTSIDE CABLING MANHOLES UC1A AND UC1B DETAIL ON ES501.
13. TURN UP AND TERMINATE OUTSIDE CABLING CONDUITS INTO BOTTOM OF CABLE ENCLOSURES IN OPS EQUIPMENT ROOM 116. SEE ENLARGED OUTSIDE CABLING ENTRANCE PATHWAY PLAN ON SHEET TN401.
14. FEED PRIMARY POWER CIRCUIT TO FIRE PUMP FROM TRANSFORMER SECONDARY LUGS.
15. FEED SECONDARY POWER CIRCUIT TO FIRE PUMP FROM GENERATOR SWITCHBOARD IN GENERATOR ROOM 115.
16. FEED SECONDARY ELECTRICAL SERVICE TO GUARD SHACK FROM SWITCHBOARD ES1.
17. PORTABLE GENERATOR QUICK DISCONNECT ENCLOSURE. SEE ELECTRICAL POWER PLAN ON SHEET EP101.
18. GENERATOR RESISTIVE LOAD BANK. SEE ELECTRICAL POWER PLAN ON SHEET EP101.
19. PROVIDE INFRASTRUCTURE ONLY (53 C FROM HANDHOLE TO POLE) FOR VSS CAMERA (PROVIDED BY OTHERS) MOUNTED TO LIGHTING POLE.
20. WASTEWATER HOLDING TANK ALARM POWER - 120V. COORDINATE EXACT LOCATION AND CONNECTION WITH TANK PROVIDER. SEE WASTEWATER HOLDING TANK DETAIL ON SHEET C-504.
21. STANDARD ELECTRICAL NON-TRAFFIC, COMPOSITE/FIBERGLASS HAND HOLE - UG-6, TYPE 7.
22. 2 x #10 + 1 x #12 GND - 27.
23. 2 - 53 EMPTY CONDUITS DIRECT BURIED FOR VSS CAMERA CABLING PROVIDED BY OTHERS.
24. TELECOMMUNICATIONS MANHOLE UC1B. SEE OUTSIDE CABLING MANHOLES UC1A AND UC1B DETAIL ON SHEET ES501.
25. TELECOMMUNICATIONS MANHOLE UC1C. SEE OUTSIDE CABLING MANHOLES UC1A AND UC1B DETAIL ON SHEET ES501.
26. BREAK INTO EXISTING UNDERGROUND CONCRETE MANHOLE TO FACILITATE ROUTING OF DUCTBANK AS SHOWN. REPAIR TO ORIGINAL STRENGTH AND INTEGRITY.

GRAPHIC SCALE

1:300 — 0 10 20 30m

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ACTIVITY: D. EVANS VIA EMAIL
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DES	CAB	DRW	CAB	CHK	SEK
PM/DM	MJD/ATP				
BRANCH MANAGER	SEK				
DES PROD DIR	WBF				
FIRE PROTECTION ENGINEER	DSN				

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VARIOUS LOCATIONS

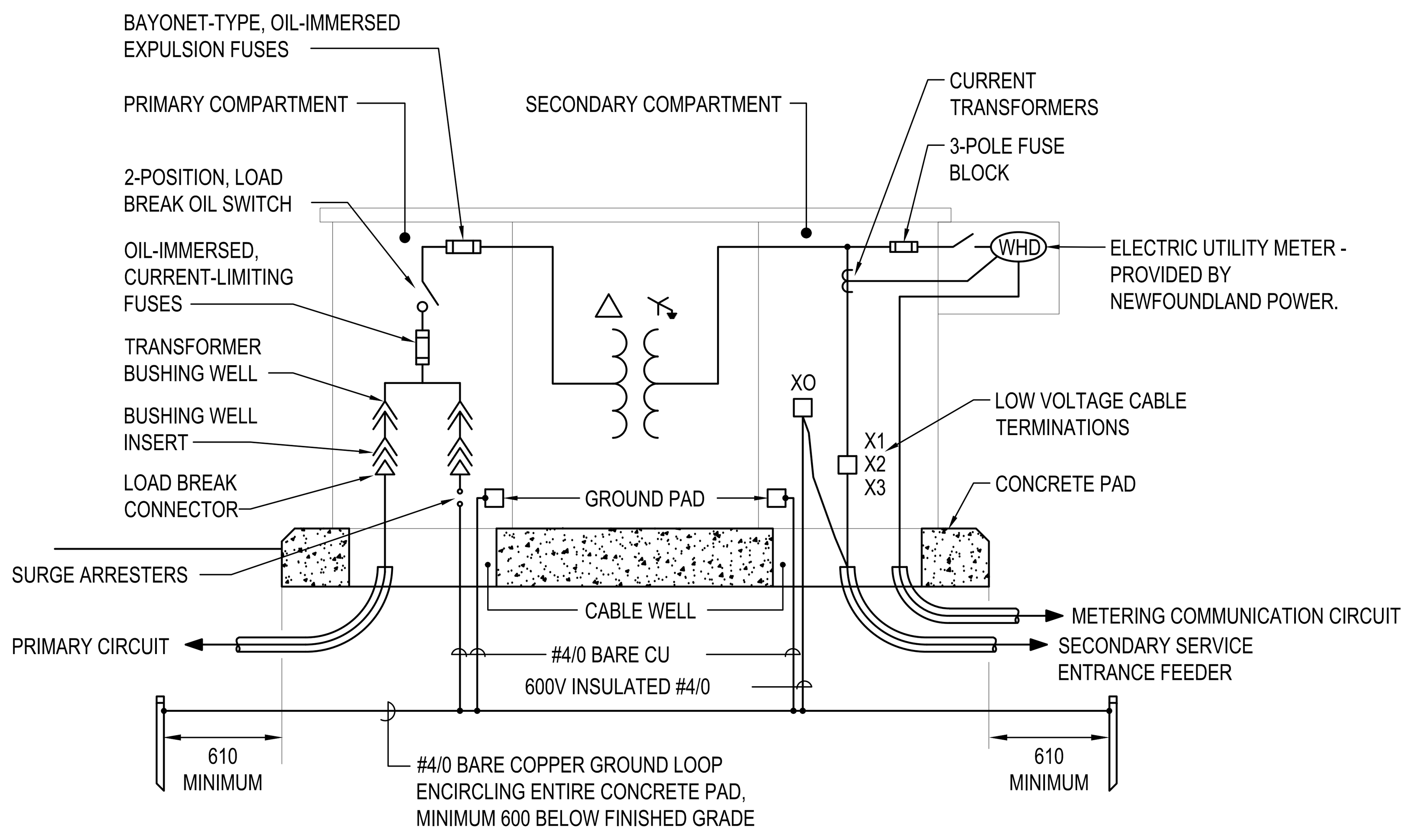
JOINT MARITIME FACILITY

ELECTRICAL SITE PLAN

SCALE: AS NOTED
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157199
SHEET 137 OF 170
ES102

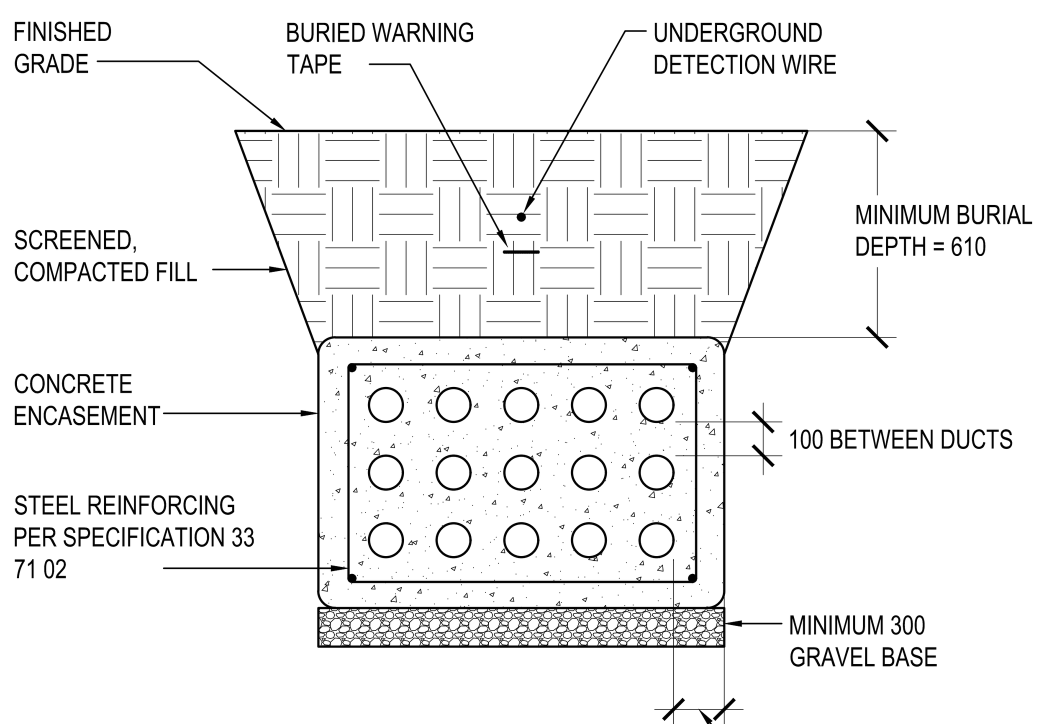
NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1360_MJMF\B_Design\DWGS\Sheet\13_ELEC\P1360_1783264_ES102.dwg LAYOUT NAME: ES102 - Electrical Site Plan PLOTTED: Tuesday, January 20, 2026 - 12:15pm USER: eric.herbon



C1 PAD-MOUNTED TRANSFORMER DETAIL

SCALE: NONE

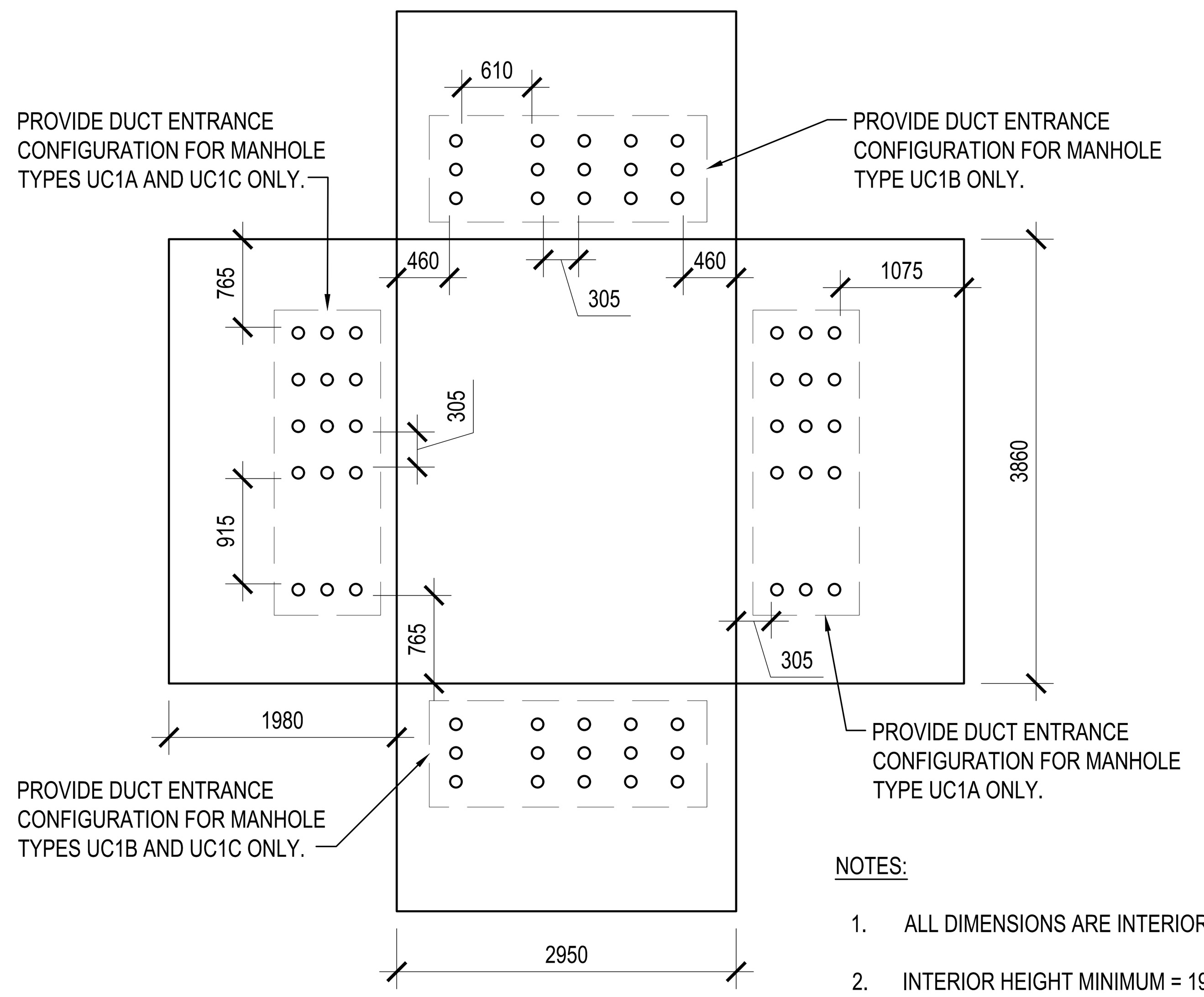


NOTES:

1. DUCT DIAMETER IS TYPICALLY 100 FOR COMMUNICATIONS AND <600V ELECTRICAL CIRCUITS, AND 150 FOR ELECTRICAL >600V. SEE CABLE SCHEDULE ON SHEET ES102 FOR SPECIFIC CIRCUIT AND DUCT DESCRIPTIONS.
2. QUANTITY OF VERTICAL AND HORIZONTAL DUCTS VARY. SEE SITE PLAN FOR DUCTBANK SECTION DESCRIPTIONS.

A1 UNDERGROUND DUCTBANK DETAIL

SCALE: NONE



NOTES:

1. ALL DIMENSIONS ARE INTERIOR TO MANHOLE.
2. INTERIOR HEIGHT MINIMUM = 1980 MIN.
3. PROVIDE CONSTRUCTION OF MANHOLE IN MANNER SIMILAR TO TYPE UG1 SHOWN ON ES502.
4. ALL DUCTS = 100 IN DIA.

A3 COMMUNICATIONS MANHOLES UC1A, UC1B AND UC1C DETAIL

SCALE: NONE

SYM | DESCRIPTION | DATE | APPR

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ACTIVITY
D. EVANS VIA EMAIL

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DES CAB DRW CAB CHK SEK
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BRANCH MANAGER SEK
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FIRE PROTECTION ENGINEER DSN

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NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
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VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

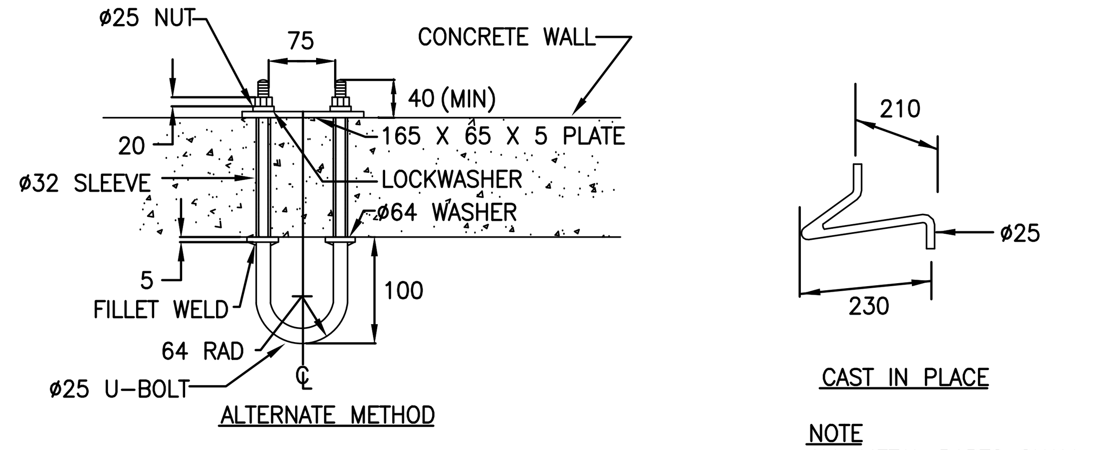
ELECTRICAL SITE DETAILS

SCALE: NONE
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157200
SHEET 138 OF 170

ES501

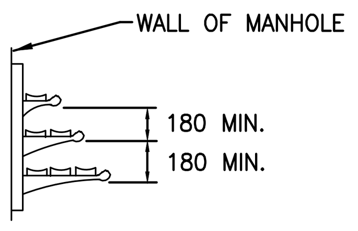
NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018

FILE NAME: Y:\OCONUS\~MultipleSites\2026_1783264_P1560_JMF\B_Design\DWGS\Sheet\13_ELEC\P1560_1783264_ES501.dwg LAYOUT NAME: ES501 - Electrical Site Details PLOTTED: Tuesday, January 20, 2026 - 12:15pm USER: eric.herlon

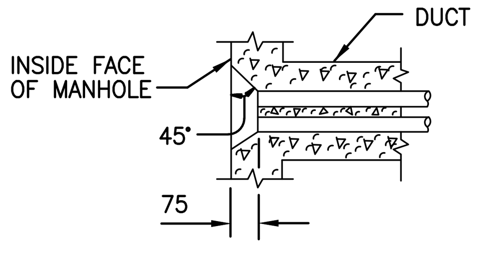


NOTE
ALL METAL PARTS SHALL BE HOT DIP GALVANIZED

DETAIL OF PULLING-IN IRON



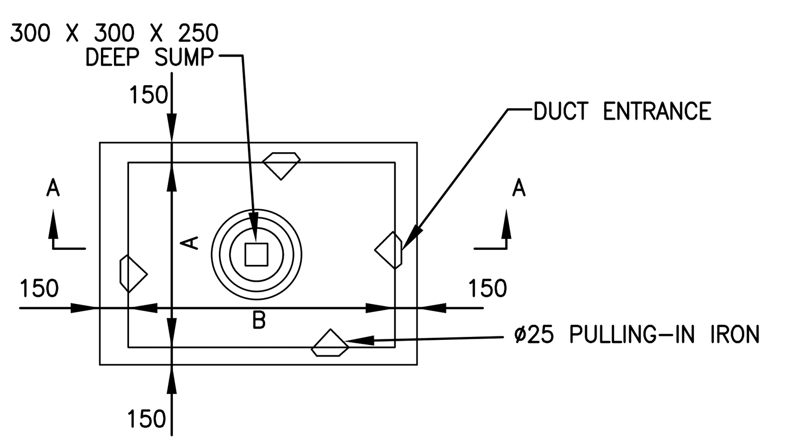
TYPICAL CABLE RACK



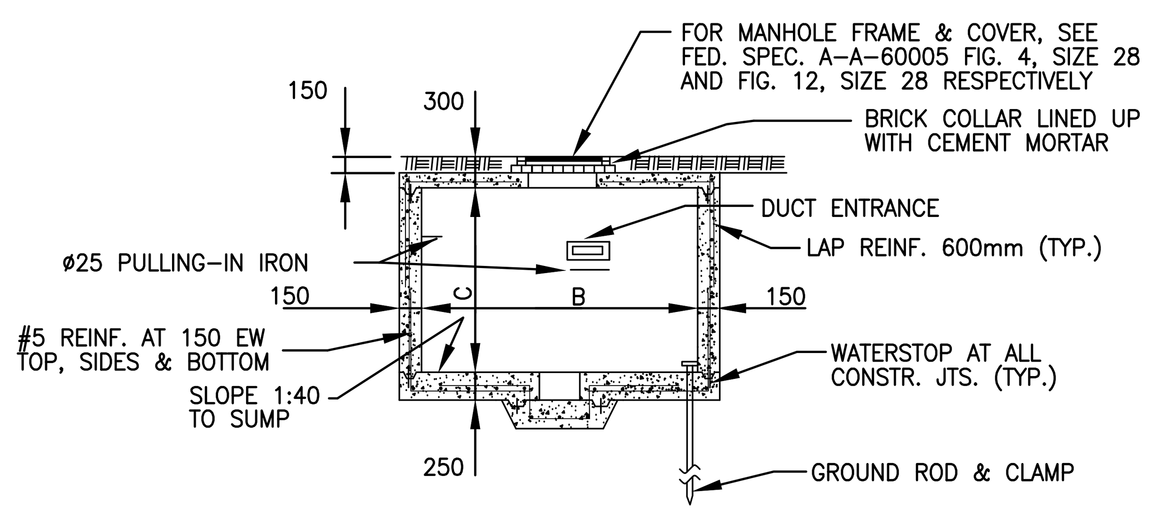
TYPICAL DUCT ENTRANCE

DETAILS
(PULLING-IN IRONS, CABLE RACK AND DUCT ENTRANCE)

SKETCH DATE	JUNE 2002	STYLE	UG-7



PLAN



SECTION A-A

NOTES:

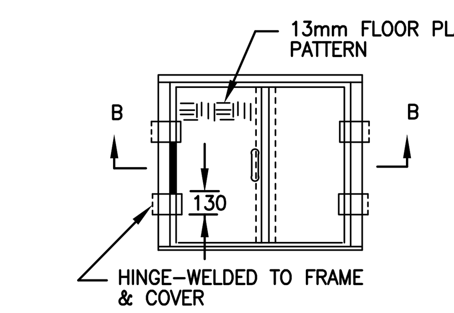
1. FOR DETAILS OF CABLE RACKS, DUCT ENTRANCE AND PULLING-IN IRONS, SEE PLATE UG-7.
2. MINIMUM CONCRETE COMPRESSIVE STRENGTH SHALL BE 21 MPa.

MANHOLE DIMENSIONS

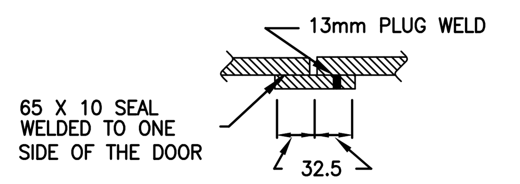
TYPE	A	B	C (AT HIGH PT.)
1	1800	1800	2000
2	1800	2500	2000

STANDARD ELECTRICAL MANHOLE (NONTRAFFIC)
TYPES 1 & 2

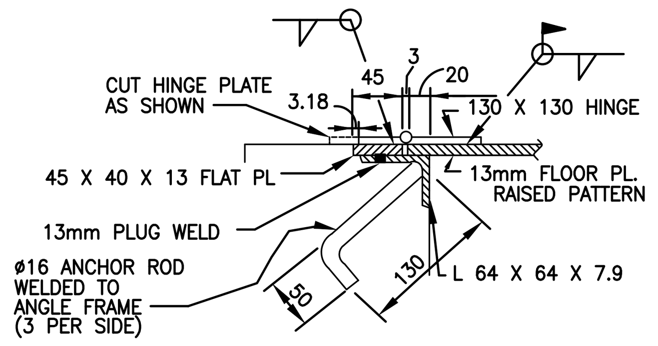
SKETCH DATE	JUNE 2002	STYLE	UG-1



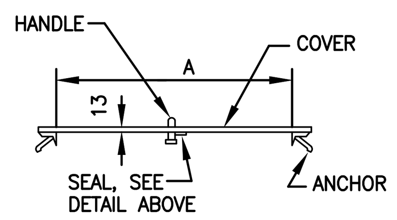
PLAN OF COVER



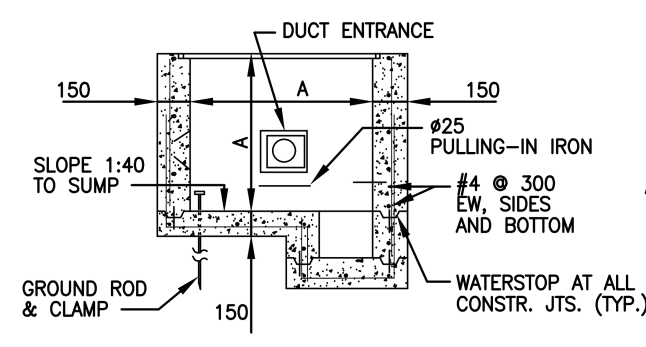
SEAL DETAIL



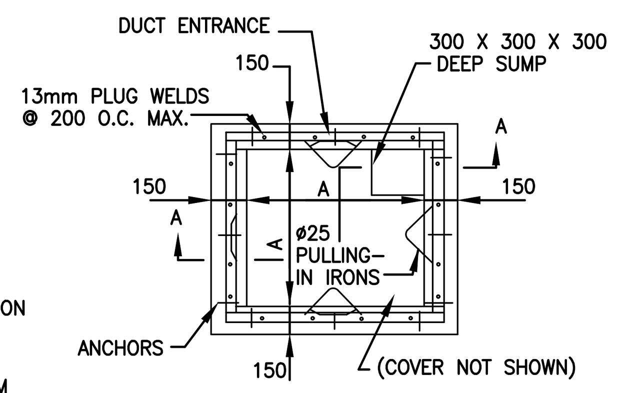
SECTION OF FRAME



SECTION B-B



SECTION A-A



PLAN

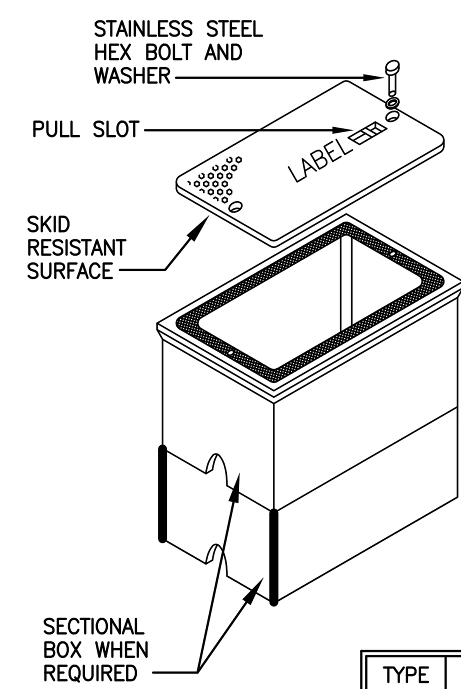
NOTES:

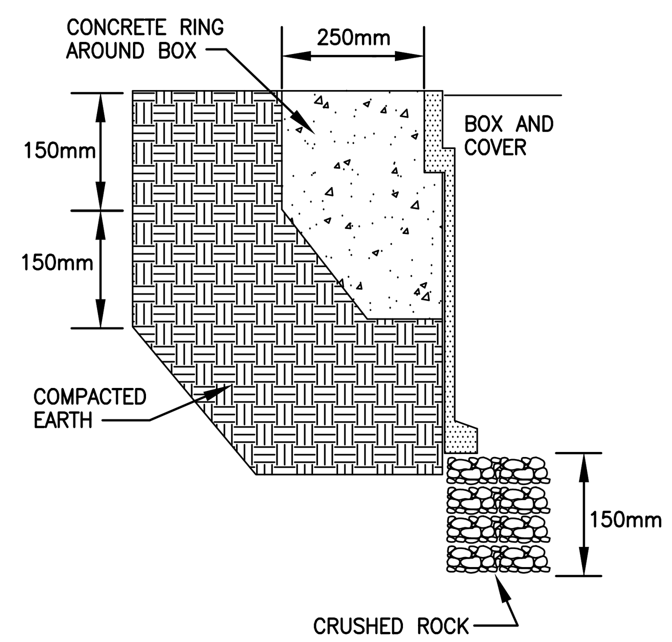
1. FOR DETAILS OF CABLE RACKS, DUCT ENTRANCE AND PULLING-IN IRONS, SEE PLATE UG-7.
2. MINIMUM CONCRETE COMPRESSIVE STRENGTH SHALL BE 21 MPa.

HANDHOLE TYPE	DIMENSIONS A
1	1000
2	1250

STANDARD ELECTRICAL HANDHOLE (NONTRAFFIC)
TYPES 1 & 2

SKETCH DATE	JUNE 2002	STYLE	UG-4





TYPE	HANDHOLE SIZING
5	300mm X 300mm X 600mm DEEP
6	300mm X 500mm X 600mm DEEP
7	300mm X 600mm X 600mm DEEP
8	600mm X 1000mm X 600mm DEEP
9	800mm X 1250mm X 600mm DEEP

HANDHOLE REQUIREMENTS

1. HOUSING SHALL BE A POLYMER CONCRETE REINFORCED WITH A HEAVY WEAVE FIBERGLASS REINFORCING WITH COMPRESSIVE STRENGTH OF NO LESS THAN 70 MPa.
2. COVER AND BOX SHALL WITHSTAND A SERVICE LOAD OF NO LESS THAN 6800 kg OVER A 250 x 250 AREA.
3. PROVIDE STAINLESS STEEL BOLTS AND INSERTS.
4. PROVIDE WITH (2) 64mm MOUSEHOLES.
5. PROVIDE LABEL "ELECTRICAL" FOR POWER HANDHOLES OR "TELEPHONE" FOR TELEPHONE HANDHOLES, OR AS INDICATED.

STANDARD ELECTRICAL HANDHOLE (NONTRAFFIC)
(COMPOSITE/FIBERGLASS) TYPES 5, 6, 7, 8 & 9

SKETCH DATE	JUNE 2002	STYLE	UG-6

APPROVED
Wm B forbes
Digitally signed by signature
Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

ACTIVITY
D. EVANS VIA EMAIL
SATISFACTORY TO DATE 01/15/2026

DES CAB | DRW CAB | CHK SEK
PM/DM MJD/ATP
BRANCH MANAGER SEK
DES PROD DIR WBF
FIRE PROTECTION ENGINEER DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY

ELECTRICAL SITE DETAILS

SCALE: NONE
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157201
SHEET 139 OF 170

ES502

SYMBOL LEGEND

A	ANGLE
AB	ANGLE BRACE
CL	CURRENT LIMITING
D	DOUBLE
DE	DEADEND
F	FLAT (HORIZONTAL)
FB	FLAT BRACE
HD	HEAVY DUTY
I	INSULATED
PP	PHASE TO PHASE
N	NEUTRAL
R	RIDGE OR POLE TOP PIN
S	SECONDARY, OPEN WIRE
ST	SECONDARY, TRIPLEX
SQ	SECONDARY, QUADRUPLEX
T	TRANSFORMER
TERM	TERMINAL
UG	UNDERGROUND
V	VERTICAL
X	CROSSARM, 2.4m
X10	CROSSARM, 3m

GENERAL NOTES

1. SYMBOLS COMPRISING THE OVERHEAD SKETCHES ARE NOT INTENDED TO BE "ALL INCLUSIVE" FOR USE ON EVERY DISTRIBUTION POLE LINE CONFIGURATION. ONLY SKETCHES WHICH REFLECT TYPICAL ARRANGEMENTS ARE INCLUDED. FOR OTHER DESIRED ARRANGEMENTS, PROVIDE SEPARATE DETAILS DRAWN TO REFLECT THE SPECIFIC CONDITIONS.
2. THE METHOD OF SHOWING INFORMATION ON SITE PLAN IS OPTIONAL; HOWEVER, IT SHALL BE CONSISTENT WITH INFORMATION CONTAINED IN THE GUIDE LEGEND (APPENDIX C) INCLUDED IN "TECHNICAL GUIDELINES AND CRITERIA FOR ELECTRICAL DESIGN". THE CHARACTERISTICS AND IDENTIFICATION OF ALL CIRCUITS SHALL BE INCLUDED ON THE SITE PLAN.
3. EACH SKETCH CONTAINS MATERIAL ITEMS WHICH COMPRISE A PART OF EACH INDIVIDUAL SYMBOL REFERENCED BY THAT SKETCH. THESE ITEMS ARE INDICATED BY CIRCLED NUMERALS WHICH ARE IDENTIFIED BY SKETCHES OH-1.5 AND OH-1.5A.
4. SPACING REQUIREMENTS RELATED TO INDIVIDUAL COMPONENTS OF A SYMBOL ARE INDICATED ON THE APPROPRIATE SKETCH. VERTICAL SPACING REQUIREMENTS BETWEEN CIRCUITS AND/OR SYSTEMS ARE INDICATED ON SKETCH OH-1.4. ALL OTHER SEPARATIONS BETWEEN CIRCUITS, EQUIPMENT, ETC., SHALL CONFORM TO THE NATIONAL ELECTRICAL SAFETY CODE, IEEE C2.
5. FOR NEW CONSTRUCTION OR OPERATING VOLTAGES GREATER THAN 5KV, LIMIT THE NUMBER OF CONDUCTORS ON ANY CROSSARM TO A MAXIMUM OF 3.
6. USE 3m CROSSARMS FOR ALL UNDERBUILD CIRCUITS WITH OPERATING VOLTAGES GREATER THAN 15KV.

THIS INFORMATION IS FOR DESIGNER USE AND SHALL NOT BE INCLUDED ON CONSTRUCTION DRAWINGS.

SYMBOL LEGEND & GENERAL NOTES

SKETCH DATE JUNE 2002 | STYLE OH-1.1

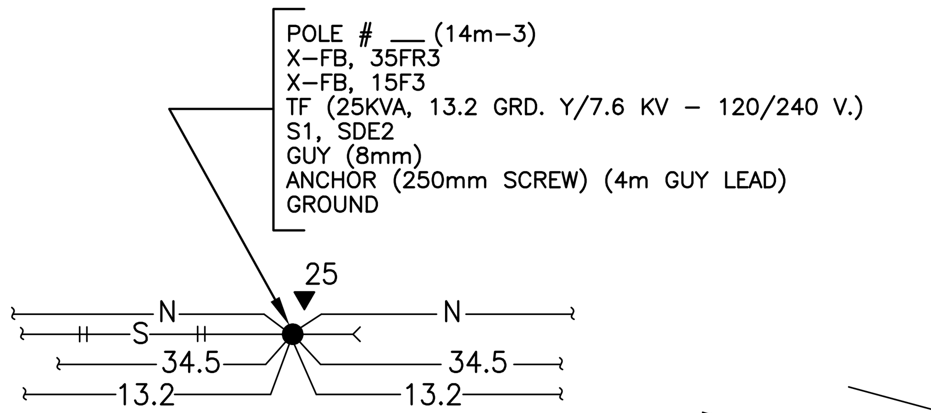
LIST OF SYMBOLS

SKETCH NUMBER	CATEGORY
OH-2 THRU OH-10	CROSSARM SYMBOLS
OH-11 THRU OH-14	HORIZONTAL (TANGENT OR ANGLE) CONSTRUCTION SYMBOLS
OH-15 THRU OH-20	HORIZONTAL DEADEND CONSTRUCTION SYMBOLS
OH-21 THRU OH-25	VERTICAL CONSTRUCTION SYMBOLS
OH-26 THRU OH-29	TRANSFORMER SYMBOLS
OH-30 THRU OH-31	UNDERGROUND TERMINAL SYMBOLS
OH-32 THRU OH-33	GUY SYMBOLS
OH-34 THRU OH-35	CONDUIT RISER SYMBOLS
OH-36 THRU OH-40	SECONDARY SYMBOLS
OH-41	GROUND SYMBOL

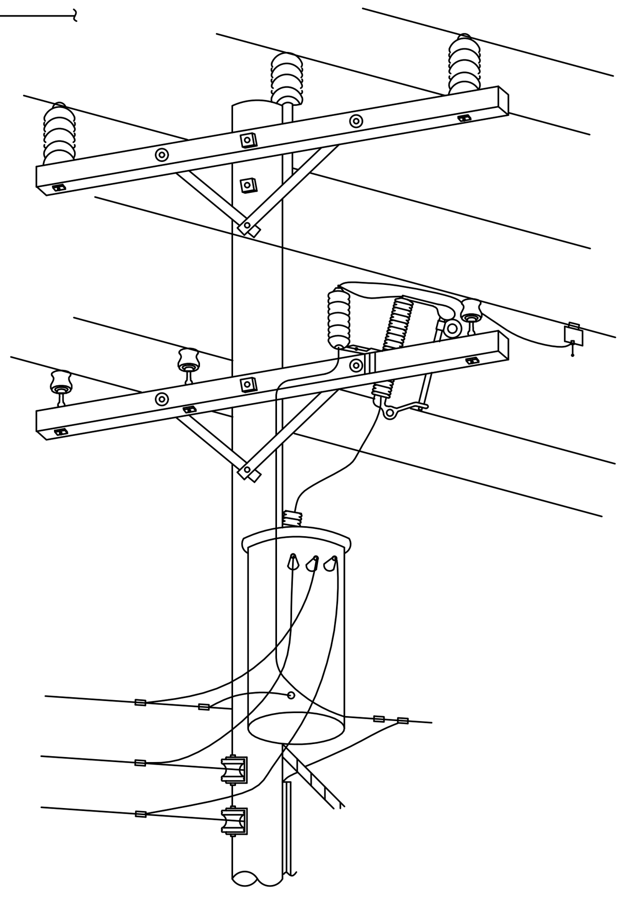
THIS INFORMATION IS FOR DESIGNER USE AND SHALL NOT BE INCLUDED ON CONSTRUCTION DRAWINGS.

LIST OF SYMBOLS

SKETCH DATE JUNE 2002 | STYLE OH-1.2



PLAN VIEW



NOTES

1. THE SYMBOLS LISTED ABOVE INCLUDE MATERIALS (DESCRIBED BY OVERHEAD SKETCHES) WHICH ARE GRAPHICALLY ILLUSTRATED BY THIS PICTURE. SEE SKETCH OH-1.3A FOR EXPLANATORY NOTES.
2. EACH SKETCH CONTAINS MATERIAL ITEMS WHICH COMPRISE A PART OF EACH INDIVIDUAL SYMBOL REFERENCED BY THAT SKETCH, THESE ITEMS ARE INDICATED BY CIRCLED NUMERALS WHICH ARE IDENTIFIED BY SKETCHES OH-1.5 AND OH-1.5A.
3. SPACING REQUIREMENTS RELATED TO INDIVIDUAL COMPONENTS OF A SYMBOL ARE INDICATED ON THE APPROPRIATE SKETCH. VERTICAL SPACING REQUIREMENTS BETWEEN CIRCUITS AND/OR SYSTEMS ARE INDICATED ON SKETCH OH-1.4. ALL OTHER SEPARATIONS BETWEEN CIRCUITS, EQUIPMENT, ETC., SHALL CONFORM TO THE NATIONAL ELECTRICAL SAFETY CODE, IEEE C2.

METHOD OF SHOWING SYMBOLS

SKETCH DATE JUNE 2002 | STYLE OH-1.3

EXPLANATORY NOTES – METHOD OF SHOWING SYMBOLS

1. SYMBOLS ARE SHOWN IN THE BASIC ORDER AS THEY APPEAR ON THE POLE, BY STARTING AT THE TOP AND WORKING DOWN.
2. NUMERALS PRECEDING THE SYMBOL INDICATE THE MINIMUM REQUIRED VOLTAGE (KV) RATING (5,15,35) OF THE ASSEMBLY, IF APPLICABLE.
3. NUMERAL FOLLOWING THE SYMBOL INDICATES THE NUMBER OF CONDUCTORS ASSOCIATED WITH THE ASSEMBLY, IF APPLICABLE.
4. NUMERAL IN PARENTHESIS FOLLOWING THE SYMBOL DENOTES THE NUMBER OF ASSEMBLIES REQUIRED, IF MORE THAN ONE.
5. DATA IN PARENTHESIS FOLLOWING THE SYMBOL PROVIDES INFORMATION RELATIVE TO THE SYMBOL.

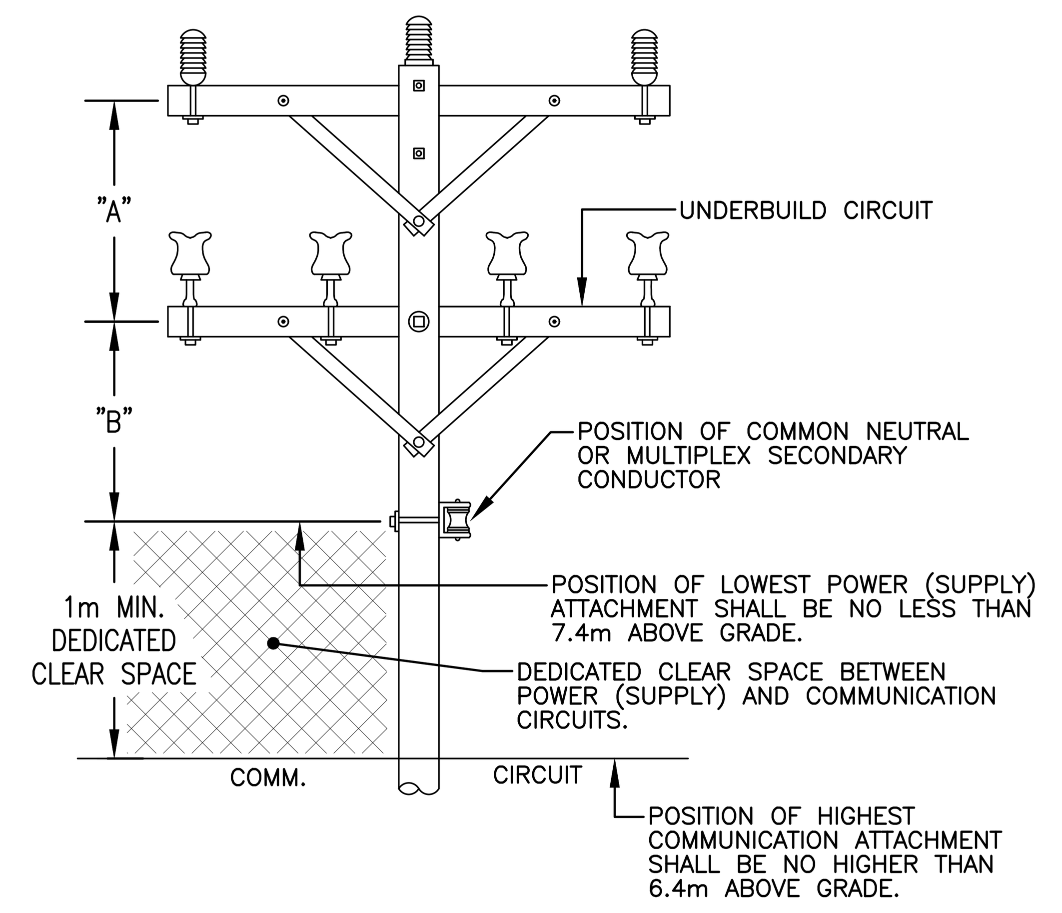
EXPLANATION OF SYMBOLS LISTED FOR POLE ON SKETCH OH-1.3

PROVIDE 14m LONG, CLASS 3 POLE CONTAINING:

- X-FB – 2.4m CROSSARM WITH FLAT BRACE
- 35FR3 – 35KV INSULATORS, FLAT (MOUNTED HORIZONTAL ON CROSSARM), RIDGE PIN (CENTER PHASE ON POLE TOP PIN), THREE CONDUCTORS
- X-FB – 2.4m CROSSARM WITH FLAT BRACE
- 15F3 – 15KV INSULATORS, FLAT (MOUNTED HORIZONTAL ON CROSSARM), THREE CONDUCTORS. NOTE: THIS SYMBOL CALLS FOR THREE CROSSARM MOUNTED PINS IN LIEU OF RIDGE PIN ON CENTER PHASE.
- TF – TRANSFORMER ON FLAT (HORIZONTAL) CONSTRUCTION. DATA IN PARENTHESIS DESCRIBES THE TRANSFORMER CHARACTERISTICS.
- S1 – SECONDARY, ONE CONDUCTOR, TANGENT CONSTRUCTION (COMMON NEUTRAL).
- SDE2 – SECONDARY DEADEND, TWO CONDUCTORS, OPEN WIRE
- GUY (8mm) – DOWN GUY – WIRE SIZE 8mm
- ANCHOR (250mm SCREW) – 250mm SCREW TYPE ANCHOR WITH 4m GUY LEAD. NOTE: NO PLATE IS INCLUDED FOR THE ANCHOR SYMBOL.
- GROUND – NO EXPLANATION NECESSARY

EXPLANATION OF NOTES/SYMBOLS

SKETCH DATE JUNE 2002 | STYLE OH-1.3A



NOTE

1. FOR HORIZONTAL SPACING REQUIREMENTS FOR CONDUCTORS ON SAME SUPPORT, REFER TO THE NATIONAL ELECTRICAL SAFETY CODE, IEEE C2.

ø-ø VOLTAGE	0-15KV	15-50KV
SPACING "A"	1m	1.2m *
SPACING "B"	1m	1m

* PROVIDE 1.5m CLEARANCE WHEN OPERATING VOLTAGE OF UNDERBUILD CIRCUIT IS GREATER THAN 15KV.

BASIC VERTICAL SPACING REQUIREMENTS

SKETCH DATE JUNE 2002 | STYLE OH-1.4

POLE LINE MATERIAL LIST

1. FLAT STEEL BRACE (TWO PIECES)
2. MACHINE BOLT, 10mm X LENGTH NEEDED WITH WASHER, NUT AND LOCKWASHER
3. 2.4m WOOD CROSSARM WITH CROSS SECTION DIMENSIONS OF 90mm X 115mm
4. MACHINE BOLT, 16mm X LENGTH NEEDED WITH WASHER, NUT AND LOCKWASHER
5. TIMBER CONNECTOR
6. LAGSCREW, 12mm X 100mm
7. ANGLE STEEL BRACE (TWO PIECES)
8. MACHINE BOLT, 12mm X LENGTH NEEDED, WITH WASHER, NUT & LOCKWASHER
9. DEADEND BOX
10. STEEL PIN
11. PIN INSULATOR
12. GRID GAIN, USED ONLY WHEN THERE IS NO POLE GAIN
13. ANGLE STEEL BRACE (ONE PIECE)
14. 3m WOOD CROSSARM WITH CROSS SECTION DIMENSIONS OF 90mm X 115mm
15. 16mm EYE NUT
16. 16mm EYE BOLT, LENGTH AS NEEDED, WITH WASHER, NUT & LOCKWASHER
17. EXTENSION LINK
18. BELL TYPE SUSPENSION INSULATOR WITH CONNECTING HARDWARE
19. STRAIN CLAMP
20. STEEL ANGLE PIN
21. CLUSTER MOUNTING BRACKET, STEEL
22. TRANSFORMER GROUNDING CONNECTION
23. STIRRUP
24. SECONDARY LEAD SUPPORT BRACKET
25. ADAPTER PLATE FOR CLUSTER MOUNTING
26. CLEVIS BRACKET FOR SPOOL INSULATOR
27. SPOOL INSULATOR
28. U BOLT CLAMP
29. PREFORMED GUY GRIP
30. GUY HOOK
31. GUY STRAIN INSULATOR
32. GUY WIRE, SIZE AS SPECIFIED
33. #4 WP CU. SOFT DRAWN GROUND WIRE
34. GROUND CLAMP
35. CONDUIT COUPLING
36. CONDUIT BEND
37. INSULATED BUSHING
38. PERFORATED STRAPPING, 40mm WIDE
39. HOT LINE CLAMP
40. FUSED CUTOUT, AS SPECIFIED
41. SURGE ARRESTER, AS SPECIFIED
42. POLE TOP PIN (RIDGE PIN) – 610mm LONG
43. CROSSARM ANGLE PIN
44. ANGLE POLE TOP PIN
45. WEATHERPROOF SOFT DRAWN WIRE-SIZE
    (a) TO MATCH OR EXCEED AMPACITY OF CONNECTING CABLE, OR
    (b) AT 125% OF TRANSFORMER FULL LOAD CURRENT, BUT NOT LESS THAN NO. 4 AWG

POLE LINE MATERIAL LIST

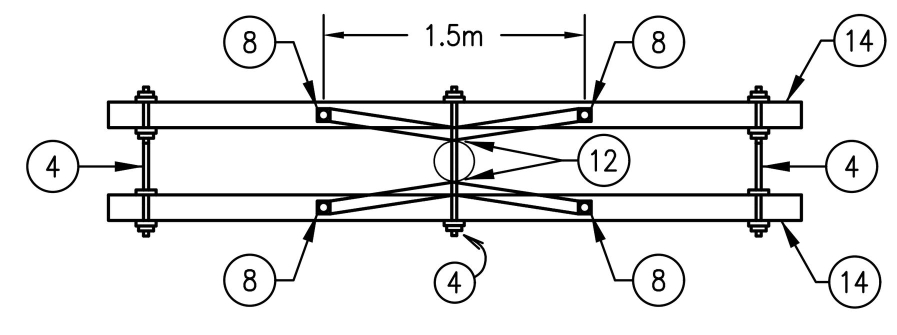
SKETCH DATE JUNE 2002 | STYLE OH-1.5

POLE LINE MATERIAL LIST

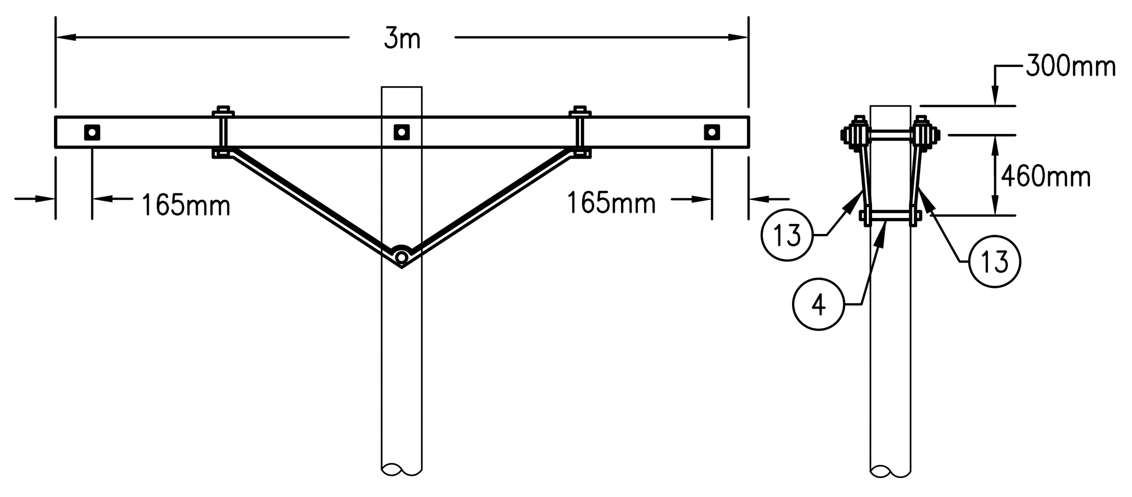
46. TRI-MOUNT BRACKET
47. TERMINATOR
48. MOUNTING BRACKET
49. CABLE GRIP HANGER
50. HOSE CLAMP
51. STUD, 20mm X 45mm
52. LINE POST INSULATOR
53. TRIPLE INSULATOR BRACKET
54. ANGLE CLAMP
55. INSULATOR, LINE POST CLAMP
56. 1.2m CROSSARM
57. CROSSARM GAIN BRACKET
58. PULLEY BRACKET
59. WEDGE CLAMP
60. MIDSPAN SERVICE CLAMP
61. STUD, 175mm
62. SADDLE, ANGLE
63. SADDLE CROSSARM
64. FITTING, POLE TOP
65. CONNECTOR
66. SUSPENSION CLAMP
67. TIE, SERVICE CABLE
68. 1370mm FIBERGLASS STRAIN INSULATOR
69. PVC RISER SHIELD
70. PVC EXTENSION SHIELD
71. PVC BACK PLATE
72. 2.4m WOOD CROSSARM WITH CROSS SECTION DIMENSIONS OF 120mm X 145mm
73. 3m WOOD CROSSARM WITH CROSS SECTION DIMENSIONS OF 120mm X 145mm
74. BACK-UP CURRENT LIMITING FUSE

POLE LINE MATERIAL LIST

SKETCH DATE JUNE 2002 | STYLE OH-1.5A



PLAN VIEW



ELEVATIONS

DX10-AB

SKETCH DATE JUNE 2002 | STYLE OH-8

APPROVED Wm B Forbes Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'

FOR COMMANDER NAVFAC

ACTIVITY D. EVANS VIA EMAIL

SATISFACTORY TO DATE 01/15/2026

DES CAB DRW CAB CHK SEK

PM/DM MJD/ATP

BRANCH MANAGER SEK

DES PROD DIR WBF

FIRE PROTECTION ENGINEER DSN

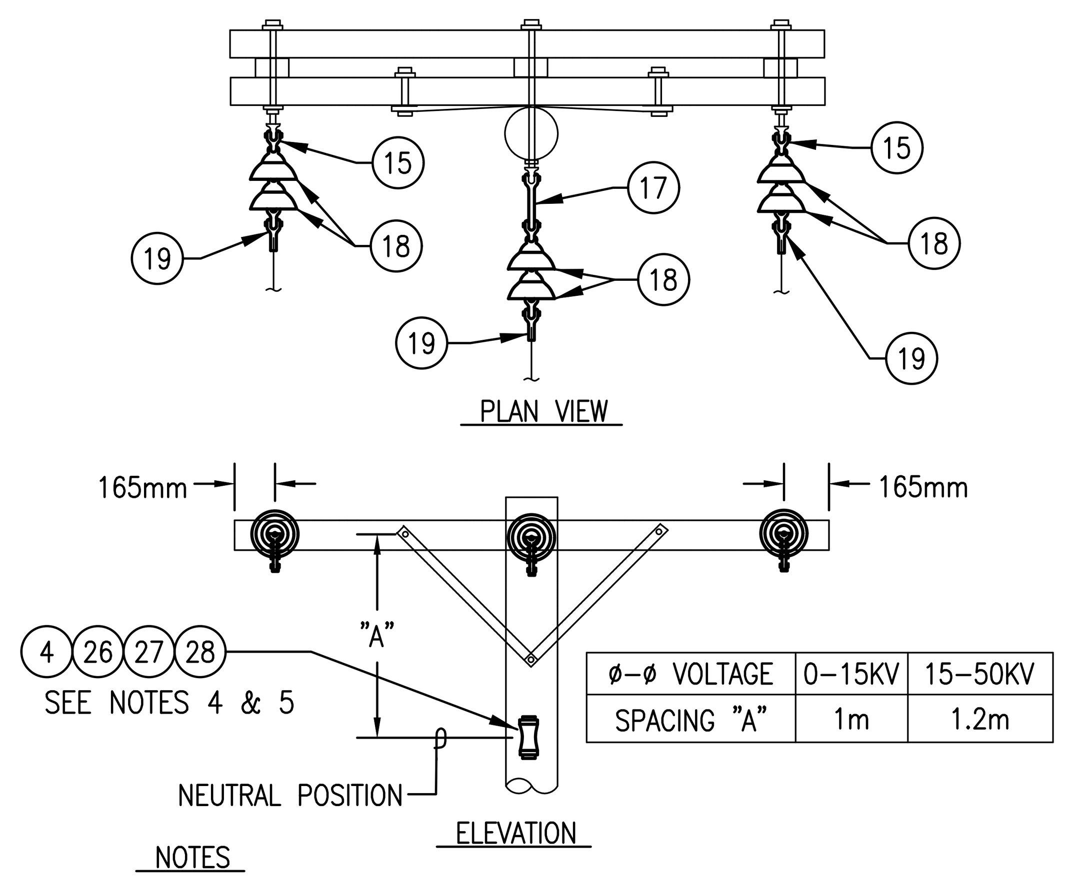
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC NORFOLK, VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS
VARIOUS LOCATIONS

JOINT MARITIME FACILITY
OVERHEAD ELECTRICAL SITE DETAILS

SCALE: NONE
EPROJECT NO.: 1838720
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 14157202
SHEET 140 OF 170
ES503

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2016

FILE NAME: Y:\DCOMO\SYS\~MultipleSites\2026_1783264_P1560_MJMF\B_Design\DWGS\Sheet\13_ELEC\P1560_1783264_ES503.dwg LAYOUT NAME: ES503 – Overhead Electrical Site Details PLOTTED: Tuesday, January 20, 2026 – 12:15pm USER: eric.herlan



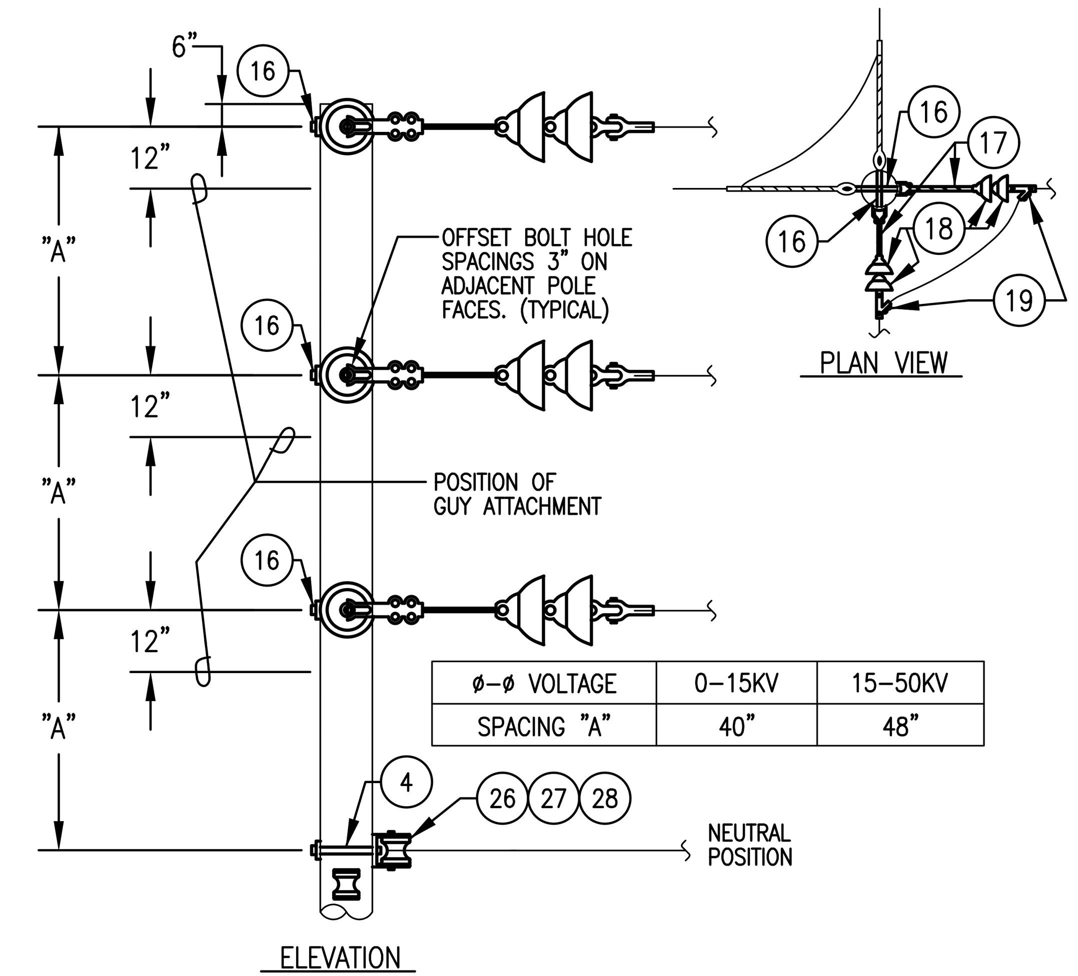
ø–ø VOLTAGE	0–15KV	15–50KV
SPACING "A"	1m	1.2m

NOTES

1. DRAWING REPRESENTS SYMBOL FDE3–N. ELIMINATE INSULATOR ASSEMBLY FOR MIDDLE PHASE ON SYMBOLS FDE2 AND FDE2–N.
2. DRAWING REPRESENTS DEADEND ASSEMBLY FOR CIRCUIT VOLTAGES >5KV AND ≤15KV. REFER TO SPECIFICATIONS SECTION 16301 FOR NUMBER AND CLASS OF INSULATORS REQUIRED FOR EACH VOLTAGE LEVEL.
3. OMIT ITEMS (4), (26) (27) AND (28) FOR NEUTRAL ON FDE2 AND FDE3.
4. FOR NEUTRAL CONDUCTOR LARGER THAN #2 AWG, SUBSTITUTE (19) FOR (26), (27) AND (28).
5. MODIFY THE NEUTRAL SPACING "A" AS INDICATED ON OTHER SKETCHES FOR TRANSFORMER AND U.G. TERMINAL INSTALLATIONS.

FDE3–N, FDE3, FDE2–N, FDE2
(0–50KV)

SKETCH DATE	JUNE 2002	STYLE	OH–15



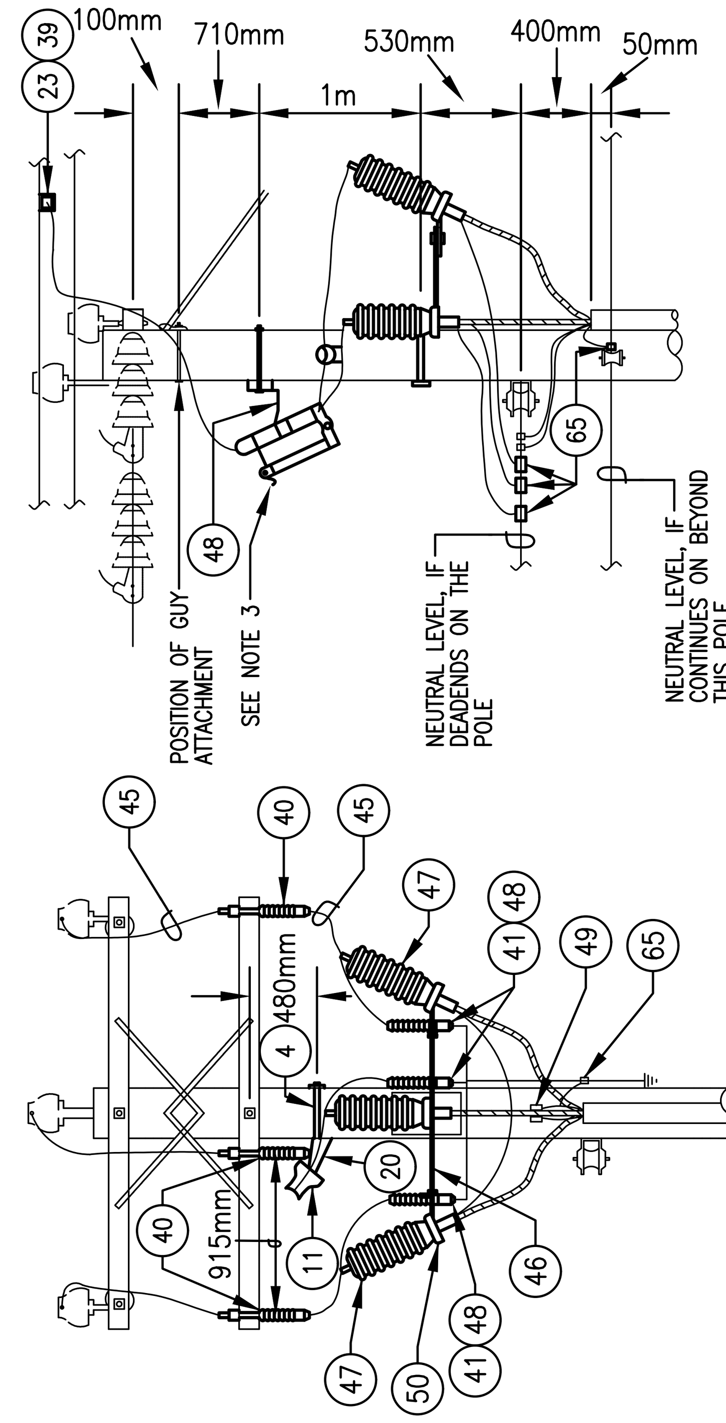
ø–ø VOLTAGE	0–15KV	15–50KV
SPACING "A"	40"	48"

NOTES

1. DRAWING REPRESENTS VDDE3–N FOR CIRCUIT VOLTAGES >5KV AND ≤15KV. MODIFY INSULATOR ASSEMBLIES AS REQUIRED TO COINCIDE WITH THE NUMBER OF PHASE CONDUCTORS. REFER TO SPECIFICATION SECTION 16301 FOR THE REQUIRED NUMBER AND CLASS OF INSULATORS.
2. OMIT ITEMS (4) (26) (27) AND (28) FOR NEUTRAL ON ALL SYMBOLS WHICH DO NOT CONTAIN "N".
3. FOR NEUTRAL CONDUCTORS LARGER THAN #1/0 AWG, PROVIDE (15) AND (19) IN LIEU OF (26) (27) AND (28). TWO INSULATOR ASSEMBLIES REQUIRED FOR NEUTRAL.

VDDE3–N, VDDE3, VDDE2–N, VDDE2, VDDE1–N, VDDE1
(0–50KV)

SKETCH DATE	JUNE 2002	STYLE	OH–24

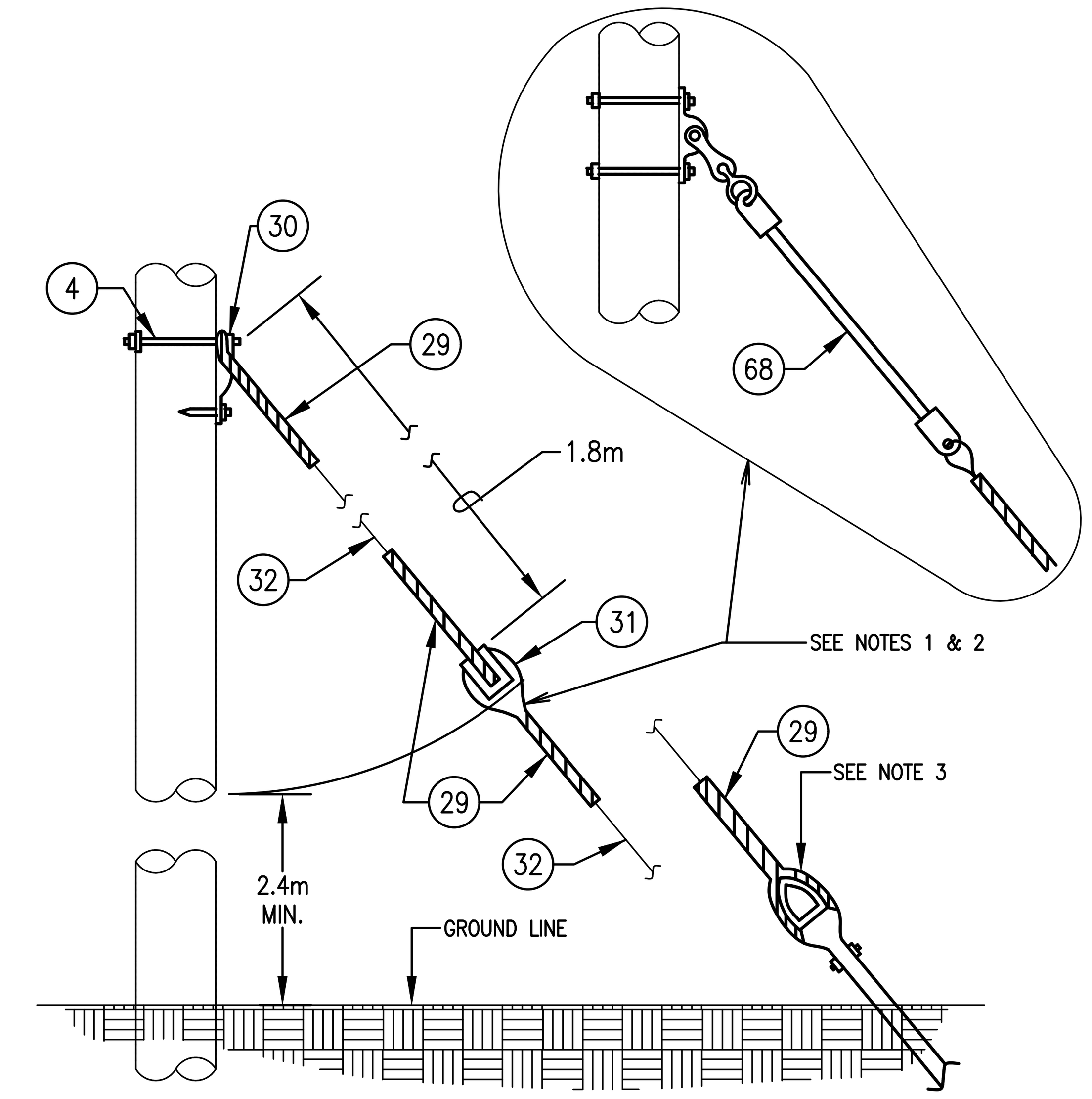


NOTES

1. MODIFY POSITION OF TERMINAL ON DEADENDS TO BE UNDER THE CONDUCTORS AND THE CUTOUTS ON THE BACKSIDE OF CROSSARM. POLE RISER MUST BE OFFSET TO CLEAR NEUTRAL CLEVIS BRACKET.
2. CONNECT BOTTOM LEAD OF ARRESTER DIRECTLY TO POLE GROUND. INTERCONNECT CABLE INSULATION SHIELD DRAIN WIRES AND CONDUIT RISER GROUND TO MULTI-GROUNDED NEUTRAL (IF EXISTING) AND POLE GROUND.
3. CUTOUT PROVIDES A FUSE OR A SOLID BLADE (NON-FUSED) OPTION. COORDINATE WITH SPECIFIC DESIGN REQUIREMENTS PROVIDED.

U.G. TERMINAL
(0–25KV)

SKETCH DATE	JUNE 2002	STYLE	OH–31

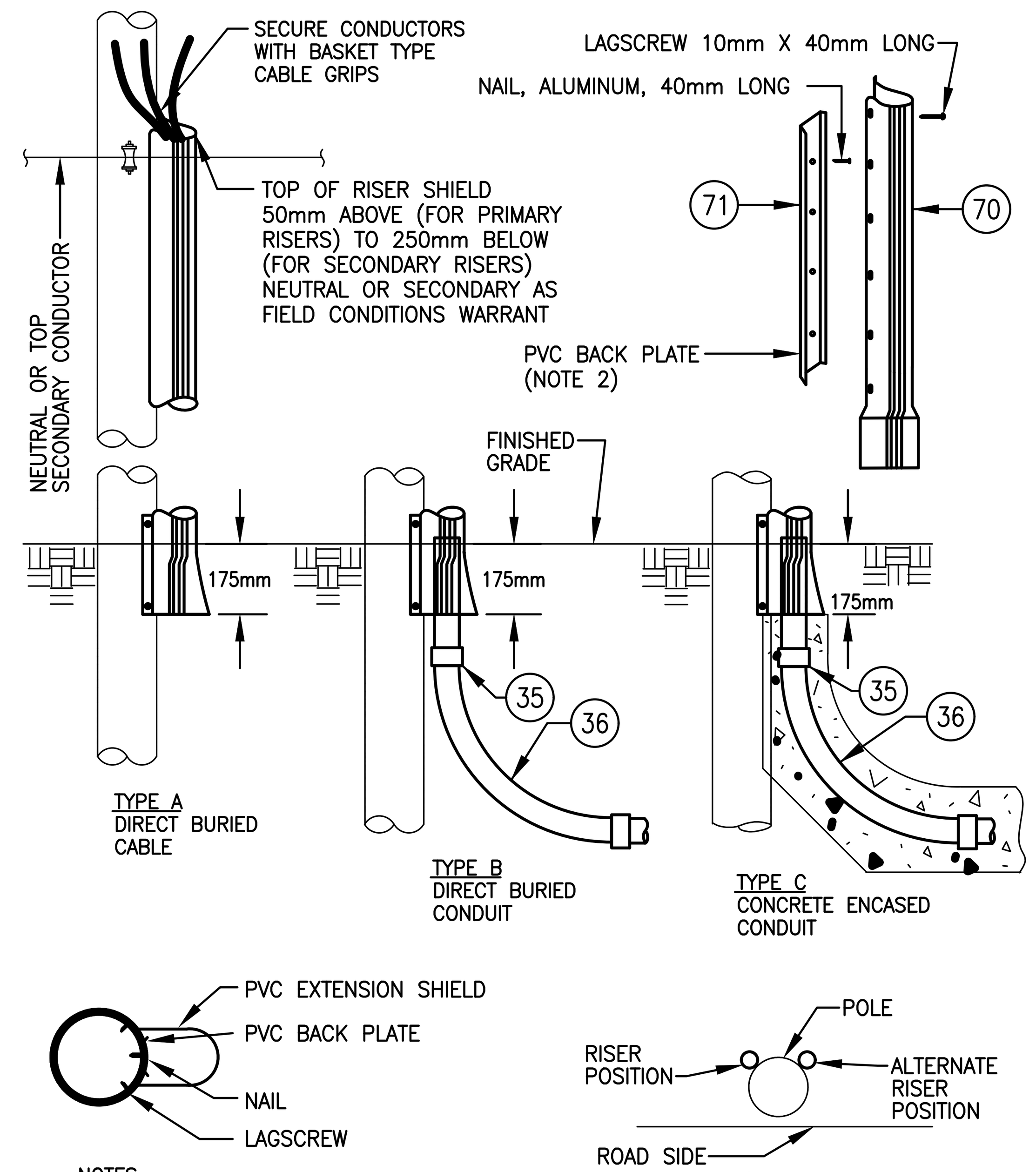


NOTES

1. DRAWING REPRESENTS SYMBOL FOR "GUY–I". OMIT ITEM (31) FOR THE "GUY" SYMBOL.
2. ON CIRCUIT OPERATING VOLTAGES GREATER THAN 15KV, SUBSTITUTE (68) FOR (31).
3. COORDINATE INSTALLATION WITH ANCHOR AS SPECIFIED.
4. UTILIZE ITEM (68) WHEN GUYING ATTACHMENT IS LOCATED IN THE PRIMARY AREA OF THE POLE AS INDICATED BY SPECIFIC DESIGN REQUIREMENTS PROVIDED.
5. BOND ALL GUYS (SUPPLY & COMMUNICATION) AND CONNECT TO POLE GROUND AND SYSTEM NEUTRAL (IF EXISTING).

GUY–I
GUY

SKETCH DATE	JUNE 2002	STYLE	OH–32

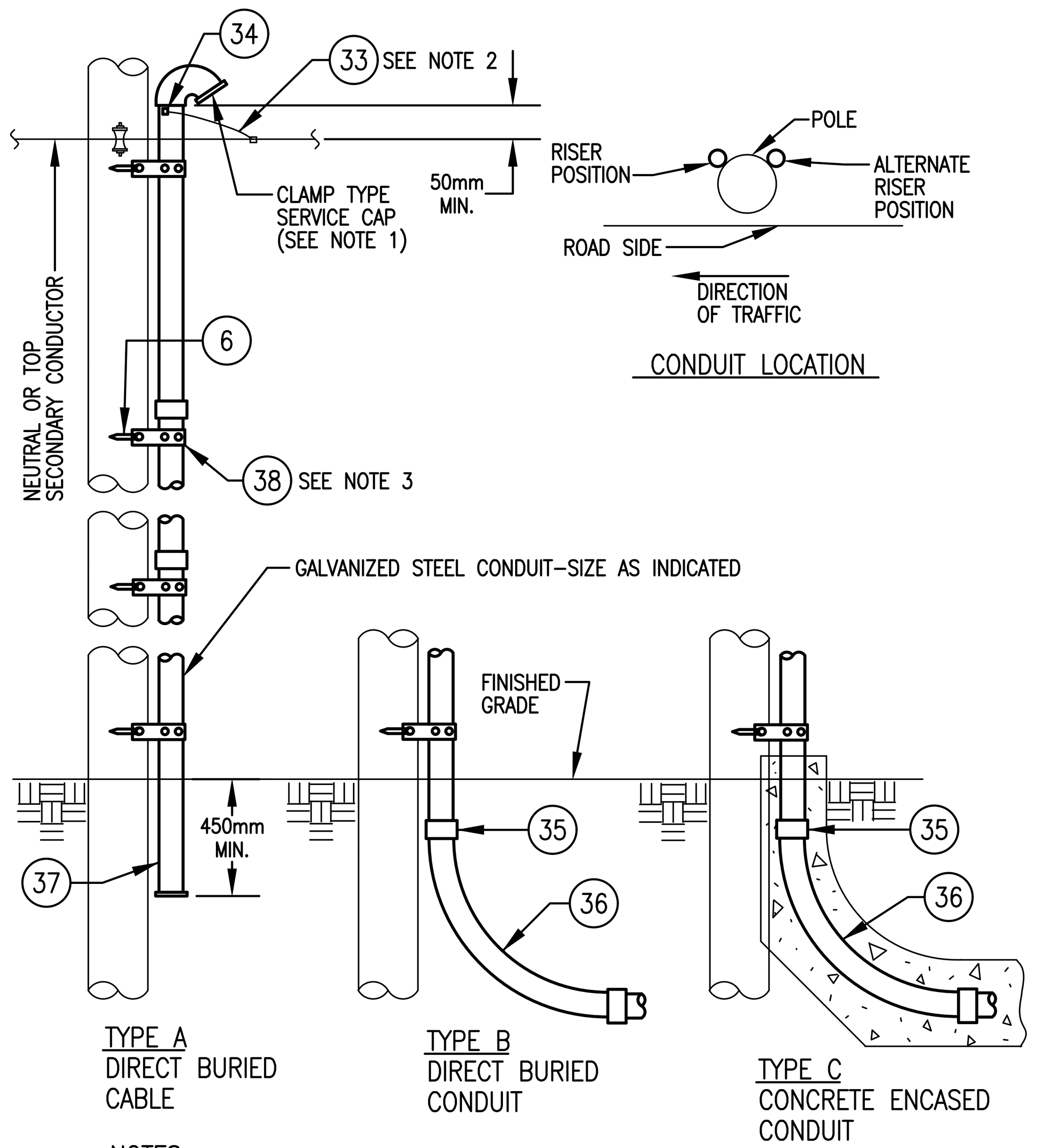


NOTES

1. INSTALL RISER ON SIDE OR QUARTER OF POLE AWAY FROM CLIMBING SURFACE.
2. INSTALL BACK PLATE WITH THE RISER SHIELD FROM THE BOTTOM OF THE POLE.

PVC RISER SHIELD
(SIZE & TYPE AS INDICATED)

SKETCH DATE	JUNE 2002	STYLE	OH–34

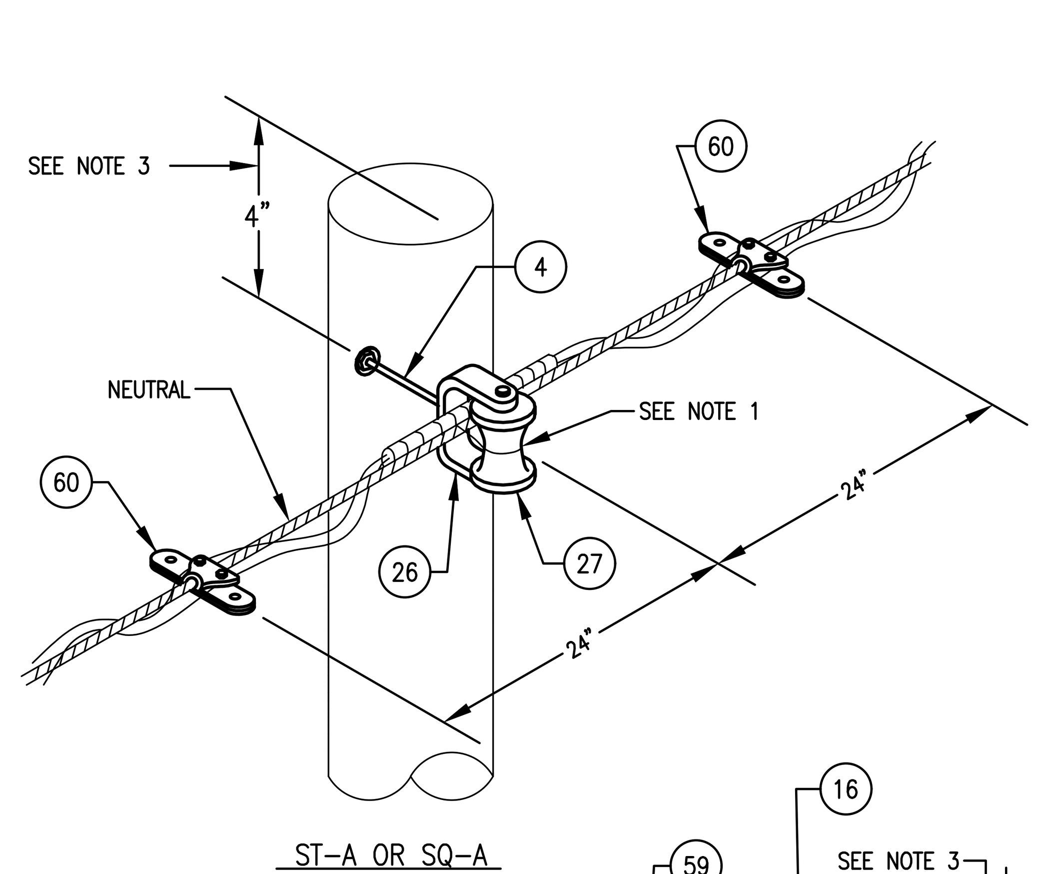


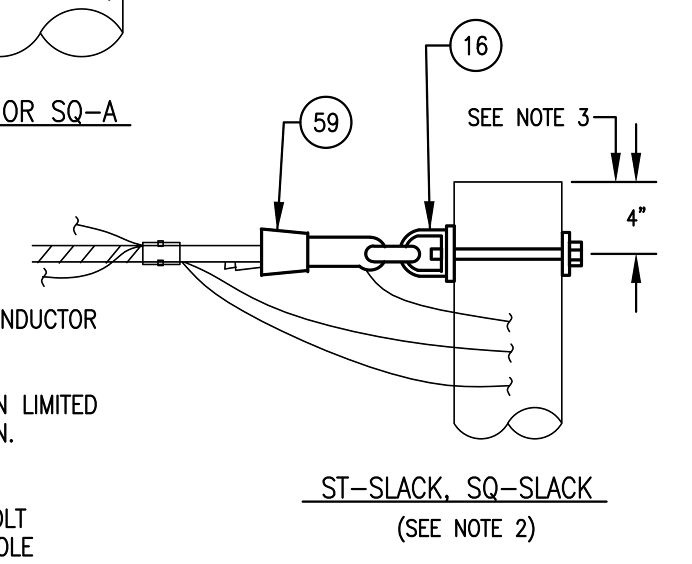
NOTES

1. ON CONDUIT RISER FOR PRIMARY CIRCUITS, ELIMINATE SERVICE CAP AND PROVIDE GROUNDING TYPE INSULATING BUSHING.
2. BOND CONDUIT TO POLE GROUND AND SYSTEM NEUTRAL (IF EXISTING). SEE GROUNDING NOTES ON SKETCH OH–41.
3. SPACE STRAPS AT MAXIMUM OF 1.2m INTERVALS.

CONDUIT RISER
(SIZE & TYPE AS INDICATED)

SKETCH DATE	JUNE 2002	STYLE	OH–35



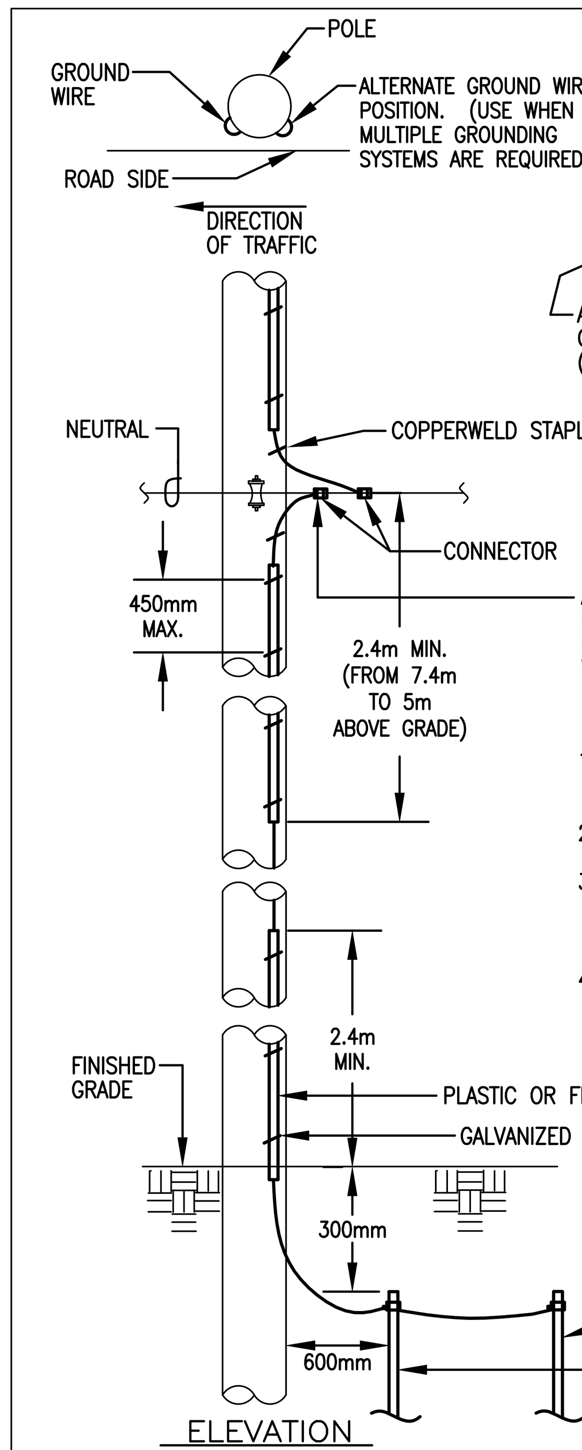


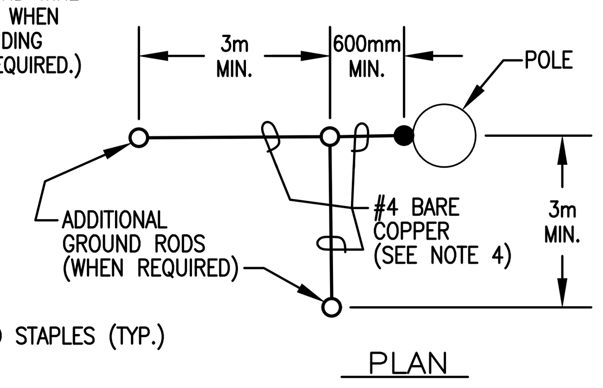
NOTES

1. TIE ONLY THE NEUTRAL CONDUCTOR TO THE SPOOL INSULATOR.
2. SLACK SPAN CONSTRUCTION LIMITED TO MAXIMUM OF 100' SPAN.
3. WHEN USED AT TOP OF POLE, INSTALL MACHINE BOLT 4" DOWN FROM TOP OF POLE

ST–SLACK, SQ–SLACK, ST–A, SQ–A
(0–600V)

SKETCH DATE	JUNE 2002	STYLE	OH–38





NOTES

1. USE ADDITIONAL GROUND RODS, WHEN REQUIRED, TO OBTAIN RESISTANCE VALUE, AS REQUIRED BY IEEE C2.
2. MAKE GROUNDING CONNECTIONS TO ALL EQUIPMENT ON POLE AS REQUIRED BY IEEE C2.
3. FOR SYSTEMS OTHER THAN 4 WIRE, MULTI-GROUNDED, USE SEPARATE GROUNDING CONDUCTORS AND ELECTRODES FOR PRIMARY AND SECONDARY SYSTEM DEVICE CONNECTIONS.
4. #4 600V INSULATED COPPER GROUNDING CONDUCTOR REQUIRED ON PRIMARY SYSTEM WHEN SEPARATE SYSTEMS ARE REQUIRED.

GROUND

SKETCH DATE	JUNE 2002	STYLE	OH–41

SYM | DESCRIPTION | DATE | APPR

SEAL

A/E INFO

APPROVED: Wm B Forbes — Digitally signed by signature Date: 2026.01.28 16:59:01-05'00'
FOR COMMANDER NAVFAC

ACTIVITY: D. EVANS VIA EMAIL

SATISFACTORY TO	DATE 01/15/2026
DES CAB	DRW CAB	CHK SEK
PM/DM	MJD/ATP
BRANCH MANAGER	SEK
DES PROD DIR	WBF
FIRE PROTECTION ENGINEER	DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC
NORFOLK, VA
PLANNING, DESIGN AND CONSTRUCTION
VARIOUS LOCATIONS

VARIOUS LOCATIONS
JOINT MARITIME FACILITY
OVERHEAD ELECTRICAL SITE DETAILS

SCALE:	NONE
EPROJECT NO.:	1838720
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14157203
SHEET 141	OF 170

ES504

NAVFAC METRIC DRAWFORM REVISION: 01 OCTOBER 2018